

64. Graphics LCD Controller (GLCDC)

64.1 Overview

The multifunctional Graphics LCD Controller (GLCDC) supports multiple data formats and panels. Key GLCDC features include:

- GLCDC0 and GLCDC1 bus master function for accessing graphics data
- Superimposition of three planes (single color background plane, graphics 1 plane, and graphics 2 plane)
- Supports many types of 32- and 16-bit/pixel graphics data and 8-, 4-, and 1-bit LUT data formats
- Digital interface signal output supporting the video image size up to WXGA (1280 x 800).

Figure 64.1 shows a block diagram of the GLCDC. See also Figure 62.1 for the block diagram and I/O pins.

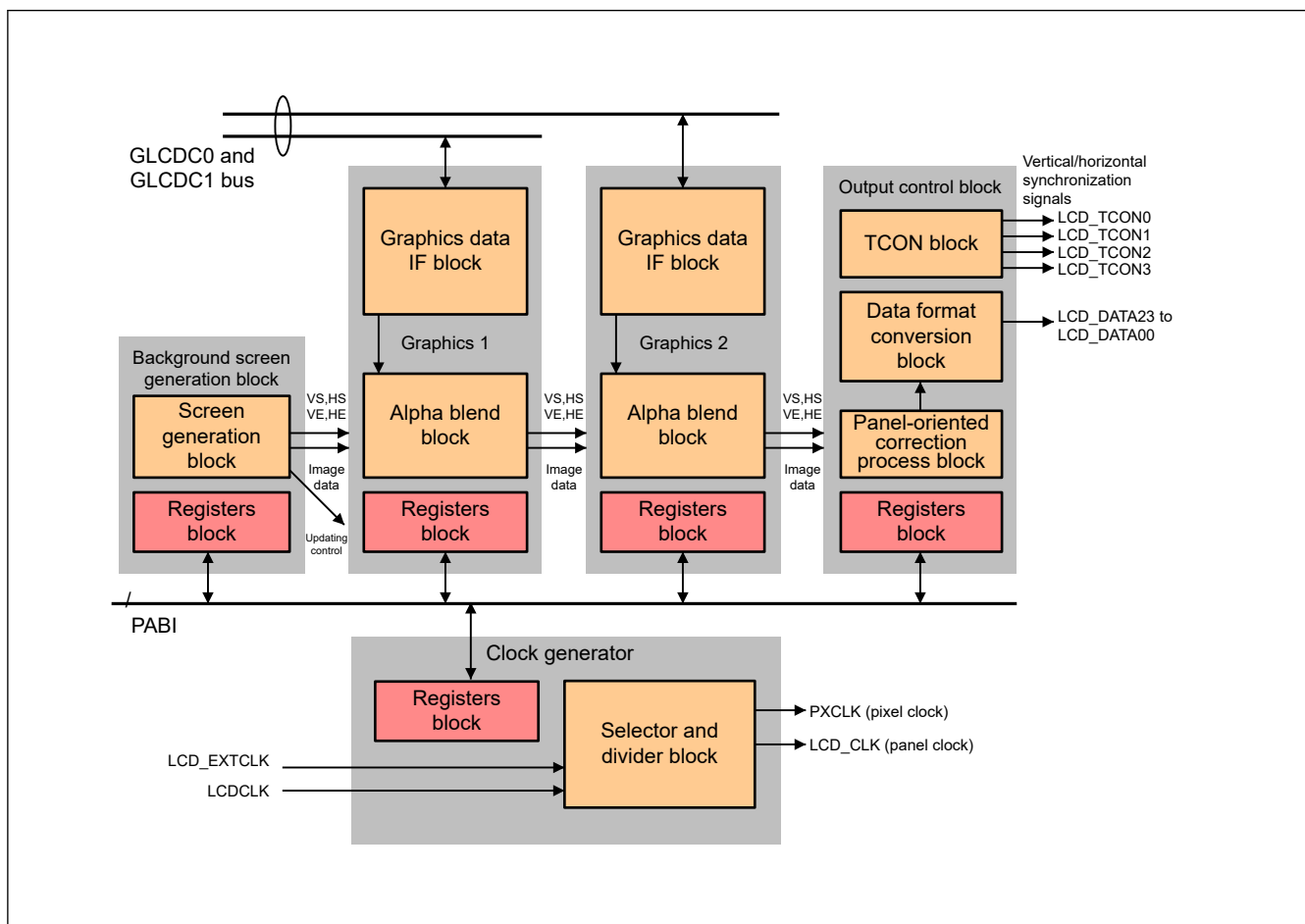


Figure 64.1 GLCDC block diagram

Table 64.1 GLCDC I/O pins

Function	Pin name	I/O	Description
GLCDC	LCD_EXTCLK	Input	Panel clock source input pin
	LCD_CLK	Output	Panel clock output pin
	LCD_DATA23 to LCD_DATA00	Output	LCD signal output pins RGB888 signal R/G/B: 8 bits each (unsigned) RGB666 signal R/G/B: 6 bits each (unsigned) RGB565 signal R/B: 5 bits each (unsigned) G: 6 bits (unsigned) Serial RGB signal R/G/B/-: 8 bits (unsigned)
	LCD_TCON3	Output	LCD_TCON3 output signal pin
	LCD_TCON2	Output	LCD_TCON2 output signal pin
	LCD_TCON1	Output	LCD_TCON1 output signal pin
	LCD_TCON0	Output	LCD_TCON0 output signal pin

Table 64.2 provides a functional overview of GLCDC.

Table 64.2 Functional overview of GLCDC (1 of 2)

Parameter		Function
Graphics	Graphics planes	- Single color background (lowest layer) and two graphics planes - Graphics planes can be alpha-blended with the lower-layer plane
	Pixel format	- RGB-888 progressive format (-: 8 bits, R: 8 bits, G: 8 bits, B: 8 bits; 32 bits in total) - ARGB8888 progressive format (A: 8 bits, R: 8 bits, G: 8 bits, B: 8 bits; 32 bits in total) - RGB565 progressive format (A: None, R: 5 bits, G: 6 bits, B: 5 bits; 16 bits in total) - ARGB1555 progressive format (CLUT: 1 bit, R: 5 bits, G: 5 bits, B: 5 bits; 16 bits in total) - ARGB4444 progressive format (A: 4 bits, R: 4 bits, G: 4 bits, B: 4 bits; 16 bits in total) - CLUT8 progressive format (color palette address: 8 bits) - CLUT4 progressive format (color palette address: 4 bits) - CLUT1 progressive format (color palette address: 1 bit) - CLUT memory: 512 words × 32 bits per graphics plane (ARGB8888)
	Frame buffer control	The following parameters can be set for the frame buffer: - Base address: Start address of the frame buffer, aligned with a 64-byte boundary (burst transfer size) - Macro line offset: Offset address from the start address to the next macro line, aligned with a 64-byte boundary (burst transfer size) - Frame offset: Offset address from the start address to the next frame, aligned with a 64-byte boundary (burst transfer size) - Number of data transfers: Number of data transfers of a macro line - Number of macro lines: Number of macro lines in a single frame
	Alpha blending	- Alpha blending in rectangular area (blending ratio: 256 gradation levels) - Alpha blending in pixel units (blending ratio: 256 gradation levels) - RGB-index chroma key (replaced with the specified color when the object color agrees with the preset value)
Internal video image format		- Total number of vertical lines: Up to 2048 lines - Number of valid vertical lines: 16 to 2044 lines (resolution: 1 line) - Vertical front porch: 2 lines (minimum) - Vertical back porch: 1 line (minimum) - Vertical pulse width: 1 line - Total number of horizontal pixels: Up to 2048 pixels - Number of valid horizontal pixels: 16 to 2040 pixels (resolution: 2 pixels) - Horizontal front porch: 3 pixels (minimum) (preliminary) - Horizontal back porch: 1 pixel (minimum) (preliminary) - Horizontal pulse width: 4 pixels

Table 64.2 Functional overview of GLCDC (2 of 2)

Parameter		Function
Data format conversion	Output video image size	From 16 lines × 16 pixels to 2044 lines × 2040 pixels
	Data format	- RGB888 (parallel 24 bits) - RGB666 (parallel 18 bits) - RGB565 (parallel 16 bits) - RGB888 (serial 8 bits); clock cycle is four times the pixel clock - Bit endian order change and B/R signal swap
	Dither processing	- Reduces 10-bit signals (output after panel-oriented correction) to 8-, 6-, or 5-bit signals (output data format) - Supports the following modes: - Truncate mode - Round-off mode 2 × 2 pattern dither mode
Timing control signal	Signal generation	5 timing signals (STVA, STVB, STHA, STHB, and DE) can be generated from HS / VS: - Vertical synchronization signal (variable) - Horizontal synchronization signal (variable) - Data enable signal
	Signal select	Signals generated by the signal generation circuit can be output from LCD_TCONn pins (n = 0 to 3)
Output control panel-oriented correction processing	Brightness and contrast	10-bit internal processing; the sequence of this processing and gamma correction can be swapped. - Brightness: DC value adjustment range: from -512[LSB] to +512[LSB] - Contrast: gain value adjustment range: from 0 to 2 times (from 0/128 to 255/128)
	Gamma correction	- Sixteen areas; input/output: 10 bits - Gain value adjustment range in the area: From 0 to 2 times (from 0/1024 to 2047/1024)
Interrupts		- Number of specified lines interrupt (GLCDC_VPOS) - Graphics 1 buffer underflow interrupt (GLCDC_L1UNDF) - Graphics 2 buffer underflow interrupt (GLCDC_L2UNDF)
Module-stop function		Module-stop state can be set to reduce power consumption.
TrustZone Filter		Security and Privilege attribution can be set.

64.1.1 GLCDC Configuration

Figure 64.2 shows the configuration of the GLCDC.

The GLCDC includes the following blocks:

- Background screen generation block: Generates the background screen (including the blanking interval), selects the background color, and generates the synchronization signals for controlling the screens.
- Graphics data interface blocks (2 blocks): Convert the graphics data/CLUT data read through the GLCDC0 and GLCDC1 bus into ARGB (8888) data for internal processing, and transfer the clocks (PCLKA → PXCLK).
- Alpha blending blocks (2 blocks): Superimpose graphics data on the lower-layer screen and perform alpha blending based on the register settings and the alpha blending values for the current screen graphics data.
- TCON block: Generates the vertical and horizontal synchronization and enable signals suited for the specifications of the connected panel, from the internal vertical and horizontal synchronization signals.
- Data format conversion block: Processes data into the specific internal RGB888 format into the data of the specific format suited for the specifications of the panel with dither correction for output image data length.
- Panel-oriented correction processing block: Corrects brightness and contrast, and performs gamma correction suited for the characteristics of the connected panel, allowing either brightness and contrast correction or gamma correction to be performed first.
- Clock generator block: Generates the pixel and the panel clocks on a specific frequency from either LCD_EXTCLK or LCDCLK.

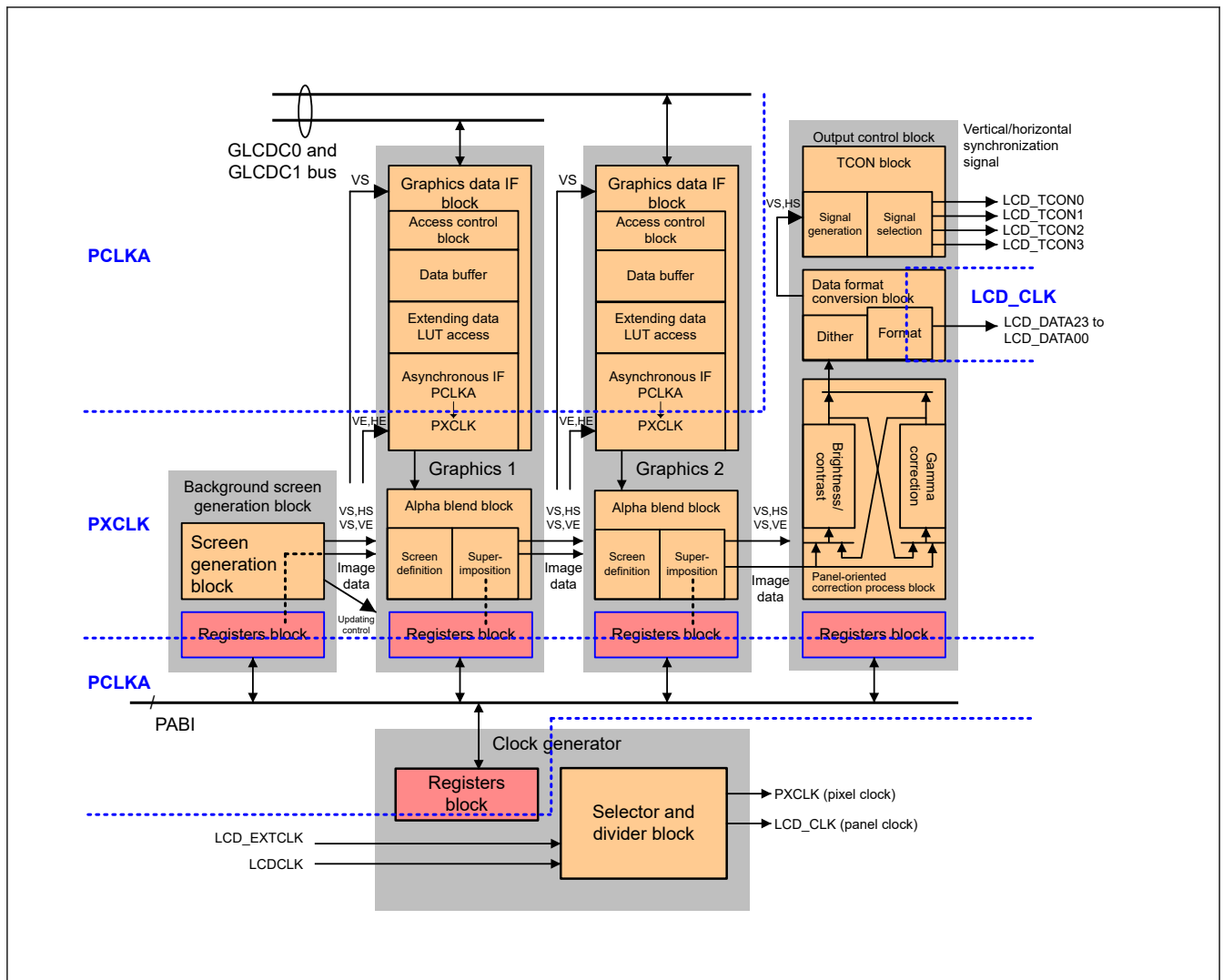


Figure 64.2 GLCDC configuration

64.1.2 Screen Format

Figure 64.3 shows the screen format overview of the GLCDC.

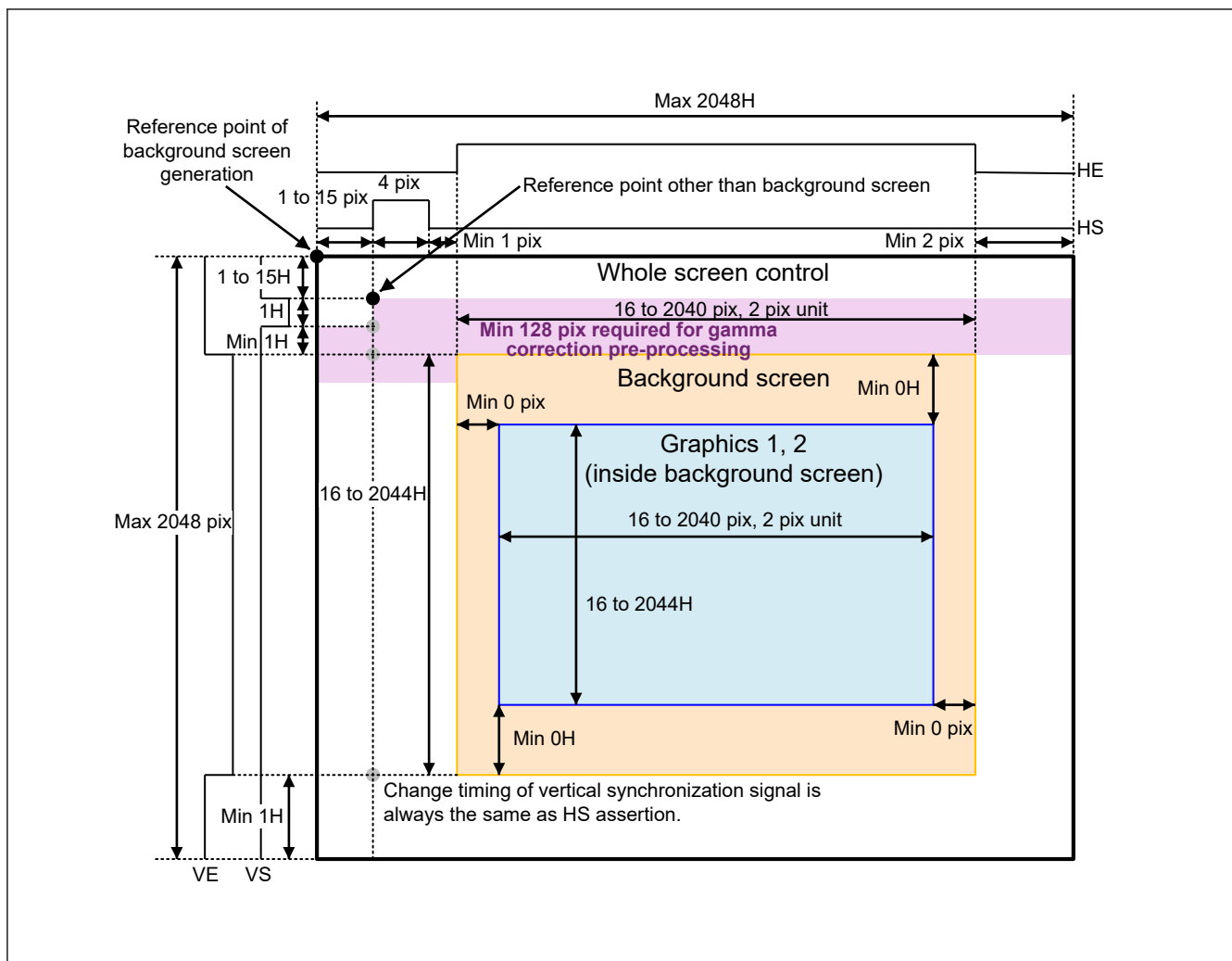


Figure 64.3 Screen format

The background screen generation block generates the essential timing signals for operations in the module as a whole. The graphics data interface blocks, alpha blending blocks, gamma correction block, output control block, and TCON block operate based on the synchronization and enable signals, which are sequentially transferred from the background screen generation block.

64.1.3 Graphics and Color Palette (CLUT) Data Formats

The GLCDC handles three display screens, one of which is a background screen. For the background screen, frame data is RGB888 graphics data stored in the registers, and for the other two screens, frame data is stored in the memory connected to the GLCDC0 and GLCDC1 bus as 32- or 16-bit/pixel graphics data or 8-, 4-, or 1-bit/pixel color palette (CLUT) data. The frame data of the relevant screen is read by the graphics data interface block, read into the GLCDC, and extended (converted) into ARGB8888 data for superimposition and blending. [Figure 64.4](#) shows the frame data formats.

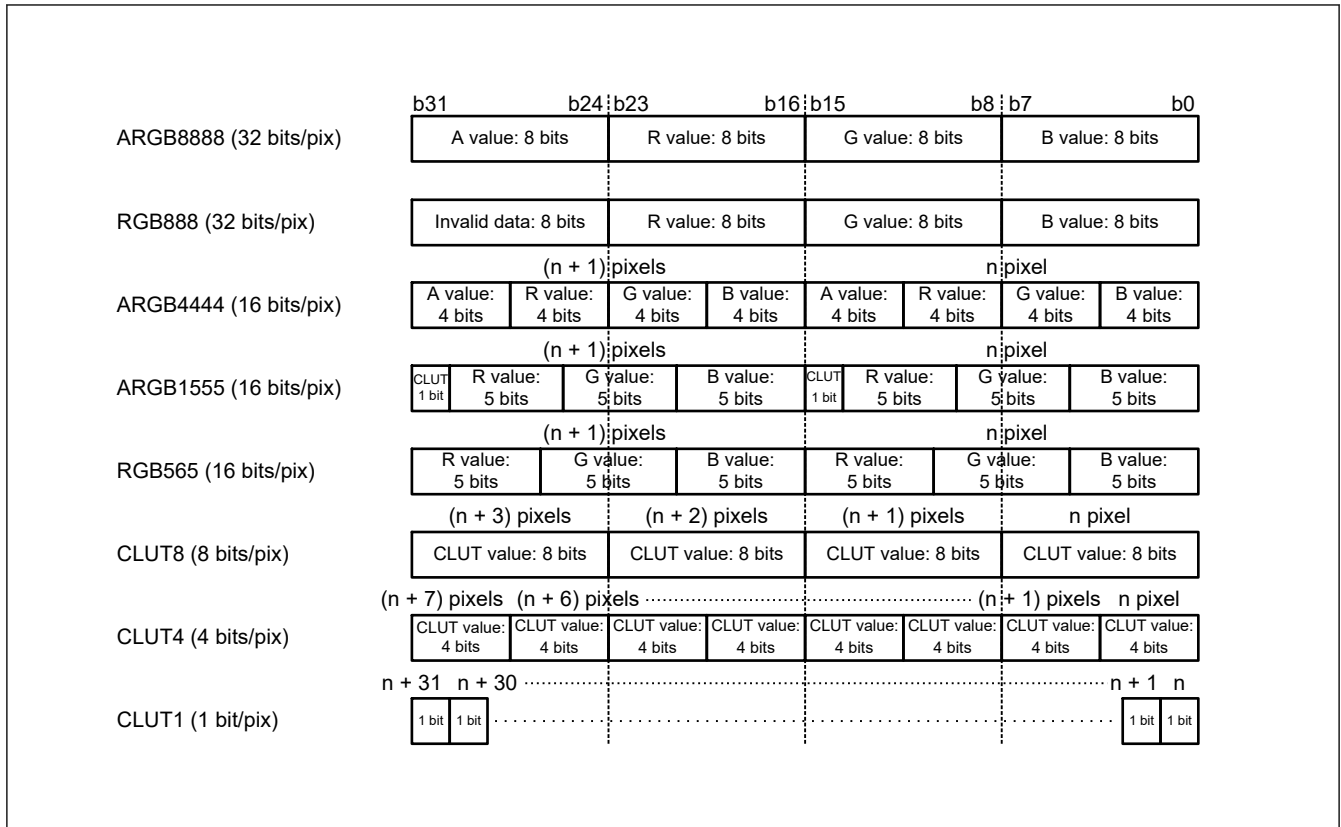


Figure 64.4 Frame data format

The A value (alpha blending value) represents the blending ratio of pixel data for displaying the lower-layer and current graphics screens after superimposition. The blending process is performed in accordance with the equations given in [section 64.1.7. Graphics Data Interface](#). The CLUT value represents the color palette memory address (plane 0 or 1 is selected in the registers), and the data in the color palette is ARGB8888 (32 bits/pixel) graphics data. Addresses 0x01/0x00 are accessed with CLUT1, 0x0F to 0x00 with CLUT4, and 0xFF to 0x00 with CLUT8. Addresses 0x80/0x00 are accessed with ARGB1555 data.

64.1.4 Output Control for Data Format

The GLCDC can output data in accordance with the following formats and register settings:

- Data formats
 - Parallel: RGB888, RGB666, and RGB565
 - Serial: RGB888
 - Lower bits are processed in one of the following modes when 10-bit signals are reduced to 8-, 6-, and 5-bit signals:
 - Truncate mode
 - Round-off mode
 - 2×2 pattern dither processing + truncate mode
- Pixel arrangement
 - RGB
 - BGR
- Scan direction select for serial RGB888
 - Forward scan
 - Reverse scan
- Data output delay of serial RGB888

- 0 to 3 clock cycles.
- Pin assignment
 - Little endian
 - Big endian

For details on dither processing, see [section 64.2.48. OUT_PDTHA : Output Control Block Panel Dither Correction Register](#).

Bit assignment of LCD signals for parallel RGB888 format

[Table 64.3](#) shows RGB signal inputs assigned to the LCD signal outputs for the parallel RGB888 output format.

Table 64.3 Bit assignment of RGB signal inputs for parallel RGB888 format

Pin assignment	Pixel arrangement			
	RGB, little endian	RGB, big endian	BGR, little endian	BGR, big endian
LCD_DATA23	R[7]	R[0]	B[7]	B[0]
LCD_DATA22	R[6]	R[1]	B[6]	B[1]
LCD_DATA21	R[5]	R[2]	B[5]	B[2]
LCD_DATA20	R[4]	R[3]	B[4]	B[3]
LCD_DATA19	R[3]	R[4]	B[3]	B[4]
LCD_DATA18	R[2]	R[5]	B[2]	B[5]
LCD_DATA17	R[1]	R[6]	B[1]	B[6]
LCD_DATA16	R[0]	R[7]	B[0]	B[7]
LCD_DATA15	G[7]	G[0]	G[7]	G[0]
LCD_DATA14	G[6]	G[1]	G[6]	G[1]
LCD_DATA13	G[5]	G[2]	G[5]	G[2]
LCD_DATA12	G[4]	G[3]	G[4]	G[3]
LCD_DATA11	G[3]	G[4]	G[3]	G[4]
LCD_DATA10	G[2]	G[5]	G[2]	G[5]
LCD_DATA09	G[1]	G[6]	G[1]	G[6]
LCD_DATA08	G[0]	G[7]	G[0]	G[7]
LCD_DATA07	B[7]	B[0]	R[7]	R[0]
LCD_DATA06	B[6]	B[1]	R[6]	R[1]
LCD_DATA05	B[5]	B[2]	R[5]	R[2]
LCD_DATA04	B[4]	B[3]	R[4]	R[3]
LCD_DATA03	B[3]	B[4]	R[3]	R[4]
LCD_DATA02	B[2]	B[5]	R[2]	R[5]
LCD_DATA01	B[1]	B[6]	R[1]	R[6]
LCD_DATA00	B[0]	B[7]	R[0]	R[7]

Note: R[7:0], G[7:0], and B[7:0] are RGB pixel data that is internally processed.

Bit assignment of LCD signals for parallel RGB666 format

[Table 64.4](#) shows RGB signal inputs assigned to the LCD signal outputs for the parallel RGB666 output format.

Table 64.4 Bit assignment of RGB signal inputs for parallel RGB666 format (1 of 2)

Pin assignment	Pixel arrangement			
	RGB, little endian	RGB, big endian	BGR, little endian	BGR, big endian
LCD_DATA23	Fixed to 0	Fixed to 0	Fixed to 0	Fixed to 0

Table 64.4 Bit assignment of RGB signal inputs for parallel RGB666 format (2 of 2)

Pin assignment	Pixel arrangement			
	RGB, little endian	RGB, big endian	BGR, little endian	BGR, big endian
LCD_DATA22	Fixed to 0	Fixed to 0	Fixed to 0	Fixed to 0
LCD_DATA21	Fixed to 0	Fixed to 0	Fixed to 0	Fixed to 0
LCD_DATA20	Fixed to 0	Fixed to 0	Fixed to 0	Fixed to 0
LCD_DATA19	Fixed to 0	Fixed to 0	Fixed to 0	Fixed to 0
LCD_DATA18	Fixed to 0	Fixed to 0	Fixed to 0	Fixed to 0
LCD_DATA17	R[7]	R[2]	B[7]	B[2]
LCD_DATA16	R[6]	R[3]	B[6]	B[3]
LCD_DATA15	R[5]	R[4]	B[5]	B[4]
LCD_DATA14	R[4]	R[5]	B[4]	B[5]
LCD_DATA13	R[3]	R[6]	B[3]	B[6]
LCD_DATA12	R[2]	R[7]	B[2]	B[7]
LCD_DATA11	G[7]	G[2]	G[7]	G[2]
LCD_DATA10	G[6]	G[3]	G[6]	G[3]
LCD_DATA09	G[5]	G[4]	G[5]	G[4]
LCD_DATA08	G[4]	G[5]	G[4]	G[5]
LCD_DATA07	G[3]	G[6]	G[3]	G[6]
LCD_DATA06	G[2]	G[7]	G[2]	G[7]
LCD_DATA05	B[7]	B[2]	R[7]	R[2]
LCD_DATA04	B[6]	B[3]	R[6]	R[3]
LCD_DATA03	B[5]	B[4]	R[5]	R[4]
LCD_DATA02	B[4]	B[5]	R[4]	R[5]
LCD_DATA01	B[3]	B[6]	R[3]	R[6]
LCD_DATA00	B[2]	B[7]	R[2]	R[7]

Note: R[7:2], G[7:2], and B[7:2] are RGB pixel data that is internally processed.

Bit assignment of LCD signals for parallel RGB565 format

Table 64.5 shows RGB signal inputs assigned to the LCD signal outputs for the parallel RGB565 output format.

Table 64.5 Bit assignment of RGB signal inputs for parallel RGB565 format (1 of 2)

Pin assignment	Pixel arrangement			
	RGB, little endian	RGB, big endian	BGR, little endian	BGR, big endian
LCD_DATA23	Fixed to 0	Fixed to 0	Fixed to 0	Fixed to 0
LCD_DATA22	Fixed to 0	Fixed to 0	Fixed to 0	Fixed to 0
LCD_DATA21	Fixed to 0	Fixed to 0	Fixed to 0	Fixed to 0
LCD_DATA20	Fixed to 0	Fixed to 0	Fixed to 0	Fixed to 0
LCD_DATA19	Fixed to 0	Fixed to 0	Fixed to 0	Fixed to 0
LCD_DATA18	Fixed to 0	Fixed to 0	Fixed to 0	Fixed to 0
LCD_DATA17	Fixed to 0	Fixed to 0	Fixed to 0	Fixed to 0
LCD_DATA16	Fixed to 0	Fixed to 0	Fixed to 0	Fixed to 0
LCD_DATA15	R[7]	R[3]	B[7]	B[3]
LCD_DATA14	R[6]	R[4]	B[6]	B[4]
LCD_DATA13	R[5]	R[5]	B[5]	B[5]

Table 64.5 Bit assignment of RGB signal inputs for parallel RGB565 format (2 of 2)

Pin assignment	Pixel arrangement			
	RGB, little endian	RGB, big endian	BGR, little endian	BGR, big endian
LCD_DATA12	R[4]	R[6]	B[4]	B[6]
LCD_DATA11	R[3]	R[7]	B[3]	B[7]
LCD_DATA10	G[7]	G[2]	G[7]	G[2]
LCD_DATA09	G[6]	G[3]	G[6]	G[3]
LCD_DATA08	G[5]	G[4]	G[5]	G[4]
LCD_DATA07	G[4]	G[5]	G[4]	G[5]
LCD_DATA06	G[3]	G[6]	G[3]	G[6]
LCD_DATA05	G[2]	G[7]	G[2]	G[7]
LCD_DATA04	B[7]	B[3]	R[7]	R[3]
LCD_DATA03	B[6]	B[4]	R[6]	R[4]
LCD_DATA02	B[5]	B[5]	R[5]	R[5]
LCD_DATA01	B[4]	B[6]	R[4]	R[6]
LCD_DATA00	B[3]	B[7]	R[3]	R[7]

Note: R[7:3], G[7:2], and B[7:3] are RGB pixel data that is internally processed.

Bit assignment of RGB signal inputs for serial RGB888 format

Table 64.6 and Table 64.7 show RGB signal inputs assigned to the LCD signal outputs for the serial RGB888 output format.

Table 64.6 Bit assignment of RGB signal inputs for serial RGB888 format, RGB arrangement

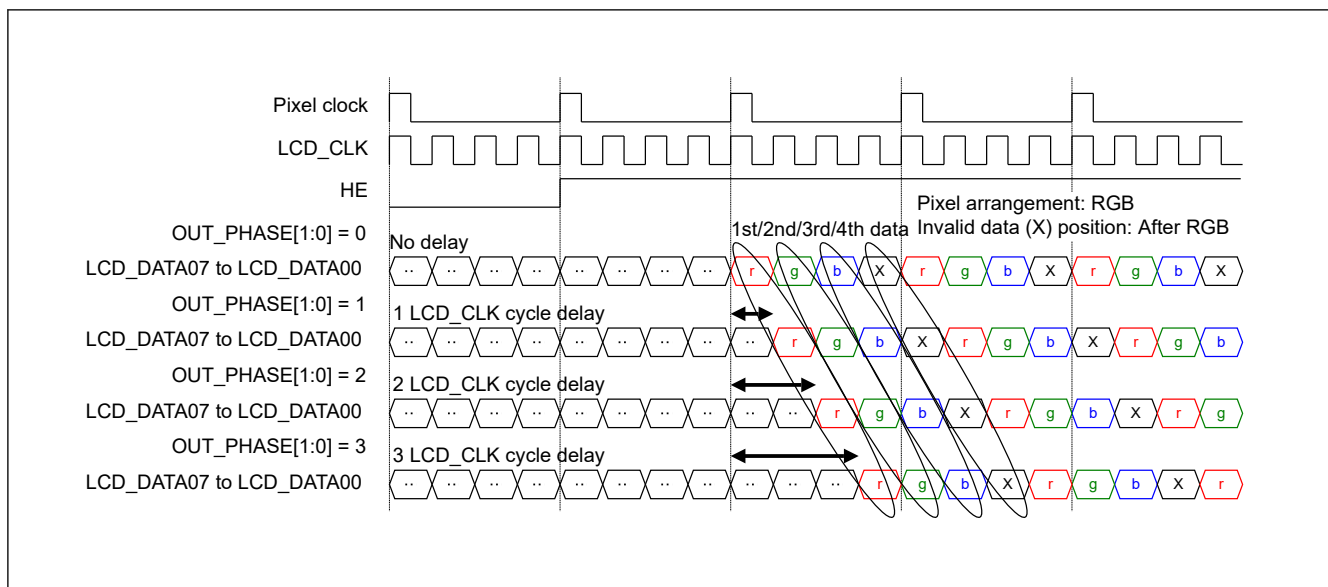
Pin assignment	Pixel arrangement, scan direction select, and cycle															
	RGB, little endian								RGB, big endian							
	Reverse scan				Forward scan				Reverse scan				Forward scan			
	1st	2nd	3rd	4th	1st	2nd	3rd	4th	1st	2nd	3rd	4th	1st	2nd	3rd	4th
LCD_DATA23	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:
LCD_DATA08	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
LCD_DATA07	Undefined	B[7]	G[7]	R[7]	R[7]	G[7]	B[7]	Undefined	Undefined	B[0]	G[0]	R[0]	R[0]	G[0]	B[0]	Undefined
LCD_DATA06	Undefined	B[6]	G[6]	R[6]	R[6]	G[6]	B[6]	Undefined	Undefined	B[1]	G[1]	R[1]	R[1]	G[1]	B[1]	Undefined
LCD_DATA05	Undefined	B[5]	G[5]	R[5]	R[5]	G[5]	B[5]	Undefined	Undefined	B[2]	G[2]	R[2]	R[2]	G[2]	B[2]	Undefined
LCD_DATA04	Undefined	B[4]	G[4]	R[4]	R[4]	G[4]	B[4]	Undefined	Undefined	B[3]	G[3]	R[3]	R[3]	G[3]	B[3]	Undefined
LCD_DATA03	Undefined	B[3]	G[3]	R[3]	R[3]	G[3]	B[3]	Undefined	Undefined	B[4]	G[4]	R[4]	R[4]	G[4]	B[4]	Undefined
LCD_DATA02	Undefined	B[2]	G[2]	R[2]	R[2]	G[2]	B[2]	Undefined	Undefined	B[5]	G[5]	R[5]	R[5]	G[5]	B[5]	Undefined
LCD_DATA01	Undefined	B[1]	G[1]	R[1]	R[1]	G[1]	B[1]	Undefined	Undefined	B[6]	G[6]	R[6]	R[6]	G[6]	B[6]	Undefined
LCD_DATA00	Undefined	B[0]	G[0]	R[0]	R[0]	G[0]	B[0]	Undefined	Undefined	B[7]	G[7]	R[7]	R[7]	G[7]	B[7]	Undefined

Note: R[7:0], G[7:0], and B[7:0] are RGB pixel data that is internally processed.

Table 64.7 Bit assignment of RGB signal inputs for serial RGB888 format, BGR arrangement

Pin assignment	Pixel arrangement, scan direction select, and cycle															
	BGR, little endian								BGR, big endian							
	Reverse scan				Forward scan				Reverse scan				Forward scan			
	1st	2nd	3rd	4th	1st	2nd	3rd	4th	1st	2nd	3rd	4th	1st	2nd	3rd	4th
LCD_DATA23	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:
LCD_DATA08	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
LCD_DATA07	Undefined	R[7]	G[7]	B[7]	B[7]	G[7]	R[7]	Undefined	Undefined	R[0]	G[0]	B[0]	B[0]	G[0]	R[0]	Undefined
LCD_DATA06	Undefined	R[6]	G[6]	B[6]	B[6]	G[6]	R[6]	Undefined	Undefined	R[1]	G[1]	B[1]	B[1]	G[1]	R[1]	Undefined
LCD_DATA05	Undefined	R[5]	G[5]	B[5]	B[5]	G[5]	R[5]	Undefined	Undefined	R[2]	G[2]	B[2]	B[2]	G[2]	R[2]	Undefined
LCD_DATA04	Undefined	R[4]	G[4]	B[4]	B[4]	G[4]	R[4]	Undefined	Undefined	R[3]	G[3]	B[3]	B[3]	G[3]	R[3]	Undefined
LCD_DATA03	Undefined	R[3]	G[3]	B[3]	B[3]	G[3]	R[3]	Undefined	Undefined	R[4]	G[4]	B[4]	B[4]	G[4]	R[4]	Undefined
LCD_DATA02	Undefined	R[2]	G[2]	B[2]	B[2]	G[2]	R[2]	Undefined	Undefined	R[5]	G[5]	B[5]	B[5]	G[5]	R[5]	Undefined
LCD_DATA01	Undefined	R[1]	G[1]	B[1]	B[1]	G[1]	R[1]	Undefined	Undefined	R[6]	G[6]	B[6]	B[6]	G[6]	R[6]	Undefined
LCD_DATA00	Undefined	R[0]	G[0]	B[0]	B[0]	G[0]	R[0]	Undefined	Undefined	R[7]	G[7]	B[7]	B[7]	G[7]	R[7]	Undefined

Note: R[7:0], G[7:0], and B[7:0] are RGB pixel data that is internally processed.

**Figure 64.5 Serial RGB888 (4x speed) output timing**

64.1.5 Output Control for Panel-Oriented Correction Process

The following panel-oriented correction processes are provided:

- Brightness correction
- Contrast correction
- RGB gamma correction

Brightness correction always precedes contrast correction, but RGB gamma correction can either precede or follow brightness and contrast correction, based on the register settings. In panel-oriented correction, 10-bit RGB data obtained by extending 8-bit RGB data output from graphics 2 is used, and 10-bit RGB data is also output to the output control (data format conversion) block. Figure 64.6 shows the configuration of the panel-oriented correction circuit.

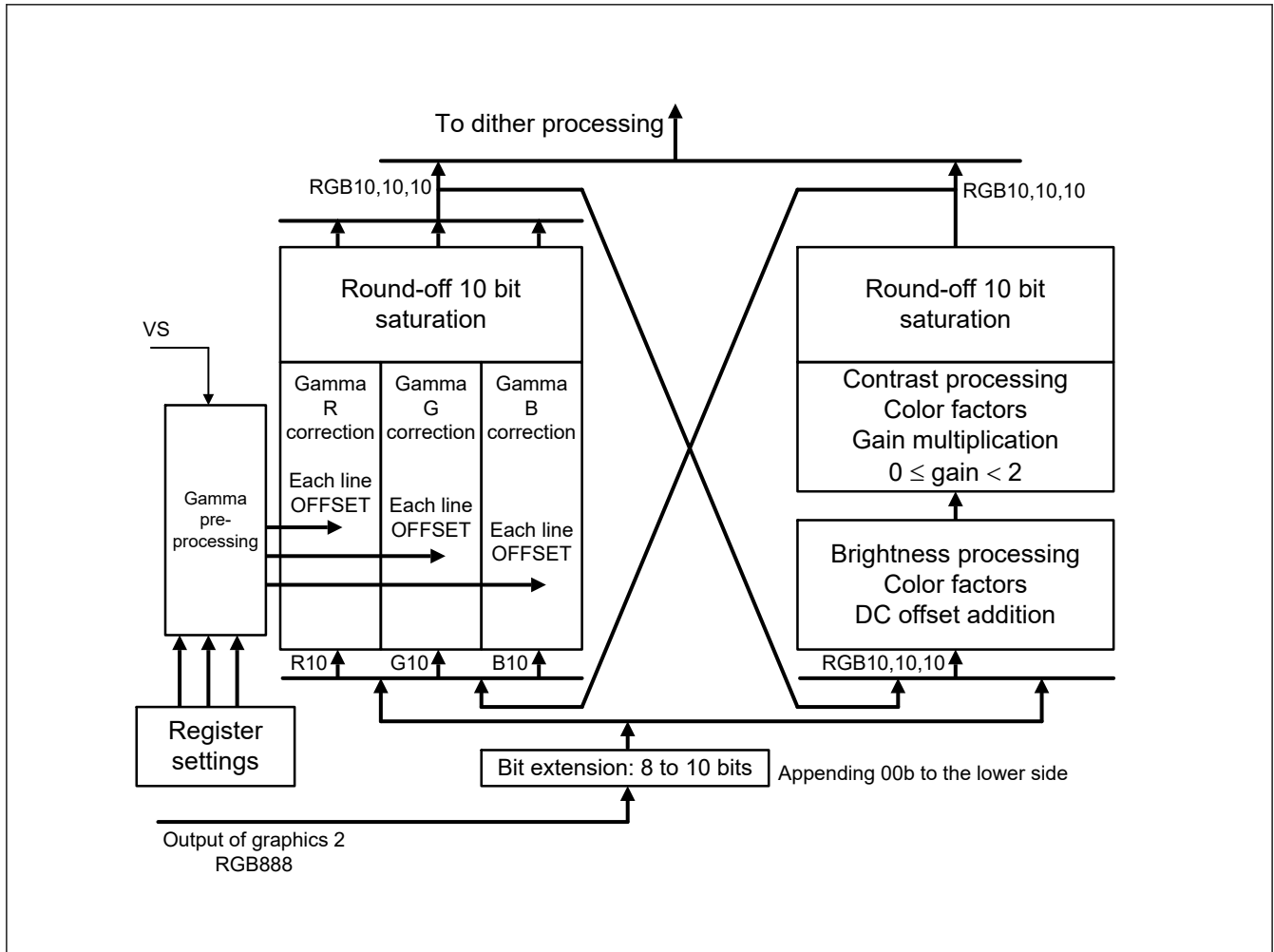


Figure 64.6 Configuration of panel-oriented correction circuit

64.1.6 Output Control for TCON

Any of the four signals generated from the internal vertical and horizontal synchronization signals (STVA, STVB, STHA, and STHB) that have passed through the data format conversion block can be selected for output on four pins, LCD_TCON0, LCD_TCON1, LCD_TCON2, and LCD_TCON3. The generated signals are completely independent of the image data. They are not affected by the output image format or any internal data process, and no register settings for signal generation affect the output image format or any internal data process. The data enable signal DE, which is to be generated by the TCON block, is the logical AND of the two signals STVB and STHB, which were previously generated by the TCON block. Consequently, three signals in total can actually be generated if DE is to be output.

64.1.7 Graphics Data Interface

Two circuit systems are provided for reading graphics data (graphics 1 and 2), each of which incorporates an access control block, data buffer, CLUT memory, data extension block, and asynchronous interface block.

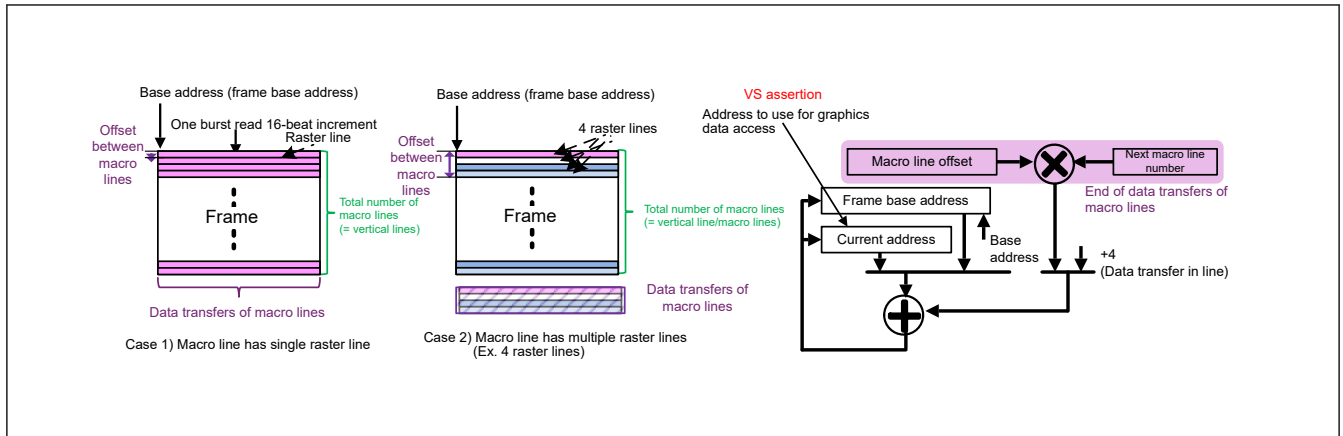


Figure 64.7 Calculation of graphics data access address

Graphics (or CLUT) data is accessed and output to the pixel operation block as ARGB8888 data (32 bits/pixel).

The GLCDC0 and GLCDC1 bus is accessed in word (32-bit) units in 16-beat increment burst reads, in accordance with the preset parameters in the two-dimensional addressing mode, in which the macro line structure is accounted for as shown in Figure 64.7, and data is stored in the data buffer. Even if invalid data is at the raster line end, all the data is stored in the data buffer, and the invalid data is skipped when data is read internally. Regardless of the format of the graphics data, data is extended to ARGB8888 data before being output to the alpha blending blocks.

64.1.8 Blending

The following processes are performed for the graphics areas specified in the registers. The lower-layer graphics plane is output without any processing to the display area outside the graphics area.

- Display plane selection
- RGB-index chroma-key
- Alpha blending

Figure 64.8 shows the relationship between the graphics display selection and rectangular alpha blending area.

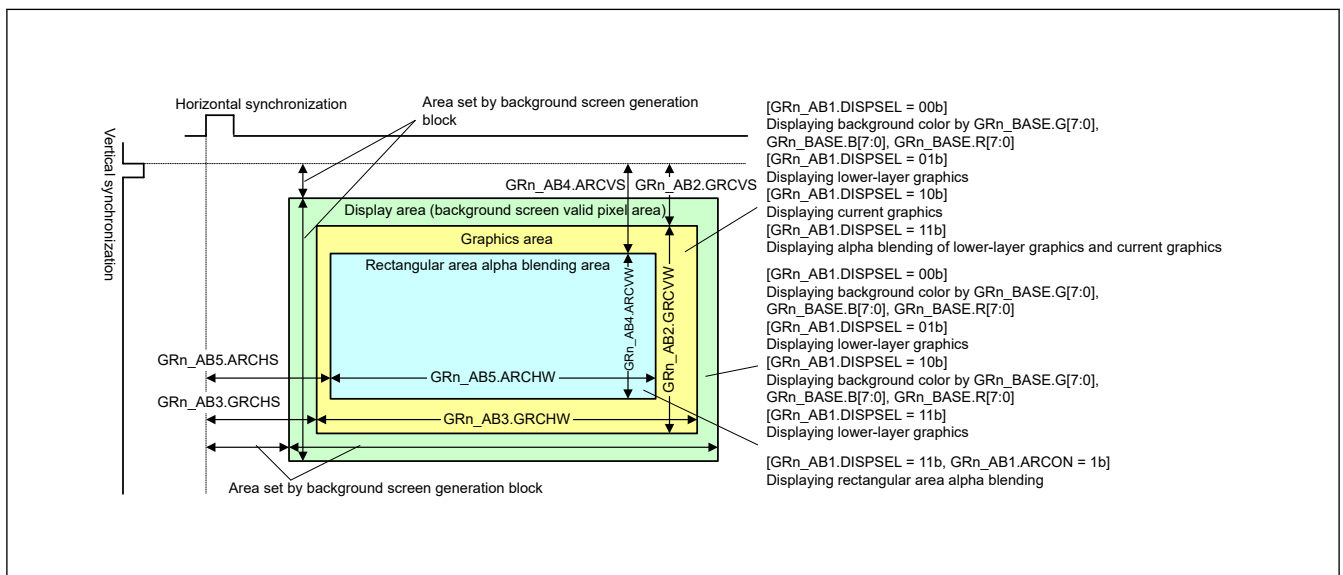


Figure 64.8 Graphics display selection

(1) Display plane selection

Based on the register settings, the following data is output to the graphics area:

- Background color:

- RGB data specified in the registers.
- Lower-layer graphics:
 - RGB data input from the previous module
 - Output from the background screen generation block for graphics 1
 - Output from graphics 1 for graphics 2
- Current graphics:
 - RGB data obtained by the graphics data interface block extending the graphics data read by the graphics 1 or 2 module through the GLCDC0 and GLCDC1 bus or CLUT data to ARGB8888 format data.
- Blending:
 - RGB data obtained by blending the lower-layer graphics data and current graphics data obtained by the graphics data interface block extension to ARGB8888 format data, based on the alpha blending values or the register settings.

It is possible to modify the register values related to these functions while display operation is in progress, and to allow reflection of the updates to the internal operations when the VS (vertical synchronization signal) is asserted, if reflection of the register settings to the internal operations on vertical synchronization is enabled.

(2) RGB-index chroma-key

If the RGB value of the ARGB8888 data input from the graphics data interface block agrees with the value set in the GRn_AB8 register, the ARGB8888 data is entirely replaced with the value preset in the GRn_AB9 register. All the ARGB8888 data input from the graphics data interface block is subject to this process. If your application excludes CLUT memory output from this process, you must disable the corresponding process in GRn_AB7.CKON.

(3) Alpha blending

If blending is selected in the selected display plane, lower-layer graphics and current graphics are alpha-blended based on the register settings using either of the following two functions.

Alpha-blending in a rectangular area

The following process is performed for the rectangular area (in the graphics area) preset in the registers:

1. Lower-layer graphics and current graphics planes are blended in accordance with the A value set in the registers.
2. After the number of frames set in the registers passes, the A value is updated to the A value + Δ (register setting).
3. The lower-layer graphics and current graphics planes are blended in accordance with this updated A value.
4. The process of updating the A value after the number of frames set in the registers passes is repeated (A value: min/max value saturation).

Alpha-blending in pixel units

The lower-layer graphics and current graphics planes are blended in accordance with the A value of the ARGB8888 data input from the graphics data interface block.

Figure 64.9 shows the update of the alpha blending value in the rectangular alpha blending area.

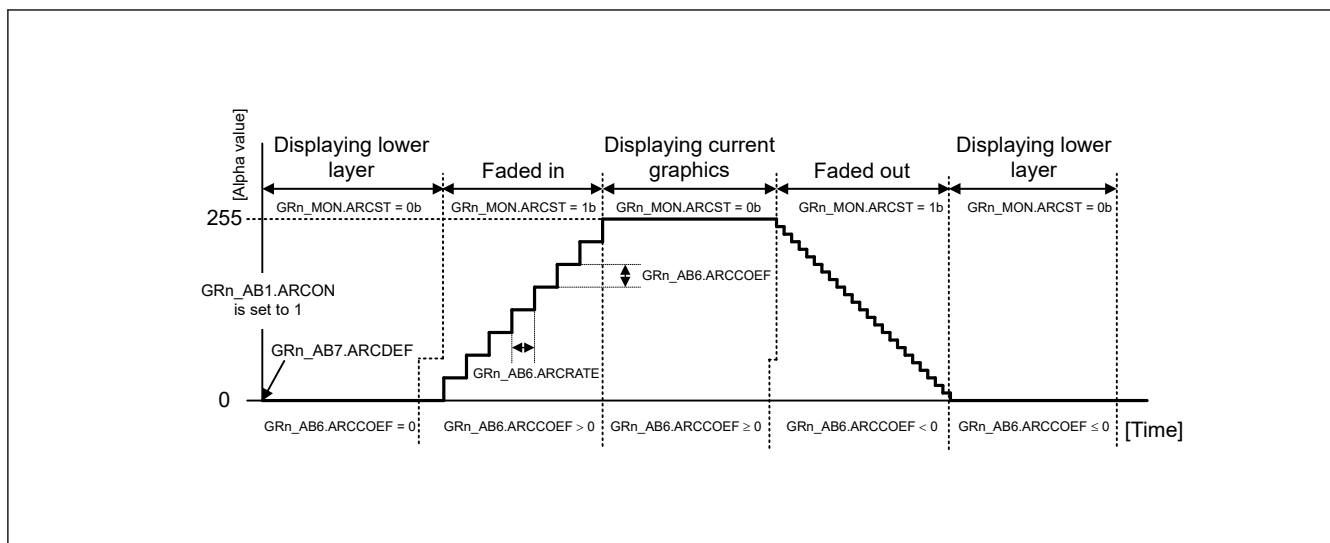


Figure 64.9 Updating of alpha blending value

Alpha blending is based on the following formulas:

When A value = 255

Rout/Gout/Bout = current graphics data

When A value ≠ 255

$R_{out} = (R_{in1} \times A + R_{in0} \times (256 - A)) / 256$

$G_{out} = (G_{in1} \times A + G_{in0} \times (256 - A)) / 256$

$B_{out} = (B_{in1} \times A + B_{in0} \times (256 - A)) / 256$

where,

A: alpha blending value

Rin1/Gin1/Bin1: current graphics data

Rin0/Gin0/Bin0: lower-layer graphics data

64.2 Register Descriptions

64.2.1 GRm_CLUTn[y] : Color Palette (m = 1 to 2, n = 0 to 1, y = 0 to 255)

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x0+0x4 × y (GR1_CLUT0[y])
0x0400+0x4 × y (GR1_CLUT1[y])
0x0800+0x4 × y (GR2_CLUT0[y])
0x0C00+0x4 × y (GR2_CLUT1[y])

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	A[7:0]								R[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	G[7:0]								B[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	B[7:0]	B Value of Color Palette n Plane for Graphics m Plane B value of color palette n plane for graphics m plane. Unsigned 8-bit integer.	R/W

Bit	Symbol	Function	R/W
15:8	G[7:0]	G Value of Color Palette n Plane for Graphics m Plane G value of color palette n plane for graphics m plane. Unsigned 8-bit integer.	R/W
23:16	R[7:0]	R Value of Color Palette n Plane for Graphics m Plane R value of color palette n plane for graphics m plane. Unsigned 8-bit integer.	R/W
31:24	A[7:0]	Alpha Blending Value of Color Palette n Plane for Graphics m Plane Alpha blending value of color palette n plane for graphics m plane. Unsigned 8-bit integer.	R/W

Note: S-TYPE-3, P-TYPE-3

B[7:0] bits (B Value of Color Palette n Plane for Graphics m Plane)

The B[7:0] bits set the B value when this color palette is used.

G[7:0] bits (G Value of Color Palette n Plane for Graphics m Plane)

The G[7:0] bits set the G value when this color palette is used.

R[7:0] bits (R Value of Color Palette n Plane for Graphics m Plane)

The R[7:0] bits set the R value when this color palette is used.

A[7:0] bits (Alpha Blending Value of Color Palette n Plane for Graphics m Plane)

The A[7:0] bits set the alpha blending value when this color palette is used.

All the planes can always be accessed through the register access bus (PABI), regardless of the plane specified to be used by the graphics data access block. The updates are reflected to the internal operations directly, not in synchronization with the vertical synchronization signal. To keep reflection of the CLUT memory contents to the internal operations in synchronization with the vertical synchronization signal, first write data through the register access bus to the plane that is not being used for internal operations, and then modify the GRn_CLUTINT.SEL bit for controlling the plane that is to be used.

64.2.2 BG_EN : Background Plane Setting Operation Control Register

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x1000

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	SWRS T
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	VEN	—	—	—	—	—	—	—	EN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	EN	Background Plane Operation Enable 0: Disable background plane operation 1: Enable background plane operation	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
8	VEN	Control of GLCDC Internal Register Value Reflection to Internal Operations This bit is set to 0 by an internal source. 0: Disable GLCDC register values from being reflected in internal operations at start of screen generation 1: Enable GLCDC register values to be reflected in internal operations at start of screen generation	R/W
15:9	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
16	SWRST	Software Reset Control 0: Place entire module in software reset state 1: Release entire module from software reset state	R/W
31:17	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

EN bit (Background Plane Operation Enable)

The EN bit enables or disables the operation of the background plane generation module. When the operation is stopped (this bit is cleared to 0) after the operation is enabled (set to 1), the operation is stopped at the end of screen generation, unlike when a software reset occurs.

When setting this bit to 1, set the BG_EN.VEN bit to 1 simultaneously to enable register value reflection to internal operations. When clearing this bit to 0, also clear the BG_EN.VEN bit to 0 simultaneously. Before clearing this bit to 0, confirm that the BG_MON.VEN bit is cleared to 0, to make sure that the signal for controlling reflection of the register values to internal operations is negated. Operation is not guaranteed if this bit and the BG_EN.VEN bit are set to 1, or if the settings in other registers are modified before the BG_MON.EN bit is cleared to 0.

VEN bit (Control of GLCDC Internal Register Value Reflection to Internal Operations)

The VEN bit enables or disables reflection of the GLCDC internal register to the GLCDC internal operation on assertion of the vertical synchronization signal (input). When this bit is set to 1, the signal of GLCDC internal register values reflection control is set to 1 at the immediate start of the screen, and are automatically cleared to 0 at the end of the vertical valid pixel of the same screen. Set this bit to 1 only when 0. Operation is not guaranteed if this bit is set to 1 when 1. Also, while this bit is 1, do not modify any register whose value is reflected to the internal operations at the start of the screen in the GLCDC or on assertion of the vertical synchronization signal (input). Otherwise, operation is not guaranteed.

SWRST bit (Software Reset Control)

The SWRST bit controls a software reset of the entire GLCDC, not only the background plane generation module. When this bit is cleared to 0, the GLCDC enters the reset state from any operation state. This bit must be set to 1 before the registers are set or operation enabled. Although the registers (except the CLUT memory and the some of the operation control registers) can be set while this bit is set to 1 (immediately after), before accessing the CLUT memory, enabling operation, or reflecting the register values to the internal operation by the vertical synchronization signal, confirm that PXCLK/LCD_CLK and PCLKA are supplied and that the BG_MON.SWRST bit, which monitors the entire module software reset state, is set to 1. Operation is not guaranteed if the software accesses CLUT memory, enables operation, or reflects the register vales to the internal operation on the vertical synchronization signal while PXCLK/LCD_CLK and PCLKA are not supplied or the BG_MON.SWRST bit is not set to 1. The peripheral bus clock (PCLKA) must be supplied to the GLCDC when this bit is used to apply or cancel a software reset. If PCLKA is not supplied, writing to this bit is impossible.

The bits in this register control the GLCDC states. The internal states can be read from the associated bits in the status monitor registers and BG_EN.VEN (this register). Because the GLCDC, which operates on multiple clock signals, requires a certain period for state transition, you must confirm that the internal state has stabilized (state transition is complete) when settings are modified. Operation is not guaranteed if settings are modified again before the internal state stabilizes (state transition is complete). However, clearing a software reset to 0 immediately makes the whole GLCDC reset, setting it to 1 releases the reset state. These operations do not require the clock supply of PXCLK.

64.2.3 BG_PERI : Background Plane Setting Free-Running Period Register

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x1004

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	FV[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	1
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	FH[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	1

Bit	Symbol	Function	R/W
10:0	FH[10:0]	Background Plane Horizontal Synchronization Signal Period Period based on pixel clocks (PXCLK). The valid range is 0x17 to 0x7FF. 0x017: 24 cycles (pixels) 0x3FF: 1024 cycles (pixels) 0x7FF: 2048 cycles (pixels)	R/W
15:11	—	These bits are read as 0. The write value should be 0.	R/W
26:16	FV[10:0]	Background Plane Vertical Synchronization Signal Period Period based on lines. The valid range is 0x013 to 0x7FF. 0x013: 20 lines 0x3FF: 1024 lines 0x7FF: 2048 lines	R/W
31:27	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: Rewriting these bits is prohibited during operation. Set the required settings before enabling operation. Operation is not guaranteed if the bit is rewritten during operation.

FH[10:0] bits (Background Plane Horizontal Synchronization Signal Period)

The FH[10:0] bits set the horizontal synchronization signal period of the background plane. This field contains 11 bits and can be set to any number from 0x000 to 0x7FF. However, the valid range is 0x017 to 0x7FF.

FV[10:0] bits (Background Plane Vertical Synchronization Signal Period)

The FV[10:0] bits set the vertical synchronization signal period of the background plane. This field contains 11 bits and can be set to any number from 0x000 to 0x7FF. However, the valid range is 0x013 to 0x7FF.

64.2.4 BG_SYNC : Background Plane Setting Synchronization Position Register

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x1008

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	VP[3:0]>			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	HP[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
3:0	HP[3:0]	Background Plane Horizontal Synchronization Signal Assertion Position Position based on pixel clocks (PXCLK). 0x0: Setting prohibited Others: HP th cycle (pixel)	R/W
15:4	—	These bits are read as 0. The write value should be 0.	R/W
19:16	VP[3:0]	Background Plane Vertical Synchronization Assertion Position Position based on lines. 0x0: Setting prohibited Others: VP th line	R/W
31:20	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: Rewriting these bits is prohibited during operation. Set the required settings before enabling operation. Operation is not guaranteed if the bit is rewritten during operation.

HP[3:0] bits (Background Plane Horizontal Synchronization Signal Assertion Position)

The HP[3:0] bits set the horizontal synchronization signal assertion position of the background plane. The signal is held asserted for a 4-pixel clock width.

VP[3:0] bits (Background Plane Vertical Synchronization Assertion Position)

The VP[3:0] bits set the vertical synchronization signal assertion position of the background plane. The signal is held asserted for a 1H width, and the assertion timing within a single horizontal line is specified in BG_SYNC.HP[3:0].

64.2.5 BG_VSIZE : Background Plane Setting Full Image Vertical Size Register

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x100C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	VP[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	VW[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0

Bit	Symbol	Function	R/W
10:0	VW[10:0]	Background Plane Vertical Valid Pixel Width Width based on lines. 0x010: 16 lines ⋮ 0x7FC: 2044 lines Others: Setting prohibited	R/W
15:11	—	These bits are read as 0. The write value should be 0.	R/W
26:16	VP[10:0]	Background Plane Vertical Valid Pixel Start Position Position based on of lines. 0x003: 3rd line ⋮ 0x7EF: 2031 lines Others: Setting prohibited	R/W
31:27	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: Rewriting these bits is prohibited during operation. Set the required settings before enabling operation. Operation is not guaranteed if the bit is rewritten during operation.

VW[10:0] bits (Background Plane Vertical Valid Pixel Width)

The VW[10:0] bits set the vertical valid pixel width of the background plane. This field contains 11 bits and can be set to any number from 0x000 to 0x7FF. However, the valid range is 0x010 to 0x7FC. Operation is not guaranteed if a value outside the valid range is set.

VP[10:0] bits (Background Plane Vertical Valid Pixel Start Position)

The VP[10:0] bits set the vertical valid pixel start position of the background plane. This field contains 11 bits and can be set to any number from 0x000 to 0x7FF. However, the valid range is 0x003 to 0x7EF. Operation is not guaranteed if a value outside the valid range is set.

Specify the vertical valid pixel area between the assertion position of the vertical synchronization signal + 2 and the (background plane end - 1)th line. Operation is not guaranteed if the area is specified outside this range.

64.2.6 BG_HSIZE : Background Plane Setting Full Image Horizontal Size Register

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x1010

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	HP[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	HW[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0

Bit	Symbol	Function	R/W
10:0	HW[10:0]	Background Plane Horizontal Valid Pixel Width Width based on pixel clocks (PXCLK). 0x010: 16 cycles ⋮ 0x7F8: 2040 cycles Others: Setting prohibited	R/W
15:11	—	These bits are read as 0. The write value should be 0.	R/W
26:16	HP[10:0]	Background Plane Horizontal Valid Pixel Start Position Position based on pixel clocks (PXCLK). 0x006: 6th cycle (pixel) ⋮ 0x7EE: 2030 cycles Others: Setting prohibited	R/W
31:27	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: Rewriting these bits is prohibited during operation. Set the required settings before enabling operation. Operation is not guaranteed if the bit is rewritten during operation.

HW[10:0] bits (Background Plane Horizontal Valid Pixel Width)

The HW[10:0] bits set the horizontal valid pixel width of the background plane. This field contains 11 bits and can be set to any number from 0x000 to 0x7FF. However, the valid range is 0x010 to 0x7F8. Operation is not guaranteed if a value outside the valid range is set. When serial RGB is selected as the output format for the output control block, add two to the horizontal valid pixel width and set the resulting value to these bits.

HP[10:0] bits (Background Plane Horizontal Valid Pixel Start Position)

The HP[10:0] bits set the horizontal valid pixel start position of the background plane. This field contains 11 bits and can be set to any number from 0x000 to 0x7FF. However, the valid range is 0x006 to 0x7EE. Operation is not guaranteed if a value outside the valid range is set.

Specify the horizontal valid pixel area between the assertion position of the horizontal synchronization signal + 5 and pixel number (line end – 2). Operation is not guaranteed if the area is specified outside this range.

The background plane generation module outputs the values that are specified in the Background Color Register (BG_BGC) for the area defined by the Full Image Vertical Size Register (BG_VSIZE) and Full Image Horizontal Size Register (BG_HSIZE), and it outputs 0x00 as the RGB values for the blanking interval area.

64.2.7 BG_BGC : Background Plane Setting Background Color Register

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x1014

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	R[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	G[7:0]								B[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	B[7:0]	Background Plane Valid Pixel Area B Value B value for background plane valid pixel area. Unsigned 8-bit integer.	R/W
15:8	G[7:0]	Background Plane Valid Pixel Area G Value G value for background plane valid pixel area. Unsigned 8-bit integer.	R/W
23:16	R[7:0]	Background Plane Valid Pixel Area R Value R value for background plane valid pixel area. Unsigned 8-bit integer.	R/W
31:24	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

B[7:0] bits (Background Plane Valid Pixel Area B Value)

The B[7:0] bits set the B value of image data to be output to the valid pixel area of the background plane.

G[7:0] bits (Background Plane Valid Pixel Area G Value)

The G[7:0] bits set the G value of image data to be output to the valid pixel area of the background plane.

R[7:0] bits (Background Plane Valid Pixel Area R Value)

The R[7:0] bits set the R value of image data to be output to the valid pixel area of the background plane.

64.2.8 BG_MON : Background Plane Setting Status Monitor Register

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x1018

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	SWRST
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	VEN	—	—	—	—	—	—	—	EN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	EN	Background Plane Operation Monitor 0: Operation is stopped 1: Operation is in progress	R
7:1	—	These bits are read as 0. The write value should be 0.	R
8	VEN	Entire Module Internal Operation Reflection Control Signal Monitor 0: Signal for controlling reflection of the register values to internal operations on assertion of vertical synchronization signal is negated 1: Signal for controlling reflection of the register values to internal operations on assertion of vertical synchronization signal is asserted	R
15:9	—	These bits are read as 0.	R
16	SWRST	Entire Module SW Reset State Monitor 0: Entire module is in software reset state 1: Entire module is released from software reset state	R
31:17	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

EN bit (Background Plane Operation Monitor)

The EN bit indicates whether the background plane generation module is operating or not. To stop the operation of the background plane generation module by clearing BG_EN.EN to 0, read this bit to confirm that the operation of the background plane is complete. Clearing BG_EN.EN to 0 does not stop the operation until completion of the background plane, unlike when BG_EN.SWRST is cleared to 0.

VEN bit (Entire Module Internal Operation Reflection Control Signal Monitor)

The VEN bit indicates the value of the signal for controlling reflection of the GLCDC internal register values to the internal operations. This signal is asserted at the start of a screen immediately after setting BG_EN.VEN to 1 and negated at the VE negate timing output from the background screen generation block.

SWRST bit (Entire Module SW Reset State Monitor)

The SWRST bit indicates the software reset state of the entire module when PXCLK is supplied. This bit value indicates the result of the peripheral module clock A (PCLKA) resampling the result of the pixel clock (PXCLK) sampling the BG_EN.SWRST bit. Even if PXCLK is not supplied, clearing the BG_EN.SWRST to 0 clears this bit to 0.

64.2.9 GRn_VEN : Graphics n Register Update Control Register (n = 1 to 2)

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x1100 + 0x100 × (n-1)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	PVEN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PVEN	Control of Graphics n Register Value Reflection to Internal Pixel Operations This bit is cleared to 0 by an internal source. 0: Disable reflection of register values to internal operations on assertion of vertical synchronization signal (VS) 1: Enable reflection of register values to internal operations on assertion of the vertical synchronization signal (VS)	R/W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

PVEN bit (Control of Graphics n Register Value Reflection to Internal Pixel Operations)

The PVEN bit enables or disables reflection of the register values to the internal operations of the pixel operation circuit on assertion of the vertical synchronization signal (input). When this bit is set to 1, the register values are immediately reflected to the internal operations on assertion of the vertical synchronization signal (input), and are automatically cleared to 0. Set this bit to 1 only when 0. Operation is not guaranteed if this bit is set to 1 when 1. Also, if the signal output is asserted from the background plane generation module that controls reflection of the register values to the internal operations of all the modules, the register values are always reflected to the internal operations on assertion of the vertical synchronization signal (input), regardless of the value of this bit. While this bit is 1, do not modify any register whose value is reflected to the internal operations on assertion of the vertical synchronization signal (input) in this block. Otherwise, operation is not guaranteed.

This bit must not be 1 at the same time as the BG_EN.VEN bit (control of background plane register value reflection to internal operations) in the Operation Control Register (BG_EN), one of the background plane setting registers. Otherwise, operation is not guaranteed.

64.2.10 GRn_FLMRD : Graphics n Frame Buffer Read Control Register (n = 1, 2)

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: $0x1104 + 0x100 \times (n - 1)$

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	RENB
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	RENB	Graphics Data Read Enable Graphics data is the frame buffer data. 0: Disable reading 1: Enable reading	R/W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: This setting is reflected to the internal operations on assertion of the vertical synchronization signal (input) when GRn_VEN.PVEN = 1 or when the register value reflection control signal to internal operations for all the modules is asserted.

RENB bit (Graphics Data Read Enable)

The RENB bit enables or disables reading of the graphics data (frame buffer data in memory connected through the GLCDC0 and GLCDC1 bus). If the current graphics data is required, data read must be enabled (GRn_FLMRD.RENB = 1) before the background panel operation is enabled, register value internal operation reflection control is enabled, or the register value internal operation reflection control of graphics 1 and 2 is enabled.

If the current graphics data is not required, data read must be disabled (GRn_FLMRD.RENB = 0) before the background panel operation is enabled, register value internal operation reflection control is enabled, or the register value internal operation reflection control of graphics 1 and 2 is enabled.

64.2.11 GR_n_FLM1 : Graphics n Frame Buffer Control Register 1 (n = 1 to 2)

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x1108 + 0x100 × (n - 1)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	BSTMD[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1

Bit	Symbol	Function	R/W
1:0	BSTMD[1:0]	Burst Transfer Control for Graphics Data Access Graphics data is the frame buffer data. 1 1: 16-beat increment burst transfer (64-byte boundary) Others: Setting prohibited	R/W
31:2	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: This setting is reflected to the internal operations on assertion of the vertical synchronization signal (input) when GR_n_VEN.PVEN = 1 or when the register value reflection control signal to internal operations for all the modules is asserted.

BSTMD[1:0] bits (Burst Transfer Control for Graphics Data Access)

The BSTMD[1:0] bits control burst transfers for accessing the graphics data (frame buffer data in memory connected to memory through the GLCDC0 and GLCDC1 bus). In this GLCDC, these bits are fixed to 11b. Operation is not guaranteed if these bits are set to any other value.

64.2.12 GR_n_FLM2 : Graphics n Frame Buffer Control Register 2 (n = 1 to 2)

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x110C + 0x100 × (n - 1)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	BASE[31:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	BASE[31:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
31:0	BASE[31:0]	Base Address for Accessing Graphics Data Start address in the frame buffer where graphics data is to be stored. Fix GR _n _FLM2.BASE [5:0] to 0 during 64-byte burst transfer.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: This setting is reflected to the internal operations on assertion of the vertical synchronization signal (input) when GR_n_VEN.PVEN = 1 or when the register value reflection control signal to internal operations for all the modules is asserted.

BASE[31:0] bits (Base Address for Accessing Graphics Data)

The BASE[31:0] bits specify the base address (start address in the first frame buffer) for graphics data access. Because the GLCDC only supports the 16-beat increment burst transfer mode, in which data is aligned with a 64-byte boundary, the lower 6 bits (GRn_FLM2.BASE[5:0]) must be fixed to 0.

64.2.13 GRn_FLM3 : Graphics n Frame Buffer Control Register 3 (n = 1 to 2)

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x1110 + 0x100 × (n - 1)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	LNOFF[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	—	These bits are read as 0. The write value should be 0.	R/W
31:16	LNOFF[15:0]	Macro Line Offset Address for Accessing Graphics Data Macro line offset address for accessing graphics data (frame buffer data). Signed, 16-bit integer.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: This setting is reflected to the internal operations on assertion of the vertical synchronization signal (input) when GRn_VEN.PVEN = 1 or when the register value reflection control signal to internal operations for all the modules is asserted.

LNOFF[15:0] bits (Macro Line Offset Address for Accessing Graphics Data)

The LNOFF[15:0] bits specify the macro line offset address for accessing graphics data (offset to be added to the current address at the macro line end for calculating the start address of the next macro line). Because the GLCDC only supports the 16-beat increment burst transfer mode, in which data is aligned with a 64-byte boundary, the lower 6 bits (GRn_FLM3.LNOFF[5:0]) must be fixed to 0.

64.2.14 GRn_FLM5 : Graphics n Frame Buffer Control Register 5 (n = 1 to 2)

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x1118 + 0x100 × (n - 1)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	LNNUM[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	DATANUM[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	DATANUM[15:0]	Number of Data Transfer Times Per Line for Accessing Graphics Data Number of data transfers per macro line for accessing graphics data (frame buffer data), where one transfer is defined as 16-beat burst access (64-byte boundary). 0x0000 Once : : 0xFFFF 65536 times :	R/W
26:16	LNNUM[10:0]	Number of Lines Per Frame for Accessing Graphics Data Number of macro lines per frame for accessing graphics data (frame buffer data). 0x000: 1 macro line : 0x7FB: 2044 macro lines Others: Setting prohibited	R/W
31:27	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: This setting is reflected to the internal operations on assertion of the vertical synchronization signal (input) when GRn_VEN.PVEN = 1 or when the register value reflection control signal to internal operations for all the modules is asserted.

DATANUM[15:0] bits (Number of Data Transfer Times Per Line for Accessing Graphics Data)

The DATANUM[15:0] bits specify the number of data transfers per macro line for accessing graphics data, where one transfer is defined as a 16-beat burst access through the GLCDC0 and GLCDC1 bus. For all the data transfers, at least 2 bytes of valid pixel data are required (16 pixels in LUT1 format and 16 pixels or more in the other formats). If the number of bytes per macro line cannot be divided by 64 (4 bytes × 16-beat), DATANUM is obtained by rounding up to the whole number.

LNNUM[10:0] bits (Number of Lines Per Frame for Accessing Graphics Data)

The LNNUM[10:0] bits specify the number of lines per frame for accessing graphics data. When graphics data for the number of lines set to these bits is read, it signals the end of the frame and the base address is loaded.

The following are two use cases for macro lines. In these use cases, the frame size is 480 pixels × 272 lines and the pixel format is RGB565 (16 bpp).

Case 1) One macro line is configured to be equivalent to the frame raster width. The number of macro lines is equivalent to the number of vertical lines.

DATANUM: 0x000E (15 times = $16 \times 480 / 512$)

LNNUM: 0x10F (272 macro lines = $272 / 1$)

Case 2) One macro line is configured to be repeated during the display frame. The number of macro lines is not equivalent to the number of vertical lines. In the example, one macro line has 16 times the raster width.

DATANUM: 0x00EF (240 times = $16 \times 480 / 512 \times 16$)

LNNUM: 0x010 (17 macro lines = $272 / 16$)

64.2.15 GRn_FLM6 : Graphics n Frame Buffer Control Register 6 (n = 1, 2)

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x111C + 0x100 × (n - 1)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	FORMAT[2:0]			—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
27:0	—	These bits are read as 0. The write value should be 0.	R/W
30:28	FORMAT[2:0]	Data Format for Accessing Graphics Data Data format for accessing graphics data (frame buffer data). 0 0 0: RGB565 (16 bits/pixel) 0 0 1: RGB888 (32 bits/pixel, 8 bits on the MSB side are invalid) 0 1 0: ARGB1555 (16 bits/pixel, 1 bit of A is LUT data) 0 1 1: ARGB4444 (16 bits/pixel) 1 0 0: ARGB8888 (32 bits/pixel) 1 0 1: CLUT8 (8 bits/pixel) 1 1 0: CLUT4 (4 bits/pixel) 1 1 1: CLUT1 (1 bit/pixel)	R/W
31	—	This bit is read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: This setting is reflected to the internal operations on assertion of the vertical synchronization signal (input) when GRn_VEN.PVEN = 1 or when the register value reflection control signal to internal operations for all the modules is asserted.

FORMAT[2:0] bits (Data Format for Accessing Graphics Data)

The FORMAT[2:0] bits specify the data format for accessing graphics data. CLUT1, CLUT4, CLUT8 contain the addresses 0x01 or 0x00, 0x0F to 0x00, and 0xFF to 0x00, respectively, for accessing the color palette. ARGB1555 contains the address for accessing the color palette (0x80 or 0x00) in the MSB and RGB data in the other bits. ARGB8888 and ARGB4444 contain the upper 8-bit or 4-bit alpha blending values and RGB data. RGB888 and RGB565 contain RGB data only.

64.2.16 GRn_AB1 : Graphics n Alpha Blending Control Register 1 (n = 1, 2)

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x1120 + 0x100 × (n - 1)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	ARCON	—	—	—	ARCDISPON	—	—	—	GRCDISPON	—	—	DISPSEL[1:0]	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	DISPSEL[1:0]	Graphics Display Plane Control 0 0: Background color display (value set in the GRn_BASE register) 0 1: Lower-layer graphics display 1 0: Current graphics display 1 1: Blended display of lower-layer graphics (input image from the previous stage) and current graphics (data read from the GLCDC0 and GLCDC1 bus)	R/W
3:2	—	These bits are read as 0. The write value should be 0.	R/W
4	GRCDISPON	Graphics Image Area Border Display Control 0: Turn display off 1: Turn display on	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W
8	ARCDISPON	Image Area Border Display Control for Rectangular Area Alpha Blending 0: Turn display off 1: Turn display on	R/W
11:9	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
12	ARCON	Rectangular Area Alpha Blending Control 0: Turn blending off 1: Turn blending on	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: This setting is reflected to the internal operations on assertion of the vertical synchronization signal (input) when GRn_VEN.PVEN = 1 or when the register value reflection control signal to internal operations for all the modules is asserted.

DISPSEL[1:0] bits (Graphics Display Plane Control)

The DISPSEL[1:0] bits control the graphics display plane. When the plane is selected as the lower-layer graphics, the image input from the previous stage is displayed (the background plane for GR1, and output from GR1 for GR2); as the background color, the background color specified in the GLCDC registers is displayed; and as the current graphics, the ARGB8888 data obtained by expanding the graphics data read by the GLCDC from the GLCDC0 and GLCDC1 bus is displayed. When the current graphics display is selected (these bits are set to 10b), RGB888 data is displayed, regardless of the alpha blending value in the pixel. Table 64.8 and Figure 64.10 show the relationship between the register setting and display area.

GRCDISPON bit (Graphics Image Area Border Display Control)

The GRCDISPON bit turns on or off the border display for the graphics image area. When the display is turned on (this bit is set to 1), the graphics image area is bordered with the preset color. The border is 1 pixel wide on the outermost periphery of the area with the display data set as 0xFF for each RGB color.

ARCDISPON bit (Image Area Border Display Control for Rectangular Area Alpha Blending)

The ARCDISPON bit turns on or off the border display for the image area where rectangular area alpha blending is performed. When the display is turned on (this bit is set to 1), the image area for the rectangular alpha blending is bordered with the preset color. The border is 1 pixel wide on the outermost periphery of the area and the display data set as 0xFF for each RGB color.

ARCON bit (Rectangular Area Alpha Blending Control)

The ARCON bit turns on or off alpha blending in a rectangular area. When alpha blending is turned on (this bit is set to 1), pixels are processed in accordance with the alpha blending control specified in the relevant registers for the specified rectangular area, not in accordance with the alpha value input from the graphics data interface for each pixel. In the areas outside the rectangular area in the graphics area, pixels are processed in accordance with the alpha value input from the graphics data interface for each pixel.

Table 64.8 Display selections

GRn_AB1. DISPSEL[1:0] (display plane)	GRn_AB1. ARCON (rectangular)	GRn_AB7. CKON (chroma key)	Within rectangular alpha blending area	Outside rectangular alpha blending area within graphics area	Outside graphics area within display area	Outside display area
00b	0	0	—	Background color	Background color	Lower layer R = G = B = 0x00
01b	0	0	—	Lower layer	Lower layer	Lower layer R = G = B = 0x00
10b	0	0	—	Current	Background color	Lower layer R = G = B = 0x00
11b	0	0	—	Current + alpha blending in pixel units	Lower layer	Lower layer R = G = B = 0x00
11b	0	1	—	Current + RGB-index chroma key + alpha blending in pixel units	Lower layer	Lower layer R = G = B = 0x00
11b	1	0	Current + rectangular alpha blending	Current + alpha blending in pixel units	Lower layer	Lower layer R = G = B = 0x00
11b	1	1	Current + rectangular alpha blending	Current + RGB-index chroma key + alpha blending in pixel units	Lower layer	Lower layer R = G = B = 0x00

Horizontal synchronization

Vertical synchronization

Area set by background screen generation block

GRn_AB4.ARCVS

GRn_AB2.GRCVS

Displaying area (background screen valid pixel area)

Graphics area

Rectangular area alpha blending area

GRn_AB5.ARCHS

GRn_AB3.GRCHS

GRn_AB5.ARCHW

GRn_AB3.GRCHW

GRn_AB4.ARCVW

GRn_AB2.GRCVW

Area set by background screen generation block

[GRn_AB1.DISPSEL = 00b]
Displaying background color by GRn_BASE.G[7:0],
GRn_BASE.B[7:0], GRn_BASE.R[7:0]
[GRn_AB1.DISPSEL = 01b]
Displaying lower-layer graphics
[GRn_AB1.DISPSEL = 10b]
Displaying current graphics
[GRn_AB1.DISPSEL = 11b]
Displaying alpha blending of lower-layer graphics and current graphics

[GRn_AB1.DISPSEL = 00b]
Displaying background color by GRn_BASE.G[7:0],
GRn_BASE.B[7:0], GRn_BASE.R[7:0]
[GRn_AB1.DISPSEL = 01b]
Displaying lower-layer graphics
[GRn_AB1.DISPSEL = 10b]
Displaying background color by the GRn_BASE.G[7:0],
GRn_BASE.B[7:0], GRn_BASE.R[7:0]
[GRn_AB1.DISPSEL = 11b]
Displaying lower-layer graphics

[GRn_AB1.DISPSEL = 11b, GRn_AB1.ARCON = 1b]
Displaying rectangular area alpha blending

Figure 64.10 Selection of graphics display plane

64.2.17 GRn_AB2 : Graphics n Alpha Blending Control Register 2 (n = 1, 2)

Offset address: $0x1124 + 0x100 \times (n - 1)$

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	GRCVS[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	GRCVW[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0

Bit	Symbol	Function	R/W
10:0	GRCVW[10:0]	Vertical Width of Graphics Image Area Width in lines. 0x010: 16 lines ⋮ 0x7FC: 2044 lines Others: Setting prohibited	R/W
15:11	—	These bits are read as 0. The write value should be 0.	R/W
26:16	GRCVS[10:0]	Vertical Start Position of Graphics Image Area Position in lines. 0x002: 2nd line ⋮ 0x7EE: 2030th line Others: Setting prohibited	R/W
31:27	—	These bits are read as 0. The write value should be 0.	R/W

Note: This setting is reflected to the internal operations on assertion of the vertical synchronization signal (input) when GRn_VEN.PVEN = 1 or when the register value reflection control signal to internal operations for all the modules is asserted.

GRCVW[10:0] bits (Vertical Width of Graphics Image Area)

The GRCVW[10:0] bits specify the vertical width of the graphics image area.

GRCVS[10:0] bits (Vertical Start Position of Graphics Image Area)

The GRCVS[10:0] bits specify the vertical start position of the graphics image area, in reference to assertion of the vertical synchronization signal (VS). For the relationship with the graphics display plane, see [Figure 64.10](#).

64.2.18 GRn_AB3 : Graphics n Alpha Blending Control Register 3 (n = 1, 2)

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x1128 + 0x100 × (n - 1)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	GRCHS[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	GRCHW[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0

Bit	Symbol	Function	R/W
10:0	GRCHW[10:0]	Horizontal Width of Graphics Image Area Width in pixels. 0x010: 16 pixels ⋮ 0x7F8: 2040 pixels Others: Setting prohibited	R/W
15:11	—	These bits are read as 0. The write value should be 0.	R/(W)
26:16	GRCHS[10:0]	Horizontal Start Position of Graphics Image Area Position in pixels. 0x005: 5th pixel ⋮ 0x7ED: 2029th pixel Others: Setting prohibited	R/W
31:27	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: This setting is reflected to the internal operations on assertion of the vertical synchronization signal (input) when GRn_VEN.PVEN = 1 or when the register value reflection control signal to internal operations for all the modules is asserted.

GRCHW[10:0] bits (Horizontal Width of Graphics Image Area)

The GRCHW[10:0] bits specify the horizontal width of the graphics image area.

GRCHS[10:0] bits (Horizontal Start Position of Graphics Image Area)

The GRCHS[10:0] bits specify the horizontal start position of the graphics image area, in reference to assertion of the horizontal synchronization signal (VS). For the relationship with the graphics display plane, see [Figure 64.10](#).

64.2.19 GRn_AB4 : Graphics n Alpha Blending Control Register 4 (n = 1, 2)

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x112C + 0x100 × (n - 1)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	ARCVS[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	ARCVW[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0

Bit	Symbol	Function	R/W
10:0	ARCVW[10:0]	Vertical Width of Rectangular Area Alpha Blending Image Area Width in lines. 0x001: 1 line ⋮ 0x7FC: 2044 lines Others: Setting prohibited	R/W
15:11	—	These bits are read as 0. The write value should be 0.	R/W
26:16	ARCVS[10:0]	Vertical Start Position of Rectangular Area Alpha Blending Image Area Position in lines. 0x002: 2nd line ⋮ 0x7EE: 2030th line Others: Setting prohibited	R/W
31:27	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: This setting is reflected to the internal operations on assertion of the vertical synchronization signal (input) when GRn_VEN.PVEN = 1 or when the register value reflection control signal to internal operations for all the modules is asserted.

ARCVW[10:0] bits (Vertical Width of Rectangular Area Alpha Blending Image Area)

The ARCVW[10:0] bits specify the vertical width of the rectangular area alpha blending image area.

ARCVS[10:0] bits (Vertical Start Position of Rectangular Area Alpha Blending Image Area)

The ARCVS[10:0] bits specify the vertical start position of the rectangular area alpha blending image area, in reference to assertion of the vertical synchronization signal (VS). For the relationship with the graphics display plane, see [Figure 64.10](#).

64.2.20 GRn_AB5 : Graphics n Alpha Blending Control Register 5 (n = 1, 2)

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x1130 + 0x100 × (n - 1)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	ARCHS[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	ARCHW[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0

Bit	Symbol	Function	R/W
10:0	ARCHW[10:0]	Horizontal Width of Rectangular Area Alpha Blending Image Area Width in pixels. 0x001: 1 pixel ⋮ 0x7F8: 2040 pixels Others: Setting prohibited	R/W
15:11	—	These bits are read as 0. The write value should be 0.	R/W
26:16	ARCHS[10:0]	Horizontal Start Position of Rectangular Area Alpha Blending Image Area Position in pixels. 0x005: 5th pixel ⋮ 0x7ED: 2029th pixel Others: Setting prohibited	R/W
31:27	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: This setting is reflected to the internal operations on assertion of the vertical synchronization signal (input) when GRn_VEN.PVEN = 1 or when the register value reflection control signal to internal operations for all the modules is asserted.

ARCHW[10:0] bits (Horizontal Width of Rectangular Area Alpha Blending Image Area)

The ARCHW[10:0] bits specify the horizontal width of the rectangular area alpha blending image area.

ARCHS[10:0] bits (Horizontal Start Position of Rectangular Area Alpha Blending Image Area)

The ARCHS[10:0] bits specify the horizontal start position of the rectangular area alpha blending image area, in reference to assertion of the horizontal synchronization signal (HS). For the relationship with the graphics display plane, see [Figure 64.10](#).

64.2.21 GRn_AB6 : Graphics n Alpha Blending Control Register 6 (n = 1, 2)

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x1134 + 0x100 × (n - 1)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	ARCCOEF[8:0]								
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	ARCRATE[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	ARCRATE[7:0]	Frame Rate for Alpha Blending in Rectangular Area 0x00: 1 frame ⋮ 0xFF: 256 frames	R/W
15:8	—	These bits are read as 0. The write value should be 0.	R/W
24:16	ARCCOEF[8:0]	Alpha Coefficient for Alpha Blending in Rectangular Area Valid settings: -255 to 255. Bit [8]: Sign 0: Add 1: Subtract Bits [7:0]: Variation, as an absolute value	R/W
31:25	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: This setting is reflected to the internal operations on assertion of the vertical synchronization signal (input) when GRn_VEN.PVEN = 1 or when the register value reflection control signal to internal operations for all the modules is asserted.

ARCRATE[7:0] bits (Frame Rate for Alpha Blending in Rectangular Area)

The ARCRATE[7:0] bits specify the frame rate for alpha blending in a rectangular area.

ARCCOEF[8:0] bits (Alpha Coefficient for Alpha Blending in Rectangular Area)

The ARCCOEF[8:0] bits specify the alpha coefficient for alpha blending in a rectangular area.

In alpha blending in a rectangular area, current graphics are faded in or out with the GRn_AB7.ARCDEF[7:0], GRn_AB6.ARCCOEF[8:0], and GRn_AB6.ARCRATE[7:0] settings. If the alpha value is set in the GR_ARC_DEF[7:0] bits, the GR_ARC_DEF[7:0] bits and the alpha blending in a rectangular area are turned on. Each time the vertical synchronization signal (VS) rises the number of times set in the GR_ARC_RATE[7:0] bits, the value in GR_ARC_COEF[8:0] is added to or subtracted from the alpha value. Figure 64.11 shows change in the alpha value.

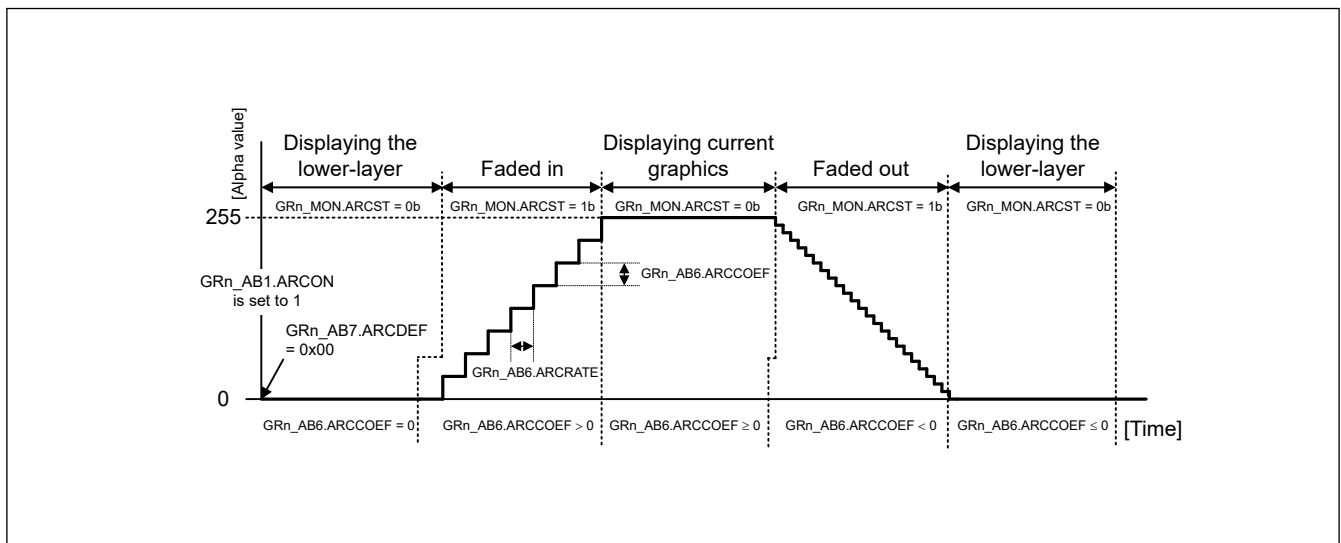


Figure 64.11 Changes in alpha value

64.2.22 GRn_AB7 : Graphics n Alpha Blending Control Register 7 (n = 1, 2)

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x1138 + 0x100 × (n - 1)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	ARCDEF[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CKON
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CKON	RGB-Index Chroma-Key Processing Control 0: Disable chroma-key processing 1: Enable chroma-key processing	R/W
15:1	—	These bits are read as 0. The write value should be 0.	R/W
23:16	ARCDEF[7:0]	Initial Alpha Value for Alpha Blending in Rectangular Area	R/W
31:24	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: This setting is reflected to the internal operations on assertion of the vertical synchronization signal (input) when GRn_VEN.PVEN = 1 or when the register value reflection control signal to internal operations for all the modules is asserted.

CKON bit (RGB-Index Chroma-Key Processing Control)

The CKON bit turns RGB-index chroma-key processing on or off. This function is enabled only if this bit is 1 when the graphics display plane is the blended display (GRn_AB1.DISPSEL[1:0] = 11b). And it is reflected to the graphics area except alpha blending in a rectangular area. For details, see [Table 64.8](#).

ARCDEF[7:0] bits (Initial Alpha Value for Alpha Blending in Rectangular Area)

The ARCDEF[7:0] bits specify the initial alpha value for alpha blending in a rectangular area. For changes in the alpha value during fade-in or fade-out of the current graphics using this bit, see [Figure 64.11](#).

64.2.23 GRn_AB8 : Graphics n Alpha Blending Control Register 8 (n = 1, 2)

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x113C + 0x100 × (n - 1)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	CKKG[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	CKKB[7:0]								CKKR[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	CKKR[7:0]	R Signal for RGB-Index Chroma-Key Processing R signal for RGB-index chroma-key processing. Unsigned 8-bit value.	R/W
15:8	CKKB[7:0]	B Signal for RGB-Index Chroma-Key Processing B signal for RGB-index chroma-key processing. Unsigned 8-bit value.	R/W
23:16	CKKG[7:0]	G Signal for RGB-Index Chroma-Key Processing G signal for RGB-index chroma-key processing. Unsigned 8-bit value.	R/W
31:24	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: This setting is reflected to the internal operations on assertion of the vertical synchronization signal (input) when GRn_VEN.PVEN = 1 or when the register value reflection control signal to internal operations for all the modules is asserted.

If RGB-index chroma-key processing is enabled when the RGB values of the current bit graphics agree with the values of the associated chroma-key bits, the image data of the current graphics (ARGB8888, including the alpha blending values) is replaced by the values in the GRn_AB9 register. In alpha blending in pixel units at later stages, the latest alpha values are used.

CKKR[7:0] bits (R Signal for RGB-Index Chroma-Key Processing)

The CKKR[7:0] bits specify the value to be compared with the R value of the current graphics in the RGB-index chromakey processing.

CKKB[7:0] bits (B Signal for RGB-Index Chroma-Key Processing)

The CKKB[7:0] bits specify the value to be compared with the B value of the current graphics in the RGB-index chromakey processing.

CKKG[7:0] bits (G Signal for RGB-Index Chroma-Key Processing)

The CKKG[7:0] bits specify the value to be compared with the G value of the current graphics in the RGB-index chroma-key processing.

64.2.24 GRn_AB9 : Graphics n Alpha Blending Control Register 9 (n = 1, 2)

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x1140 + 0x100 × (n - 1)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	CKA[7:0]								CKG[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	CKB[7:0]								CKR[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	CKR[7:0]	R Value after RGB-Index Chroma-Key Processing Replacement R value after RGB-index chroma-key processing replacement. Unsigned 8-bit value.	R/W
15:8	CKB[7:0]	B Value after RGB-Index Chroma-Key Processing Replacement B value after RGB-index chroma-key processing replacement. Unsigned 8-bit value.	R/W
23:16	CKG[7:0]	G Value after RGB-Index Chroma-Key Processing Replacement G value after RGB-index chroma-key processing replacement. Unsigned 8-bit value.	R/W
31:24	CKA[7:0]	A Value after RGB-Index Chroma-Key Processing Replacement A value after RGB-index chroma-key processing replacement.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: This setting is reflected to the internal operations on assertion of the vertical synchronization signal (input) when GRn_VEN.PVEN = 1 or when the register value reflection control signal to internal operations for all the modules is asserted.

If the RGB-index chroma-key processing is enabled and the pixel data is to be replaced (the RGB values of the current graphics agree with the GRn_AB8 values), alpha blending in pixel units is performed in later stages in accordance with this image data.

CKR[7:0] bits (R Value after RGB-Index Chroma-Key Processing Replacement)

The CKR[7:0] bits specify the R value after RGB-index chroma-key processing replacement.

CKB[7:0] bits (B Value after RGB-Index Chroma-Key Processing Replacement)

The CKB[7:0] bits specify the B value after RGB-index chroma-key processing replacement.

CKG[7:0] bits (G Value after RGB-Index Chroma-Key Processing Replacement)

The CKG[7:0] bits specify the G value after RGB-index chroma-key processing replacement.

CKA[7:0] bits (A Value after RGB-Index Chroma-Key Processing Replacement)

The CKA[7:0] bits specify the A value after RGB-index chroma-key processing replacement.

64.2.25 GRn_BASE : Graphics n Background Color Control Register (n = 1, 2)

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x114C + 0x100 × (n - 1)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	G[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	B[7:0]								R[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	R[7:0]	Background Color R Value Background color R value. Unsigned 8-bit value.	R/W
15:8	B[7:0]	Background Color B Value Background color B value. Unsigned 8-bit value.	R/W
23:16	G[7:0]	Background Color G Value Background color G value. Unsigned 8-bit value.	R/W
31:24	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: This setting is reflected to the internal operations on assertion of the vertical synchronization signal (input) when GRn_VEN.PVEN = 1 or when the register value reflection control signal to internal operations for all the modules is asserted.

When the background color is selected in the display selection (GRn_AB1.DISPSEL[1:0] = 00b), this RGB data is output to the entire display image area. When the current graphics setting is selected (GRn_AB1.DISPSEL[1:0] = 10b), the RGB data is output to the outside of the graphics image area within the display image area.

R[7:0] bits (Background Color R Value)

The R[7:0] bits specify the background color R value.

B[7:0] bits (Background Color B Value)

The B[7:0] bits specify the background color B value.

G[7:0] bits (Background Color G Value)

The G[7:0] bits specify the background color G value.

64.2.26 GRn_CLUTINT : Graphics n CLUT Table Interrupt Control Register (n = 1, 2)

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x1150 + 0x100 × (n - 1)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	SEL
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	LINE[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
10:0	LINE[10:0]	Number of Detection Lines 0x000: 1 line ⋮ 0x7FF: 2048 lines	R/W
15:11	—	These bits are read as 0. The write value should be 0./	R/W
16	SEL	CLUT Table Control 0: Select CLUT table 0 1: Select CLUT table 1	R/W
31:17	—	These bits are read as 0. The write value should be 0./	R/W

Note: S-TYPE-3, P-TYPE-3

Note: This setting is reflected to the internal operations on assertion of the vertical synchronization signal (input) when GRn_VEN.PVEN = 1 or when the register value reflection control signal to internal operations for all the modules is asserted.

LINE[10:0] bits (Number of Detection Lines)

The LINE[10:0] bits specify the number of lines to be detected. When the number of lines specified in this bit are detected, the event is recognized outside the module on the HS assertion. To retain the status of the recognized event and assert the GLCDC interrupt request signal, set the prescribed value to the State Detection Control Register (SYSCNT_DTCTEN) and Interrupt Request Enable Control Register (SYSCNT_INTEN), which are system control registers. Although this function is provided to both graphics 1 and 2, it is only enabled in graphics 2 in this GLCDC.

SEL bit (CLUT Table Control)

The SEL bit controls the CLUT plane to be used for internal operations. Access to the color palette (CLUT) through the register access bus is always valid for both planes 0 and 1, regardless of the setting in this bit, and the written value is immediately reflected to the internal operations (not in synchronization with the vertical synchronization signal).

To keep reflection of the CLUT memory contents to the internal operations in synchronization with the vertical synchronization signal, first write data through the register access bus to the plane that is not being used for the internal operations, and then modify the bits intended for controlling the plane that is to be used for the internal operations.

64.2.27 GRn_MON : Graphics n Status Monitor Register (n = 1, 2)

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x1154 + 0x100 × (n - 1)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	UNDF LST
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ARCS T
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ARCST	Status Monitor for Alpha Blending in Rectangular Area 0: Fade-in/fade-out not in progress 1: Fade-in/fade-out in progress	R
15:1	—	These bits are read as 0.	R
16	UNDFLST	Underflow Status Monitor 0: No underflow occurred in internal operations 1: Underflow occurred in internal operations	R
31:17	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

ARCST bit (Status Monitor for Alpha Blending in Rectangular Area)

The ARCST bit indicates whether or not alpha blending (fade-in/fade-out) in a rectangular area is in progress. When alpha blending in a rectangular area is turned on (GRn_AB1.ARCON is set to 1) and the register value is to be reflected to the internal operations on assertion of the vertical synchronization signal (VS), this bit sets to 1 immediately on assertion of the vertical synchronization signal. When alpha blending in a rectangular area is turned off (GRn_AB1.ARCON cleared to 0) or when the alpha blending (fade-in/fade-out) in a rectangular area is complete (alpha blended value reaches the minimum or maximum value), this bit clears to 0. If the alpha coefficient for the alpha blending in a rectangular area (GRn_AB6.ARCCOEF[8:0]) is set to 0x000 and the initial alpha value for alpha blending in a rectangular area (GRn_AB7.ARCDEF[7:0]) is set to any value other than 0xFF or 0x00, the alpha blending value does not reach the minimum or maximum value, and this bit remains 1 (no timeout processing is performed).

UNDFLST bit (Underflow Status Monitor)

The UNDFLST bit indicates whether or not an underflow has occurred in the internal operations. The underflow interrupt request flag sets when an underflow occurs, and retains its value until it is cleared by software. However, this bit monitors the internal status, and so the flag automatically clears to 0 when the graphics data bus interface initializes on assertion of the vertical synchronization signal. Even when the current graphics data is not required (GRn_AB1.DISPSEL[1:0] = 0xb), this bit sets to 1 during the period from the graphics image valid area start to the next frame vertical synchronization signal (VS) assertion timing set in the registers.

64.2.28 GAMn_LATCH : Gamma n Register Update Control Register (n = G, B, R)

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x1300 (GAMG_LATCH)
0x1340 (GAMB_LATCH)
0x1380 (GAMR_LATCH)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	VEN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	VEN	Control of Gamma Correction × Module Register Value Reflection to Internal Operations 0: Disable reflection of register values to internal operations on assertion of vertical synchronization signal (VS) 1: Enable reflection of register values to internal operations on assertion of the vertical synchronization signal (VS)	R/W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

VEN bit (Control of Gamma Correction × Module Register Value Reflection to Internal Operations)

The VEN bit enables or disables reflection of the register values to the internal operations in the gamma correction circuit on assertion of the vertical synchronization signal (input). When this bit is set to 1, the register values are immediately reflected to the internal operations on assertion of the vertical synchronization signal (input), and then this bit automatically clears to 0. Also, if the signal is asserted that controls reflection of the register values to the internal operations of all the modules output from the background plane generation module, the register values are reflected to the internal operations on assertion of the vertical synchronization signal (input), regardless of the value of this bit. While this bit is 1, do not modify any register whose value is reflected to the internal operations on assertion of the vertical synchronization signal (input) in the GLCDC. Otherwise, operation is not guaranteed.

This bit must not be 1 at the same time as the BG_EN.VEN bit (control of background plane register value reflection to internal operations) in the Operation Control Register (BG_EN), one of the background plane setting registers. Otherwise, operation is not guaranteed.

Although there are three VEN bits, one for each G, R, and B color, only the GAMG_LATCH.VEN bit controls the gamma correction with the reflection of the GAM_SW.GAMON bit. To enable gamma correction, set the GAMG_LATCH.VEN bit once after setting the GAM_SW.GAMON bit.

The VEN bit is set to 1 by writing 1, and automatically clears to 0 immediately on assertion of the vertical synchronization signal.

64.2.29 GAM_SW : Gamma Correction Block Function Switch Register

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x1304

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	GAMON
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	GAMON	Gamma Correction On/Off Control 0: Turn off gamma correction 1: Turn on gamma correction	R/W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: This setting is reflected to the internal operations on assertion of the vertical synchronization signal (input) when GAMG_LATCH.VEN = 1 or when the register value reflection control signal to internal operations for all the modules is asserted.

GAMON bit (Gamma Correction On/Off Control)

The GAMON bit turns on or off gamma correction.

64.2.30 GAMn_LUT1 : Gamma n Correction Block Table Setting Register 1 (n = G, B, R)

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x1308 (GAMG_LUT1)
0x1348 (GAMB_LUT1)
0x1388 (GAMR_LUT1)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	GAIN00[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	GAIN01[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
10:0	GAIN01[10:0]	Gain Value of Area 1 Unsigned 11-bit fixed point value. 0x000: 0.000 (0/1024) ⋮ 0x400: 1.000 (1024/1024) ⋮ 0x7FF: 1.999 (2047/1024)	R/W
15:11	—	These bits are read as 0. The write value should be 0.	R/W
26:16	GAIN00[10:0]	Gain Value of Area 0 0x000: 0.000 (0/1024) ⋮ 0x400: 1.000 (1024/1024) ⋮ 0x7FF: 1.999 (2047/1024)	R/W
31:27	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: This setting is reflected to the internal operations on assertion of the vertical synchronization signal (input) when GAMx_LATCH.VEN = 1 or when the register value reflection control signal to internal operations for all the modules is asserted.

Gamma correction for each color is carried out as follows:

- Din (input signal): 10 bits. To correct the output of graphics 2, 00b is appended as the lower 2 bits for extension
- Dout (output signal): 10 bits
- TH (threshold): 10 bits (register setting); up to 15 can be set
- GAIN (gain): 0/1024 to 2047/1024 for each area (register setting), can be set for up to 16 areas
- OFFSET (offset value): 21 bits (result of internal calculation; calculation of up to 15 points)

The following is automatically calculated from the assertion of the vertical synchronization signal (VS) to the start of valid pixel data internally:

- $D_{out} = ((D_{in} - TH(n)) \times GAIN(n) + OFFSET(n))$
- $OFFSET(n) = OFFSET(n - 1) + (TH(n) - TH(n - 1)) \times GAIN(n - 1)$, where $OFFSET(n) = 0$

Note: Because the gain is a positive number (≥ 0), the correction line shows a monotonic increase.

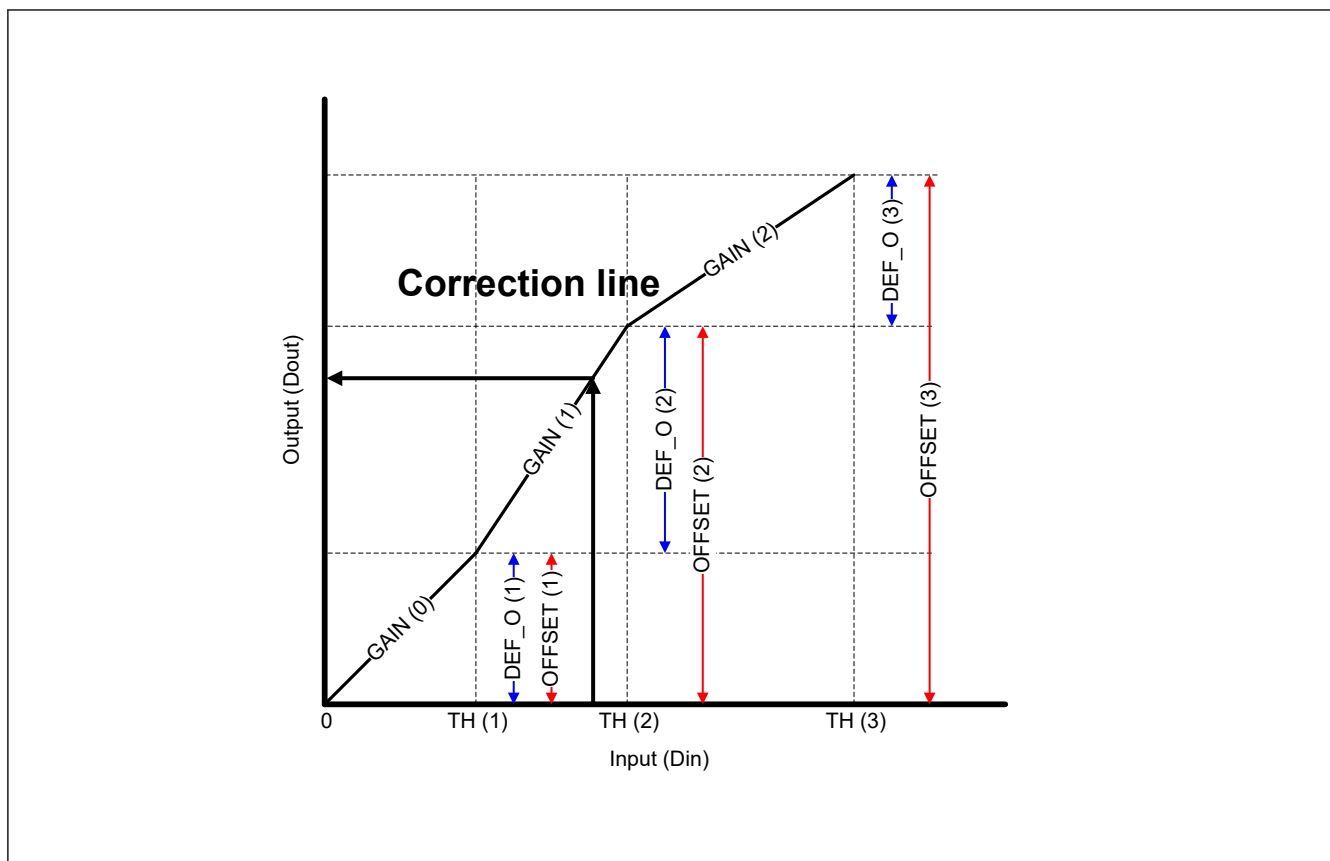


Figure 64.12 Calculation of gamma correction value

GAIN01[10:0] bits (Gain Value of Area 1)

The GAIN01[10:0] bits specify the gain value of area 1 to be used for gamma correction in terms of an unsigned 11-bit fixed point (0/1024 to 2047/1024). The location of the decimal point is between bits [10] and [11].

GAIN00[10:0] bits (Gain Value of Area 0)

The GAIN00[10:0] bits specify the gain value of area 0 to be used for gamma correction in terms of an unsigned 11-bit fixed point (0/1024 to 2047/1024). The location of the decimal point is between bits [10] and [11].

64.2.31 GAMn_LUT2 : Gamma n Correction Block Table Setting Register 2 (n = G, B, R)

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x130C (GAMG_LUT2)
0x134C (GAMB_LUT2)
0x138C (GAMR_LUT2)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	GAIN02[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	GAIN03[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
10:0	GAIN03[10:0]	Gain Value of Area 3 Unsigned 11-bit fixed point value. 0x000: 0.000 (0/1024) ⋮ 0x400: 1.000 (1024/1024) ⋮ 0x7FF: 1.999 (2047/1024)	R/W
15:11	—	These bits are read as 0. The write value should be 0.	R/W
26:16	GAIN02[10:0]	Gain Value of Area 2 Unsigned 11-bit fixed point value. 0x000: 0.000 (0/1024) ⋮ 0x400: 1.000 (1024/1024) ⋮ 0x7FF: 1.999 (2047/1024)	R/W
31:27	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: This setting is reflected to the internal operations on assertion of the vertical synchronization signal (input) when GAMx_LATCH.VEN = 1 or when the register value reflection control signal to internal operations for all the modules is asserted.

GAIN03[10:0] bits (Gain Value of Area 3)

The GAIN03[10:0] bits specify the gain value of area 3 to be used for gamma correction in terms of an unsigned 11-bit fixed point (0/1024 to 2047/1024). The location of the decimal point is between bits [10] and [11].

GAIN02[10:0] bits (Gain Value of Area 2)

The GAIN02[10:0] bits specify the gain value of area 2 to be used for gamma correction in terms of an unsigned 11-bit fixed point (0/1024 to 2047/1024). The location of the decimal point is between bits [10] and [11].

64.2.32 GAMn_LUT3 : Gamma n Correction Block Table Setting Register 3 (n = G, B, R)

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x1310 (GAMG_LUT3)
0x1350 (GAMB_LUT3)
0x1390 (GAMR_LUT3)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	GAIN04[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	GAIN05[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
10:0	GAIN05[10:0]	Gain Value of Area 5 Unsigned 11-bit fixed point value. 0x000: 0.000 (0/1024) ⋮ 0x400: 1.000 (1024/1024) ⋮ 0x7FF: 1.999 (2047/1024)	R/W
15:11	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
26:16	GAIN04[10:0]	Gain Value of Area 4 Unsigned 11-bit fixed point value. 0x000: 0.000 (0/1024) ⋮ 0x400: 1.000 (1024/1024) ⋮ 0x7FF: 1.999 (2047/1024)	R/W
31:27	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: This setting is reflected to the internal operations on assertion of the vertical synchronization signal (input) when GAMx_LATCH.VEN = 1 or when the register value reflection control signal to internal operations for all the modules is asserted.

GAIN05[10:0] bits (Gain Value of Area 5)

The GAIN05[10:0] bits specify the gain value of area 5 to be used for gamma correction in terms of an unsigned 11-bit fixed point (0/1024 to 2047/1024). The location of the decimal point is between bits [10] and [11].

GAIN04[10:0] bits (Gain Value of Area 4)

The GAIN04[10:0] bits specify the gain value of area 4 to be used for gamma correction in terms of an unsigned 11-bit fixed point (0/1024 to 2047/1024). The location of the decimal point is between bits [10] and [11].

64.2.33 GAMn_LUT4 : Gamma n Correction Block Table Setting Register 4 (n = G, B, R)

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x1314 (GAMG_LUT4)
0x1354 (GAMB_LUT4)
0x1394 (GAMR_LUT4)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	GAIN06[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	GAIN07[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
10:0	GAIN07[10:0]	Gain Value of Area 7 Unsigned 11-bit fixed point value. 0x000: 0.000 (0/1024) ⋮ 0x400: 1.000 (1024/1024) ⋮ 0x7FF: 1.999 (2047/1024)	R/W
15:11	—	These bits are read as 0. The write value should be 0.	R/W
26:16	GAIN06[10:0]	Gain Value of Area 6 Unsigned 11-bit fixed point value. 0x000: 0.000 (0/1024) ⋮ 0x400: 1.000 (1024/1024) ⋮ 0x7FF: 1.999 (2047/1024)	R/W
31:27	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: This setting is reflected to the internal operations on assertion of the vertical synchronization signal (input) when GAMx_LATCH.VEN = 1 or when the register value reflection control signal to internal operations for all the modules is asserted.

GAIN07[10:0] bits (Gain Value of Area 7)

The GAIN07[10:0] bits specify the gain value of area 7 to be used for gamma correction in terms of an unsigned 11-bit fixed point (0/1024 to 2047/1024). The location of the decimal point is between bits [10] and [11].

GAIN06[10:0] bits (Gain Value of Area 6)

The GAIN06[10:0] bits specify the gain value of area 6 to be used for gamma correction in terms of an unsigned 11-bit fixed point (0/1024 to 2047/1024). The location of the decimal point is between bits [10] and [11].

64.2.34 GAMn_LUT5 : Gamma n Correction Block Table Setting Register 5 (n = G, B, R)

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x1318 (GAMG_LUT5)
0x1358 (GAMB_LUT5)
0x1398 (GAMR_LUT5)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	GAIN08[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	GAIN09[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
10:0	GAIN09[10:0]	Gain Value of Area 9 Unsigned 11-bit fixed point value. 0x000: 0.000 (0/1024) ⋮ 0x400: 1.000 (1024/1024) ⋮ 0x7FF: 1.999 (2047/1024)	R/W
15:11	—	These bits are read as 0. The write value should be 0.	R/W
26:16	GAIN08[10:0]	Gain Value of Area 8 Unsigned 11-bit fixed point value. 0x000: 0.000 (0/1024) ⋮ 0x400: 1.000 (1024/1024) ⋮ 0x7FF: 1.999 (2047/1024)	R/W
31:27	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: This setting is reflected to the internal operations on assertion of the vertical synchronization signal (input) when GAMx_LATCH.VEN = 1 or when the register value reflection control signal to internal operations for all the modules is asserted.

GAIN09[10:0] bits (Gain Value of Area 9)

The GAIN09[10:0] bits specify the gain value of area 9 to be used for gamma correction in terms of an unsigned 11-bit fixed point (0/1024 to 2047/1024). The location of the decimal point is between bits [10] and [11].

GAIN08[10:0] bits (Gain Value of Area 8)

The GAIN08[10:0] bits specify the gain value of area 8 to be used for gamma correction in terms of an unsigned 11-bit fixed point (0/1024 to 2047/1024). The location of the decimal point is between bits [10] and [11].

64.2.35 GAMn_LUT6 : Gamma n Correction Block Table Setting Register 6 (n = G, B, R)

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x131C (GAMG_LUT6)
0x135C (GAMB_LUT6)
0x139C (GAMR_LUT6)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	GAIN10[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	GAIN11[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
10:0	GAIN11[10:0]	Gain Value of Area 11 Unsigned 11-bit fixed point value. 0x000: 0.000 (0/1024) ⋮ 0x400: 1.000 (1024/1024) ⋮ 0x7FF: 1.999 (2047/1024)	R/W
15:11	—	These bits are read as 0. The write value should be 0.	R/W
26:16	GAIN10[10:0]	Gain Value of Area 10 Unsigned 11-bit fixed point value. 0x000: 0.000 (0/1024) ⋮ 0x400: 1.000 (1024/1024) ⋮ 0x7FF: 1.999 (2047/1024)	R/W
31:27	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: This setting is reflected to the internal operations on assertion of the vertical synchronization signal (input) when GAMx_LATCH.VEN = 1 or when the register value reflection control signal to internal operations for all the modules is asserted.

GAIN11[10:0] bits (Gain Value of Area 11)

The GAIN11[10:0] bits specify the gain value of area 11 to be used for gamma correction in terms of an unsigned 11-bit fixed point (0/1024 to 2047/1024). The location of the decimal point is between bits [10] and [11].

GAIN10[10:0] bits (Gain Value of Area 10)

The GAIN10[10:0] bits specify the gain value of area 10 to be used for gamma correction in terms of an unsigned 11-bit fixed point (0/1024 to 2047/1024). The location of the decimal point is between bits [10] and [11].

64.2.36 GAMn_LUT7 : Gamma n Correction Block Table Setting Register 7 (n = G, B, R)

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x1320 (GAMG_LUT7)
0x1360 (GAMB_LUT7)
0x13A0 (GAMR_LUT7)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	GAIN012[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	GAIN13[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
10:0	GAIN13[10:0]	Gain Value of Area 13 Unsigned 11-bit fixed point value. 0x000: 0.000 (0/1024) ⋮ 0x400: 1.000 (1024/1024) ⋮ 0x7FF: 1.999 (2047/1024)	R/W
15:11	—	These bits are read as 0. The write value should be 0.	R/W
26:16	GAIN012[10:0]	Gain Value of Area 12 Unsigned 11-bit fixed point value. 0x000: 0.000 (0/1024) ⋮ 0x400: 1.000 (1024/1024) ⋮ 0x7FF: 1.999 (2047/1024)	R/W
31:27	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: This setting is reflected to the internal operations on assertion of the vertical synchronization signal (input) when GAMx_LATCH.VEN = 1 or when the register value reflection control signal to internal operations for all the modules is asserted.

GAIN13[10:0] bits (Gain Value of Area 13)

The GAIN13[10:0] bits specify the gain value of area 13 to be used for gamma correction in terms of an unsigned 11-bit fixed point (0/1024 to 2047/1024). The location of the decimal point is between bits [10] and [11].

GAIN012[10:0] bits (Gain Value of Area 12)

The GAIN012[10:0] bits specify the gain value of area 12 to be used for gamma correction in terms of an unsigned 11-bit fixed point (0/1024 to 2047/1024). The location of the decimal point is between bits [10] and [11].

64.2.37 GAMn_LUT8 : Gamma n Correction Block Table Setting Register 8 (n = G, B, R)

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x1324 (GAMG_LUT8)
0x1364 (GAMB_LUT8)
0x13A4 (GAMR_LUT8)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	GAIN14[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	GAIN15[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
10:0	GAIN15[10:0]	Gain Value of Area 15 Unsigned 11-bit fixed point value. 0x000: 0.000 (0/1024) ⋮ 0x400: 1.000 (1024/1024) ⋮ 0x7FF: 1.999 (2047/1024)	R/W
15:11	—	These bits are read as 0. The write value should be 0.	R/W
26:16	GAIN14[10:0]	Gain Value of Area 14 Unsigned 11-bit fixed point value. 0x000: 0.000 (0/1024) ⋮ 0x400: 1.000 (1024/1024) ⋮ 0x7FF: 1.999 (2047/1024)	R/W
31:27	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: This setting is reflected to the internal operations on assertion of the vertical synchronization signal (input) when GAMx_LATCH.VEN = 1 or when the register value reflection control signal to internal operations for all the modules is asserted.

GAIN15[10:0] bits (Gain Value of Area 15)

The GAIN15[10:0] bits specify the gain value of area 15 to be used for gamma correction in terms of an unsigned 11-bit fixed point (0/1024 to 2047/1024). The location of the decimal point is between bits [10] and [11].

GAIN14[10:0] bits (Gain Value of Area 14)

The GAIN14[10:0] bits specify the gain value of area 14 to be used for gamma correction in terms of an unsigned 11-bit fixed point (0/1024 to 2047/1024). The location of the decimal point is between bits [10] and [11].

64.2.38 GAMn_AREA1 : Gamma n Correction Block Area Setting Register 1 (n = G, B, R)

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x1328 (GAMG_AREA1)
0x1368 (GAMB_AREA1)
0x13A8 (GAMR_AREA1)

Bit position:	31	29											20	19											10	9											0
Bit field:	—	—	TH01[9:0]										TH02[9:0]										TH03[9:0]														
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0					

Bit	Symbol	Function	R/W
9:0	TH03[9:0]	Start Threshold of Area 3 Unsigned 10-bit integer.	R/W
19:10	TH02[9:0]	Start Threshold of Area 2 Unsigned 10-bit integer.	R/W
29:20	TH01[9:0]	Start Threshold of Area 1 Unsigned 10-bit integer.	R/W
31:30	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: This setting is reflected to the internal operations on assertion of the vertical synchronization signal (input) when $GAMx_LATCH.VEN = 1$ or when the register value reflection control signal to internal operations for all the modules is asserted.

Set the start threshold TH (n) of area n (n = 0 to 15) to satisfy the following conditions. Otherwise, operation is not guaranteed.

$$TH(n) < TH(n + 1)$$

n = 0 to 15 and TH (0) = 0x000, TH (16) = 0x3FF

TH (n) = TH (n + 1) is valid only if TH (n) = 0x3FF.

For details on calculation of the gamma correction value, see [section 64.2.30. GAMn_LUT1 : Gamma n Correction Block Table Setting Register 1 \(n = G, B, R\)](#) and [Figure 64.12](#)

TH03[9:0] bits (Start Threshold of Area 3)

The TH03[9:0] bits specify the start threshold of area 3 to be used for gamma correction in terms of an unsigned 10-bit integer.

TH02[9:0] bits (Start Threshold of Area 2)

The TH02[9:0] bits specify the start threshold of area 2 to be used for gamma correction in terms of an unsigned 10-bit integer.

TH01[9:0] bits (Start Threshold of Area 1)

The TH01[9:0] bits specify the start threshold of area 1 to be used for gamma correction in terms of an unsigned 10-bit integer.

64.2.39 GAMn_AREA2 : Gamma n Correction Block Area Setting Register 2 (n = G, B, R)

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x132C (GAMG_AREA2)
0x136C (GAMB_AREA2)
0x13AC (GAMR_AREA2)

Bit position:	31	29	20	19	10	9	0
Bit field:	—	—	TH04[9:0]	TH05[9:0]	TH06[9:0]		

Value after reset: 0

Bit	Symbol	Function	R/W
9:0	TH06[9:0]	Start Threshold of Area 6 Unsigned 10-bit integer.	R/W
19:10	TH05[9:0]	Start Threshold of Area 5 Unsigned 10-bit integer.	R/W
29:20	TH04[9:0]	Start Threshold of Area 4 Unsigned 10-bit integer.	R/W
31:30	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: This setting is reflected to the internal operations on assertion of the vertical synchronization signal (input) when $GAMx_LATCH.VEN = 1$ or when the register value reflection control signal to internal operations for all the modules is asserted.

TH06[9:0] bits (Start Threshold of Area 6)

The TH06[9:0] bits specify the start threshold of area 6 to be used for gamma correction in terms of an unsigned 10-bit integer.

TH05[9:0] bits (Start Threshold of Area 5)

The TH05[9:0] bits specify the start threshold of area 5 to be used for gamma correction in terms of an unsigned 10-bit integer.

TH04[9:0] bits (Start Threshold of Area 4)

The TH04[9:0] bits specify the start threshold of area 4 to be used for gamma correction in terms of an unsigned 10-bit integer.

64.2.40 GAMn_AREA3 : Gamma n Correction Block Area Setting Register 3 (n = G, B, R)

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x1330 (GAMG_AREA3)
0x1370 (GAMB_AREA3)
0x13B0 (GAMR_AREA3)

Bit position:	31	29	20	19	10	9	0
Bit field:	—	—	TH07[9:0]	TH08[9:0]	TH09[9:0]		
Value after reset:	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
9:0	TH09[9:0]	Start Threshold of Area 9 Unsigned 10-bit integer.	R/W
19:10	TH08[9:0]	Start Threshold of Area 8 Unsigned 10-bit integer.	R/W
29:20	TH07[9:0]	Start Threshold of Area 7 Unsigned 10-bit integer.	R/W
31:30	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: This setting is reflected to the internal operations on assertion of the vertical synchronization signal (input) when
GAMx_LATCH.VEN = 1 or when the register value reflection control signal to internal operations for all the modules is asserted.

TH09[9:0] bits (Start Threshold of Area 9)

The TH09[9:0] bits specify the start threshold of area 9 to be used for gamma correction in terms of an unsigned 10-bit integer.

TH08[9:0] bits (Start Threshold of Area 8)

The TH08[9:0] bits specify the start threshold of area 8 to be used for gamma correction in terms of an unsigned 10-bit integer.

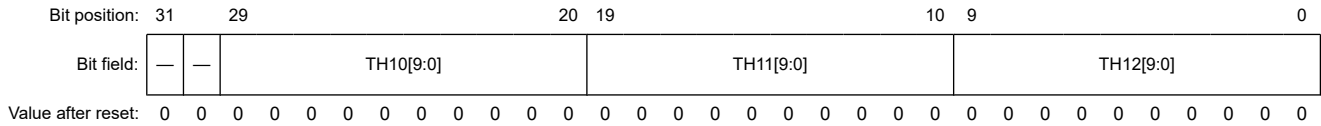
TH07[9:0] bits (Start Threshold of Area 7)

The TH07[9:0] bits specify the start threshold of area 7 to be used for gamma correction in terms of an unsigned 10-bit integer.

64.2.41 GAMn_AREA4 : Gamma n Correction Block Area Setting Register 4 (n = G, B, R)

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x1334 (GAMG_AREA4)
0x1374 (GAMB_AREA4)
0x13B4 (GAMR_AREA4)



Bit	Symbol	Function	R/W
9:0	TH12[9:0]	Start Threshold of Area 12 Unsigned 10-bit integer.	R/W
19:10	TH11[9:0]	Start Threshold of Area 11 Unsigned 10-bit integer.	R/W
29:20	TH10[9:0]	Start Threshold of Area 10 Unsigned 10-bit integer.	R/W
31:30	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: This setting is reflected to the internal operations on assertion of the vertical synchronization signal (input) when GAMx_LATCH.VEN = 1 or when the register value reflection control signal to internal operations for all the modules is asserted.

TH12[9:0] bits (Start Threshold of Area 12)

The TH12[9:0] bits specify the start threshold of area 12 to be used for gamma correction in terms of an unsigned 10-bit integer.

TH11[9:0] bits (Start Threshold of Area 11)

The TH11[9:0] bits specify the start threshold of area 11 to be used for gamma correction in terms of an unsigned 10-bit integer.

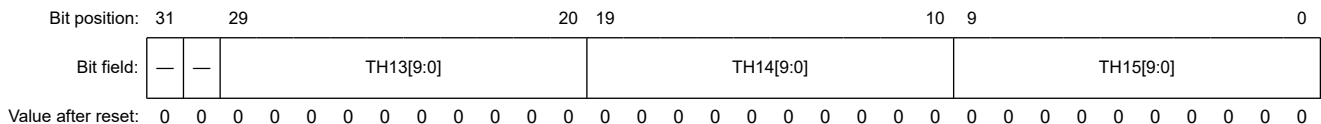
TH10[9:0] bits (Start Threshold of Area 10)

The TH10[9:0] bits specify the start threshold of area 10 to be used for gamma correction in terms of an unsigned 10-bit integer.

64.2.42 GAMn_AREA5 : Gamma n Correction Block Area Setting Register 5 (n = G, B, R)

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x1338 (GAMG_AREA5)
0x1378 (GAMB_AREA5)
0x13B8 (GAMR_AREA5)



Bit	Symbol	Function	R/W
9:0	TH15[9:0]	Start Threshold of Area 15 Unsigned 10-bit integer.	R/W
19:10	TH14[9:0]	Start Threshold of Area 14 Unsigned 10-bit integer.	R/W
29:20	TH13[9:0]	Start Threshold of Area 13 Unsigned 10-bit integer.	R/W

Bit	Symbol	Function	R/W
31:30	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: This setting is reflected to the internal operations on assertion of the vertical synchronization signal (input) when GAMx_LATCH.VEN = 1 or when the register value reflection control signal to internal operations for all the modules is asserted.

TH15[9:0] bits (Start Threshold of Area 15)

The TH15[9:0] bits specify the start threshold of area 15 to be used for gamma correction in terms of an unsigned 10-bit integer.

TH14[9:0] bits (Start Threshold of Area 14)

The TH14[9:0] bits specify the start threshold of area 14 to be used for gamma correction in terms of an unsigned 10-bit integer.

TH13[9:0] bits (Start Threshold of Area 13)

The TH13[9:0] bits specify the start threshold of area 13 to be used for gamma correction in terms of an unsigned 10-bit integer.

64.2.43 OUT_VLATCH : Output Control Block Register Update Control Register

Base address: GLCDC = 0x4034_2000
GLCDC NS = 0x5034_2000

Offset address: 0x13C0

Bit position: 31

0

Bit field:

[illegible]

Value after reset: 0

Bit	Symbol	Function	R/W
0	VEN	Control of Output Control Module Register Value Reflection to Internal Operations 0: Disable reflection of register values to internal operations on assertion of vertical synchronization signal (VS) 1: Enable reflection of register values to internal operations on assertion of vertical synchronization signal (VS).	R/W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

VEN bit (Control of Output Control Module Register Value Reflection to Internal Operations)

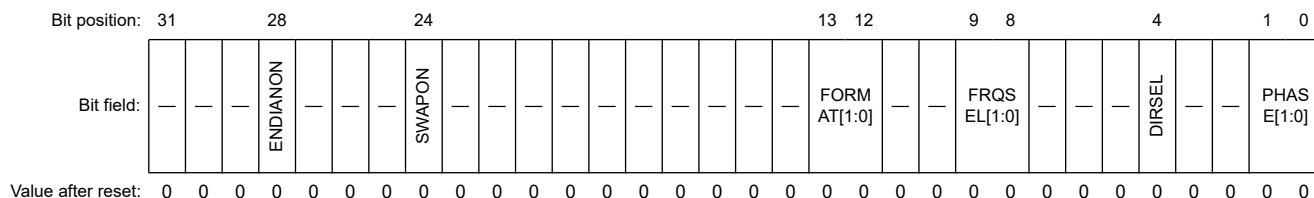
The VEN bit enables or disables reflection of the register values to the internal operations in the output control circuit on assertion of the vertical synchronization signal (input). When this bit is set to 1, the register values are immediately reflected to the internal operations on assertion of the vertical synchronization signal (input), and then this bit automatically clears to 0. Also, if the signal is asserted that controls reflection of the register values to the internal operations of all the modules output from the ground plane generation module, the register values are reflected to the internal operations on assertion of the vertical synchronization signal (input), regardless of the value of this bit. While this bit is 1, do not modify any register whose value is reflected to the internal operations on assertion of the vertical synchronization signal (input) in the GLCDC. Otherwise, operation is not guaranteed.

This bit must not be 1 at the same time as the BG_EN.VEN bit (control of background plane register value reflection to internal operations) in the Operation Control Register (BG_EN), or one of the background plane setting registers. Otherwise, operation is not guaranteed.

64.2.44 OUT SET : Output Control Block Output Interface Register

Base address: GLCDC = 0x4034_2000
GLCDC NS = 0x5034 2000

Offset address: 0x13C4



Bit	Symbol	Function	R/W
1:0	PHASE[1:0]	Data Output Delay Control in Serial RGB Format Data delay in LCD_CLK cycles. 0 0: 0 cycle 0 1: 1 cycle 1 0: 2 cycles 1 1: 3 cycles	R/W
3:2	—	These bits are read as 0. The write value should be 0.	R/W
4	DIRSEL	Scan Direction Select of Serial RGB Format 0: Forward scan 1: Reverse scan	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W
9:8	FRQSEL[1:0]	Clock Frequency Division Control 0 0: No frequency division, parallel RGB 0 1: Setting prohibited 1 0: Quarter frequency (serial RGB) 1 1: Setting prohibited	R/W
11:10	—	These bits are read as 0. The write value should be 0.	R/W
13:12	FORMAT[1:0]	Output Format Select 0 0: RGB888 — select RGB888 as dither output format 0 1: RGB666 — select RGB666 as dither output format 1 0: RGB565 — select RGB565 as dither output format 1 1: Serial RGB — select RGB888 as dither output format. Select dither output format in OUT_PDTHA.FORM[1:0]	R/W
23:14	—	These bits are read as 0. The write value should be 0.	R/W
24	SWAPON	Pixel Order Control 0: RGB order 1: BGR order	R/W
27:25	—	These bits are read as 0. The write value should be 0.	R/W
28	ENDIANON	Bit Endian Control 0: Descending order (little endian) 1: Ascending order (big endian)	R/W
31:29	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: Rewriting these bits is prohibited during operation. Set the settings before operation is enabled. Operation is not guaranteed if the bit is rewritten during operation.

For details on assignment of pixel data to the output pins (LCD_DATA23 to LCD_DATA00) by setting these bits (except for OUT_SET.FROSEL[1:0]), see [Figure 64.3](#) to [Figure 64.6](#).

PHASE[1:0] bits (Data Output Delay Control in Serial RGB Format)

The PHASE[1:0] bits control data output delay in serial RGB format. When the delay is 0 cycle (these bits are 00b), pixel data (R, B, or invalid data, depending on the setting in this register) is output one pixel clock (PXCLK) cycle after the

horizontal data enable signal (HE). When any value other than 00b is set to OUT_SET.PHASE[1:0], pixel data output is delayed for a preset number of LCD_CLK cycles.

DIRSEL bit (Scan Direction Select of Serial RGB Format)

The DIRSEL bit controls the data arrangement of the serial RGB format. When this bit is set to 1, the serial RGB data is arranged in reverse direction, and when it is 0, the serial RGB is arranged in forward direction.

FRQSEL[1:0] bits (Clock Frequency Division Control)

The FRQSEL[1:0] bits control clock frequency division of LCD_CLK (panel output clock) and PXCLK (pixel clock for internal operations). Set these bits to 10b only for the serial RGB format (OUT_SET.FORMAT[1:0] = 11b), so that PXCLK has a 1/4 frequency of the LCD_CLK frequency and synchronizes with LCD_CLK. Set these bits to 00b for the parallel RGB format (OUT_SET.FORMAT[1:0] = 10b, 01b, or 00b), so that PXCLK has the same frequency as the LCD_CLK frequency and synchronizes with LCD_CLK. Otherwise, operation is not guaranteed.

FORMAT[1:0] bits (Output Format Select)

The FORMAT[1:0] bits select the output format of RGB data. Set these bits in accordance with the output format select bits in the Panel Dither Correction Register (OUT_PDTHA.FORM[1:0]). For serial RGB format (these bits are 11b), set OUT_PDTHA.FORM[1:0] to 00b. Otherwise, operation is not guaranteed.

SWAPON bit (Pixel Order Control)

The SWAPON bit controls the pixel order of RGB data output. When this bit is set to 1, internally processed data is assigned to the output pins in BGR order, and when this bit is 0, data is assigned in the RGB order. Data is assigned to the output pins (LCD_DATA23 to LCD_DATA00) with the MSB first for the RGB parallel format, and serially for the RGB serial format.

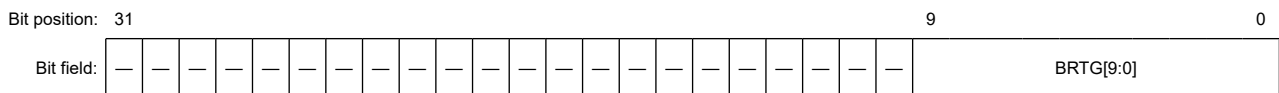
ENDIANON bit (Bit Endian Control)

The ENDIANON bit controls the bit order of RGB data output. When this bit is set to 1, internally processed data is assigned to the output pins in ascending order (big endian), and when this bit is 0, data is assigned in descending order (little endian).

64.2.45 OUT_BRIGHT1 : Output Control Block Brightness Correction Register 1

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x13C8



Value after reset: 0

Bit	Symbol	Function	R/W
9:0	BRTG[9:0]	Brightness Adjustment of G Signal Brightness (DC) adjustment of G signal. Unsigned 10-bit integer, +512 with offset.	R/W
31:10	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: This setting is reflected to the internal operations on assertion of the vertical synchronization signal (input) when OUT_VLATCH.VEN = 1 or when the register value reflection control signal to internal operations for all the modules is asserted.

BRTG[9:0] bits (Brightness Adjustment of G Signal)

The BRTG[9:0] bits specify the brightness (DC) adjustment of the G signal.

Brightness correction of the G signal is performed as follows:

- Gout: Output of brightness correction (input of contrast correction). Unsigned 10 bits
- Gin: Input of brightness correction. Unsigned 10 bits
- BRTG: Setting in this bit
- $G_{out} = G_{in} + BRTG - 512$.

64.2.46 OUT_BRIGHT2 : Output Control Block Brightness Correction Register 2

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x13CC

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	BRTB[9:0]									
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	BRTR[9:0]									
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
9:0	BRTR[9:0]	Brightness Adjustment of R Signal Brightness (DC) adjustment of R signal. Unsigned 10-bit integer, +512 with offset.	R/W
15:10	—	These bits are read as 0. The write value should be 0.	R/W
25:16	BRTB[9:0]	Brightness Adjustment of B Signal Brightness (DC) adjustment of B signal. Unsigned 10-bit integer, +512 with offset.	R/W
31:26	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: This setting is reflected to the internal operations on assertion of the vertical synchronization signal (input) when OUT_VLATCH.VEN = 1 or when the register value reflection control signal to internal operations for all the modules is asserted.

BRTR[9:0] bits (Brightness Adjustment of R Signal)

The BRTR[9:0] bits specify the brightness (DC) adjustment of the R signal.

Brightness correction of the R signal is performed as follows:

- Rout: Output of brightness correction (input of contrast correction), unsigned, 10 bits
- BRTR: Setting in this bit
- $R_{out} = R_{in} + BRTR - 512$.

BRTB[9:0] bits (Brightness Adjustment of B Signal)

The BRTB[9:0] bits specify the brightness (DC) adjustment of the B signal.

Brightness correction of the B signal is performed as follows:

- Bout: Output of brightness correction (input of contrast correction), unsigned, 10 bits
- Bin: Input of brightness correction, unsigned, 10 bits
- BRTB: Setting in this bit
- $B_{out} = B_{in} + BRTB - 512$.

64.2.47 OUT_CONTRAST : Output Control Block Contrast Correction Register

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x13D0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	CONTG[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	CONTB[7:0]								CONTR[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	CONTR[7:0]	Contrast Adjustment of R Signal Unsigned 8-bit fixed point value adjusting GAIN on R signal. 0x00: 0/128 = 0.000 ⋮ 0x80: 128/128 = 1.000 ⋮ 0xFF: 255/128 = 1.992	R/W
15:8	CONTB[7:0]	Contrast Adjustment of B Signal Unsigned 8-bit fixed point value adjusting GAIN on B signal. 0x00: 0/128 = 0.000 ⋮ 0x80: 128/128 = 1.000 ⋮ 0xFF: 255/128 = 1.992	R/W
23:16	CONTG[7:0]	Contrast Adjustment of G Signal Unsigned 8-bit fixed point value adjusting GAIN on G signal. 0x00: 0/128 = 0.000 ⋮ 0x80: 128/128 = 1.000 ⋮ 0xFF: 255/128 = 1.992	R/W
31:24	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: This setting is reflected to the internal operations on assertion of the vertical synchronization signal (input) when OUT_VLATCH.VEN = 1 or when the register value reflection control signal to internal operations for all the modules is asserted.

CONTR[7:0] bits (Contrast Adjustment of R Signal)

The CONTR[7:0] bits specify the contrast (GAIN) adjustment of R signal. The location of the decimal point is between bits [7] and [6].

CONTB[7:0] bits (Contrast Adjustment of B Signal)

The CONTB[7:0] bits specify the contrast (GAIN) adjustment of B signal. The location of the decimal point is between bits [7] and [6].

CONTG[7:0] bits (Contrast Adjustment of G Signal)

The CONTG[7:0] bits specify the contrast (GAIN) adjustment of G signal. The location of the decimal point is between bits [7] and [6].

Contrast correction of each pixel is performed as follows (x = R, G, B):

- Dxout: Output of contrast correction, unsigned, 10 bits
- Dxin: Input of contrast correction (output of brightness correction), unsigned, 10 bits
- CONTx: Setting in this bit

- $Dx_{out} = Dx_{in} \times CONTx$.

64.2.48 OUT_PDTHA : Output Control Block Panel Dither Correction Register

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x13D4

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	SEL[1:0]	—	—	—	—	FORM[1:0]
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	PA[1:0]	—	—	—	PB[1:0]	—	—	—	PC[1:0]	—	—	—	PD[1:0]
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	PD[1:0]	Pattern Value (D) of 2×2 Pattern Dither Unsigned 2-bit integer.	R/W
3:2	—	These bits are read as 0. The write value should be 0.	R/W
5:4	PC[1:0]	Pattern Value (C) of 2×2 Pattern Dither Unsigned 2-bit integer.	R/W
7:6	—	These bits are read as 0. The write value should be 0.	R/W
9:8	PB[1:0]	Pattern Value (B) of 2×2 Pattern Dither Unsigned 2-bit integer.	R/W
11:10	—	These bits are read as 0. The write value should be 0.	R/W
13:12	PA[1:0]	Pattern Value (A) of 2×2 Pattern Dither Unsigned 2-bit integer.	R/W
15:14	—	These bits are read as 0. The write value should be 0.	R/W
17:16	FORM[1:0]	Output Format Select 0 0: RGB888; select RGB888 or serial RGB as output interface format 0 1: RGB666; select RGB666 as output interface format 1 0: RGB565; select RGB565 as output interface format 1 1: Setting prohibited Select output interface format in OUT_SET.FORMAT[1:0].	R/W
19:18	—	These bits are read as 0. The write value should be 0.	R/W
21:20	SEL[1:0]	Operation Mode 0 0: Truncate 0 1: Round-off 1 0: 2×2 pattern dither 1 1: Setting prohibited	R/W
31:22	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: This setting is reflected to the internal operations on assertion of the vertical synchronization signal (input) when OUT_VLATCH.VEN = 1 or when the register value reflection control signal to internal operations for all the modules is asserted.

PA[1:0], PB[1:0], PC[1:0], PD[1:0] bits (Pattern Value (A, B, C, D) of 2×2 Pattern Dither)

The PA[1:0], PB[1:0], PC[1:0], and PD[1:0] bits specify the pattern value A, B, C, and D of 2×2 pattern dither. [Figure 64.13](#) shows the configuration of the dither correction block.

FORM[1:0] bits (Output Format Select)

The FORM[1:0] bits specify the output format of the dither process. These bits must be set in accordance with the OUT_SET.FORMAT[1:0] bits of the Output Interface Register. For serial RGB (OUT_SET.FORMAT[1:0] = 11b), set these bits to 00b. Otherwise, operation is not guaranteed.

SEL[1:0] bits (Operation Mode)

The SEL[1:0] bits specify the dither operation mode. The dither process is performed for the bits equal to or shorter than the pixel data length selected in the output format select bits (OUT_PDTHA.FORM[1:0]). OUT_PDTHA.PA[1:0], PB[1:0], PC[1:0], and PD[1:0] are used for 2×2 pattern dither.

Figure 64.13 shows the configuration of the dither correction block.

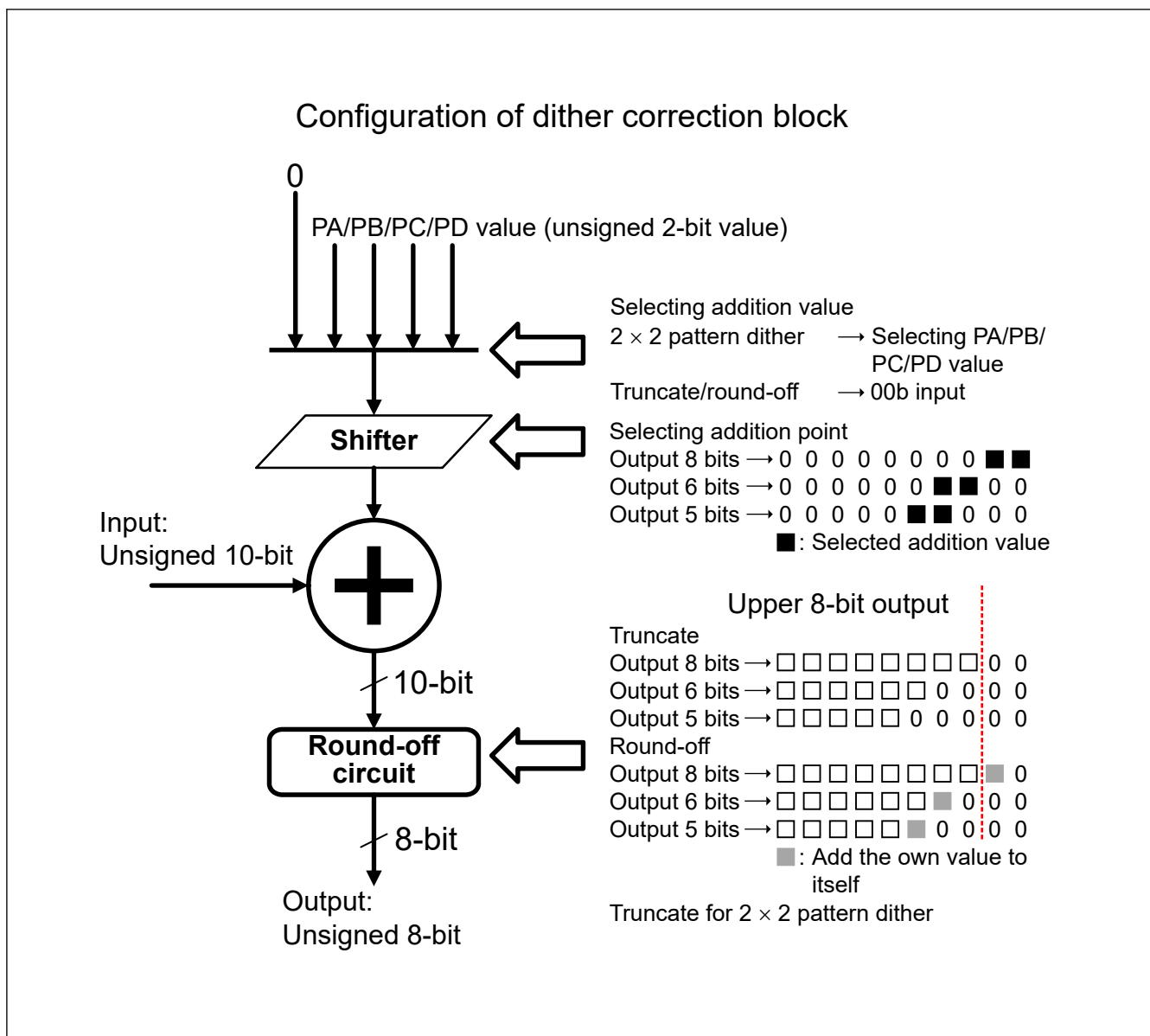


Figure 64.13 Configuration of dither correction block

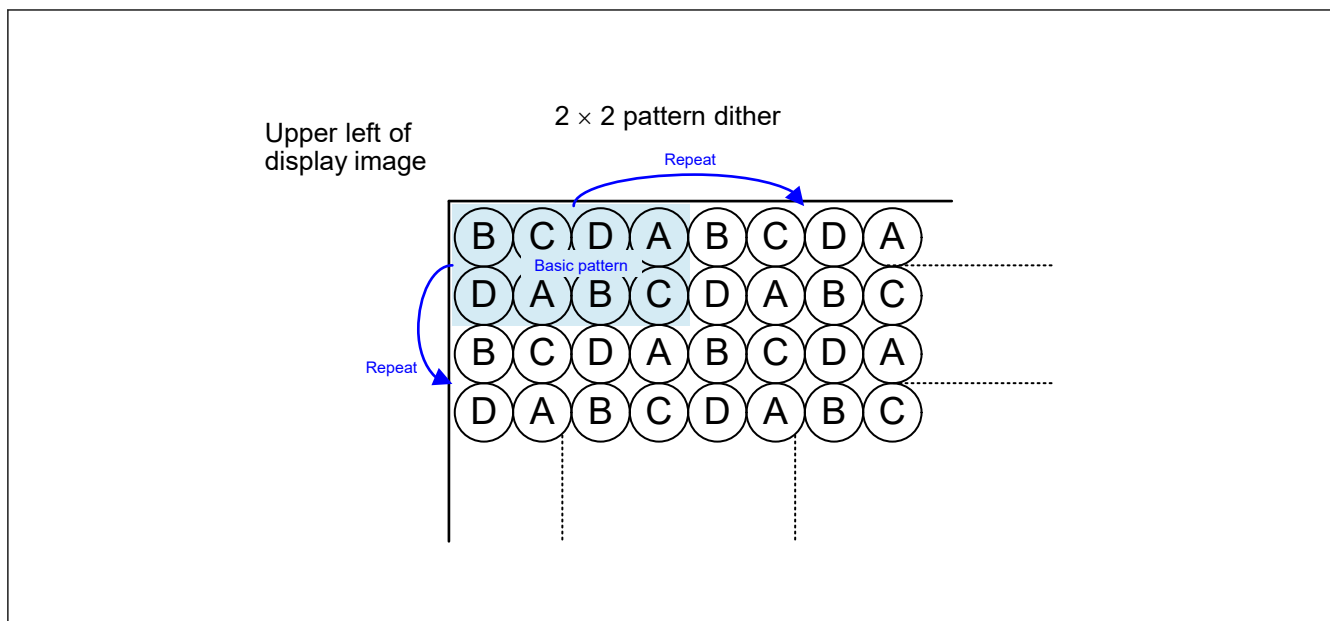
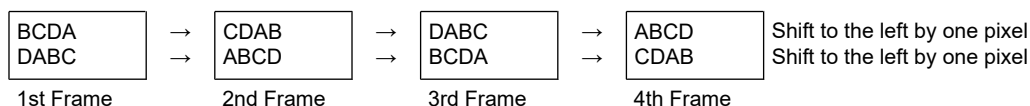


Figure 64.14 Addition value selection method for 2 × 2 pattern dither

A basic pattern is repeated as follows. Four frames constitute 1 cycle.



- A: Value obtained after OUT_PDTHA.PA[1:0] bit value is shifted in accordance with the output format
- B: Value obtained after OUT_PDTHA.PB[1:0] bit value is shifted in accordance with the output format
- C: Value obtained after OUT_PDTHA.PC[1:0] bit value is shifted in accordance with the output format
- D: Value obtained after OUT_PDTHA.PD[1:0] bit value is shifted in accordance with the output format

Renesas recommends setting the bits as follows: PA[1:0] = 11b, PB[1:0] = 00b, PC[1:0] = 10b, PD[1:0] = 01b

When 2 × 2 pattern dither (OUT_PDTHA.SEL[1:0] = 10b) is to be set, the valid pixel area of the background plane must be an integer multiple of the basic pattern. If serial RGB is selected as the output format for the output control block, add two to the horizontal valid pixel width and set the resulting value to the background plane horizontal valid pixel width bits (BG_HSIZE.HW[10:0]).

64.2.49 OUT_CLKPHASE : Output Control Block Output Phase Control Register

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x13E4

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	FRON TGAM	—	—	—	LCDE DGE	—	TCON 0EDG E	TCON 1EDG E	TCON 2EDG E	TCON 3EDG E	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	—	These bits are read as 0. The write value should be 0.	R/W
3	TCON3EDGE	LCD_TCON3 Output Phase Control 0: Synchronize output with rising edge of LCD_CLK 1: Synchronize output with falling edge of LCD_CLK	R/W
4	TCON2EDGE	LCD_TCON2 Output Phase Control 0: Synchronize output with rising edge of LCD_CLK 1: Synchronize output with falling edge of LCD_CLK	R/W
5	TCON1EDGE	LCD_TCON1 Output Phase Control 0: Synchronize output with rising edge of LCD_CLK 1: Synchronize output with falling edge of LCD_CLK	R/W
6	TCON0EDGE	LCD_TCON0 Output Phase Control 0: Synchronize output with rising edge of LCD_CLK 1: Synchronize output with falling edge of LCD_CLK	R/W
7	—	This bit is read as 0. The write value should be 0.	R/W
8	LCDEEDGE	LCD_DATA Output Phase Control 0: Synchronize output with rising edge of LCD_CLK 1: Synchronize output with falling edge of LCD_CLK	R/W
11:9	—	These bits are read as 0. The write value should be 0.	R/W
12	FRONTGAM	Correction Control 0: Process brightness/contrast correction followed by gamma correction 1: Process gamma correction followed by brightness/contrast correction	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: This setting is reflected to the internal operations on assertion of the vertical synchronization signal (input) when OUT_VLATCH.VEN = 1 or when the register value reflection control signal to internal operations for all the modules is asserted.

TCON3EDGE bit (LCD_TCON3 Output Phase Control)

The TCON3EDGE bit controls the output phase of LCD_TCON3. When this bit is set to 1, LCD_TCON3 is output in synchronization with the falling edge of LCD_CLK, and when it is set to 0, LCD_TCON3 is output in synchronization with the rising edge.

TCON2EDGE bit (LCD_TCON2 Output Phase Control)

The TCON2EDGE bit controls the output phase of LCD_TCON2. When this bit is set to 1, LCD_TCON2 is output in synchronization with the falling edge of LCD_CLK, and when it is set to 0, LCD_TCON2 is output in synchronization with the rising edge.

TCON1EDGE bit (LCD_TCON1 Output Phase Control)

The TCON1EDGE bit controls the output phase of LCD_TCON1. When this bit is set to 1, LCD_TCON1 is output in synchronization with the falling edge of LCD_CLK, and when it is set to 0, LCD_TCON1 is output in synchronization with the rising edge.

TCON0EDGE bit (LCD_TCON0 Output Phase Control)

The TCON0EDGE bit controls the output phase of LCD_TCON0. When this bit is set to 1, LCD_TCON0 is output in synchronization with the falling edge of LCD_CLK, and when it is cleared to 0, LCD_TCON0 is output in synchronization with the rising edge.

LCDEEDGE bit (LCD_DATA Output Phase Control)

The LCDEEDGE bit controls the output phase of the LCD_DATA pins (LCD_DATA23 to LCD_DATA00). When this bit is set to 1, the LCD_DATA pins are output in synchronization with the falling edge of LCD_CLK, and when it is set to 0, the LCD_DATA pins are output in synchronization with the rising edge.

FRONTGAM bit (Correction Control)

The FRONTGAM bit controls the correction sequence. When this bit is set to 1, gamma correction is followed by brightness and contrast correction, and when it is set to 0, gamma correction follows brightness and contrast correction. In both cases, each RGB data output from the graphics 2 module is extended from 8 bits to 10 bits (with 00b appended to

the lower side), and is input to the preceding stage of the correction circuit. The output is rounded to 10 bits and is input to the dither correction circuit. Although the sequence of RGB gamma correction and brightness/contrast correction can be reversed using this bit, brightness correction always precedes contrast correction.

64.2.50 TCON_TIM : TCON Reference Timing Setting Register

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x1404

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	HALF[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	OFFSET[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
10:0	OFFSET[10:0]	Horizontal Synchronization Signal Generation Reference Timing Offset from the assertion of the internal horizontal synchronization signal in pixels. 0x000: 1 pixel ⋮ 0x7FF: 2048 pixels	R/W
15:11	—	These bits are read as 0. The write value should be 0.	R/W
26:16	HALF[10:0]	Vertical Synchronization Signal Generation Change Timing Delay from the assertion of the internal horizontal synchronization signal in pixels. 0x000: 1 pixel (no delay) ⋮ 0x7FF: 2048 pixels	R/W
31:27	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: Rewriting these bits is prohibited during operation. Set the required settings before enabling operation. Operation is not guaranteed if the bit is rewritten during operation.

OFFSET[10:0] bits (Horizontal Synchronization Signal Generation Reference Timing)

The OFFSET[10:0] bits specify the reference timing to be used when the horizontal synchronization signal is generated in the TCON. Set the offset from the assertion of the internal horizontal synchronization signal (HS) in terms of pixels. [Figure 64.15](#) shows the horizontal synchronization signal generation reference timing in the TCON.

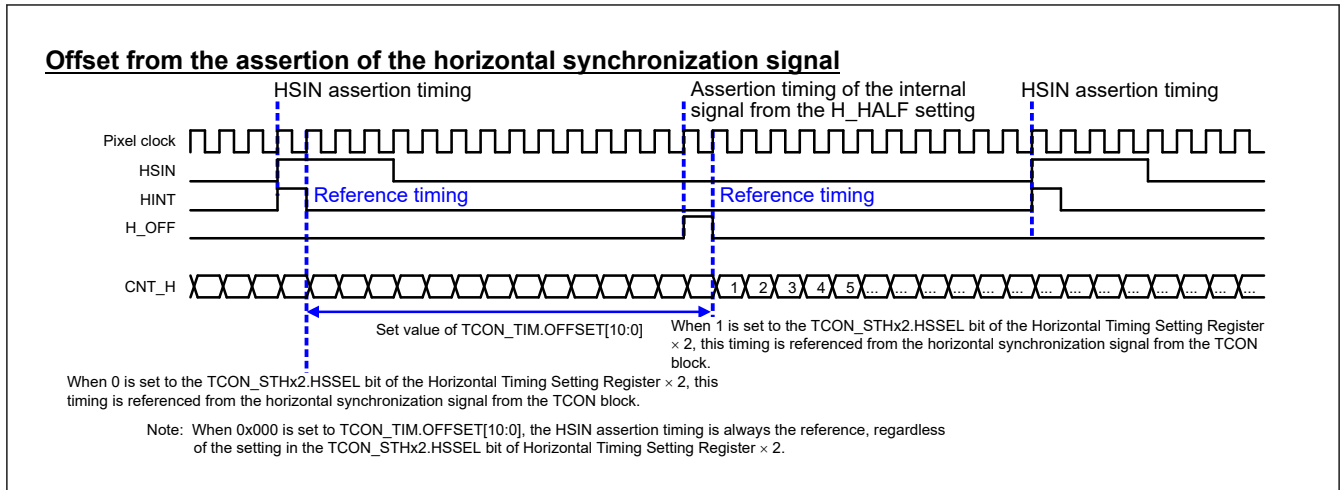


Figure 64.15 Reference timing in the TCON for horizontal synchronization signal generation

HALF[10:0] bits (Vertical Synchronization Signal Generation Change Timing)

The HALF[10:0] bits specify the vertical synchronization signal change timing when the signal is generated in the TCON. Set the change timing as a delay from the assertion of the internal horizontal synchronization signal (HS) in terms of pixels. Figure 64.16 shows the vertical synchronization signal change timing in the TCON.

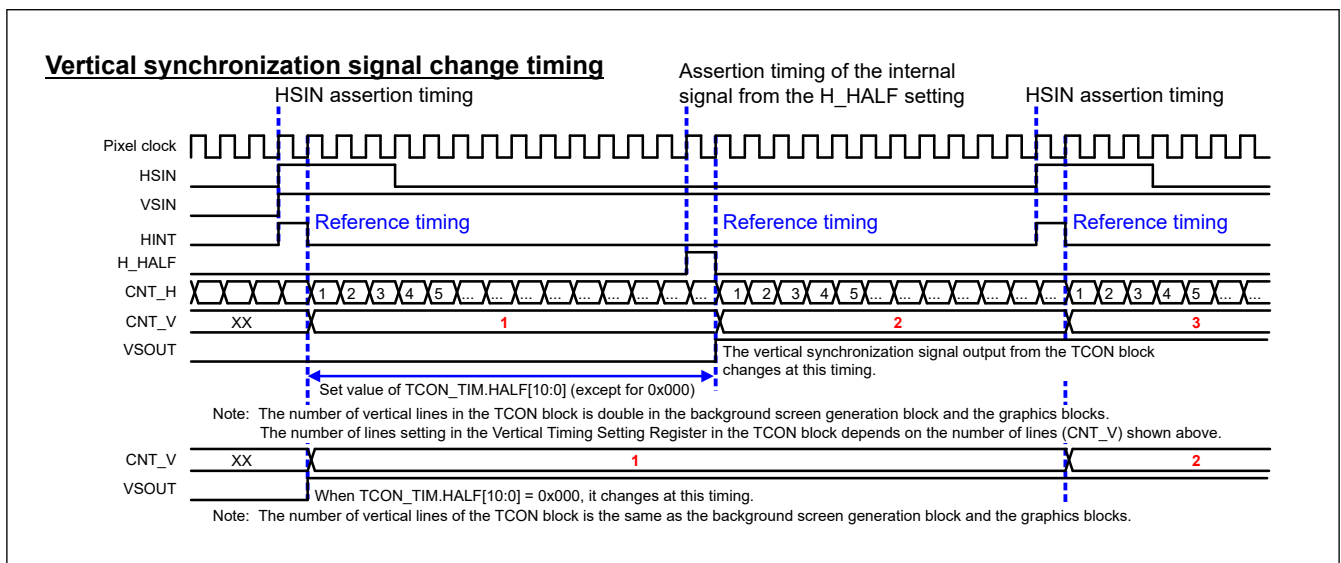


Figure 64.16 Timing in the TCON for vertical synchronization signal change

Also, Figure 64.17 shows the relationship between the TCON block and the output format block in the output control block. These blocks are based on shared control signals (same timing) and image data, and the internal delay is the same. The delay difference specified in the register settings is the source of the timing differences on the external pins.

- TCON block: 3 delay cycles of the pixel clock (PXCLK)
- Data format block:
 - Parallel RGB: 3 delay cycles of the pixel clock (PXCLK)
 - Serial RGB: 3 delay cycles of the pixel clock (PXCLK):
 - The delay of the pixel head data including invalid data
 - No delay of the serial RGB data.

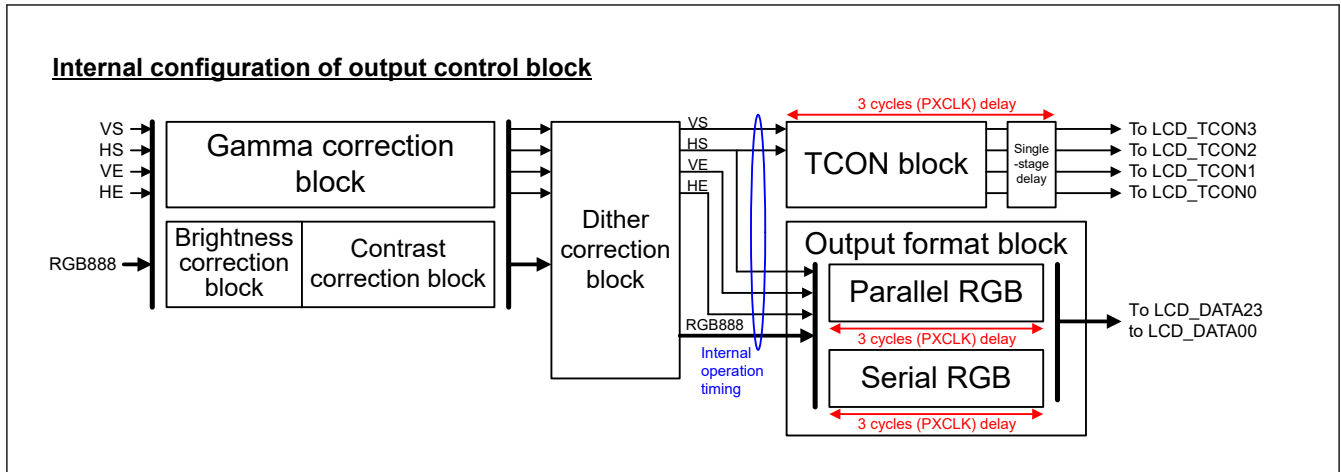


Figure 64.17 Internal configuration of output control block

64.2.51 TCON_STVx1 : TCON Vertical Timing Setting Register x1 (x = A, B)

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x1408 (TCON_STVA1)
0x1410 (TCON_STVB1)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	VS[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	VW[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
10:0	VW[10:0]	Vertical Synchronization Signal STVx1 Second Change Timing Signal assertion width in lines. 0x000: 0 line (no vertical synchronization signal assertion) ⋮ 0x7FF: 2047 lines.	R/W
15:11	—	These bits are read as 0. The write value should be 0.	R/W
26:16	VS[10:0]	Vertical Synchronization Signal STVx1 First Change Timing Signal delay in lines. 0x000: 0 line (no delay) ⋮ 0x7FF: 2047 lines	R/W
31:27	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: Rewriting these bits is prohibited during operation. Set the required settings before enabling operation. Operation is not guaranteed if the bit is rewritten during operation.

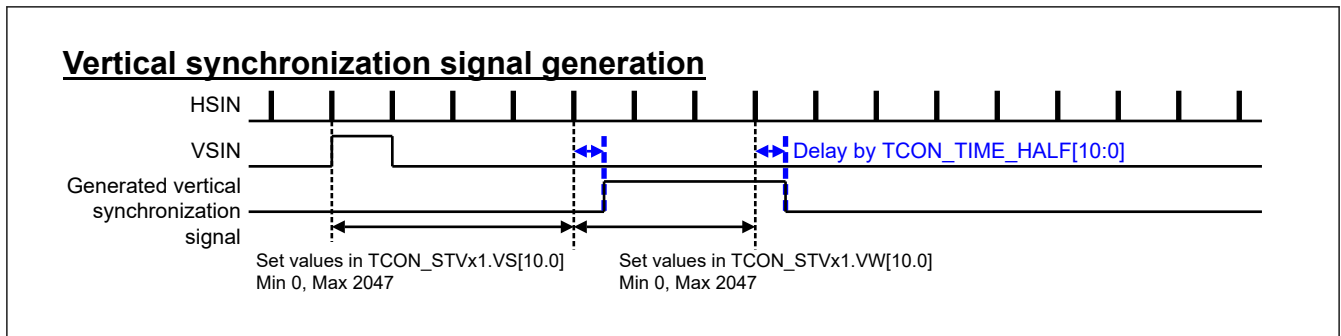
The vertical timing setting registers (TCON_STVA1/TCON_STVB1 and TCON_STVA2/TCON_STVB2) have the same configuration, and x is either A or B in the descriptions.

VW[10:0] bits (Vertical Synchronization Signal STVx1 Second Change Timing)

The VW[10:0] bits specify the second change (negation) timing of the vertical synchronization signal STVx1, which is generated in the TCON. Set the second change timing as a delay from the first change point in terms of lines. The change position in a horizontal line is defined in TCON_TIM.HALF[10:0] in the Reference Timing Setting Register (TCON_TIM), as is the first change timing.

VS[10:0] bits (Vertical Synchronization Signal STVx1 First Change Timing)

The VS[10:0] bits specify the first change (assertion) timing of the vertical synchronization signal STVx1, which is generated in the TCON. Set the change timing as a delay from the input vertical synchronization signal (VSIN) in terms of lines. The change position in a horizontal line is defined in TCON_TIM.HALF[10:0] in the Reference Timing Setting Register (TCON_TIM), as is the first change timing. Figure 64.18 shows the change timing of vertical synchronization signal to be generated.

**Figure 64.18 Generation of vertical synchronization signal****64.2.52 TCON_STVx2 : TCON Vertical Timing Setting Register x2 (x = A, B)**

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x140C (TCON_STVA2)
0x1414 (TCON_STVB2)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	INV	—	SEL[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	SEL[2:0]	Output Signal Select Control for VSOUT/VEOUT Pin Output signal select for LCD_TCON0 pin (controlled in TCON_STVA2 register) and LCD_TCON1 pin (controlled in TCON_STVB2 register). 0 0 0: STVA 0 0 1: STVB 0 1 0: STHA 0 1 1: STHB 1 0 0: Setting prohibited 1 0 1: Setting prohibited 1 1 0: Setting prohibited 1 1 1: DE	R/W
3	—	This bit is read as 0. The write value should be 0.	R/W
4	INV	Vertical Synchronization Signal STVx Polarity Inversion Control 0: Do not invert 1: Invert	R/W
31:5	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: Rewriting these bits is prohibited during operation. Set the required settings before enabling operation. Operation is not guaranteed if the bit is rewritten during operation.

The vertical timing setting registers (TCON_STVA1/TCON_STVB1 and TCON_STVA2/TCON_STVB2) have the same configuration, and x is either A or B in the descriptions.

SEL[2:0] bits (Output Signal Select Control for VSOUT/VEOUT Pin)

The SEL[2:0] bits control output signal select for the LCD_TCON0/LCD_TCON1 pin. Figure 64.19 shows the configuration of the inversion control and output signal select.

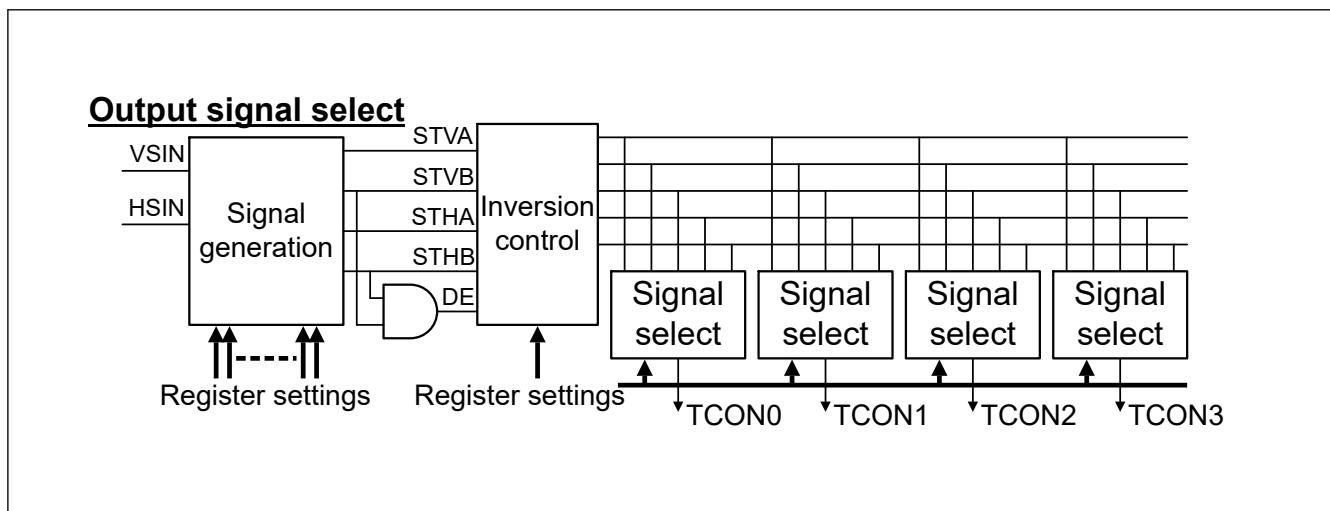


Figure 64.19 Configuration of inversion control and output signal selects

INV bit (Vertical Synchronization Signal STVx Polarity Inversion Control)

The INV bit controls polarity inversion of the vertical synchronization signal (STVx).

64.2.53 TCON_STHx1 : TCON Horizontal Timing Setting Register x1 (x = A, B)

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x1418 (TCON_STHA1)
0x1420 (TCON_STHB1)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	HS[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	HW[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
10:0	HW[10:0]	Horizontal Synchronization Signal STHx1 Second Change Timing Signal assertion width in pixels. 0x000: 0 pixel (no horizontal synchronization signal assertion) ⋮ 0x7FF: 2047 pixels	R/W
15:11	—	These bits are read as 0. The write value should be 0.	R/W
26:16	HS[10:0]	Horizontal Synchronization Signal STHx1 First Change Timing Signal delay in pixels. 0x000: 0 pixel (no delay) ⋮ 0x7FF: 2047 pixels	R/W
31:27	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: Rewriting these bits is prohibited during operation. Set the required settings before enabling operation. Operation is not guaranteed if the bit is rewritten during operation.

The horizontal timing setting registers (TCON_STHA1/TCON_STHB1 and TCON_STHA2/TCON_STHB2) have the same configuration, and x is either A or B in the descriptions.

HW[10:0] bits (Horizontal Synchronization Signal STHx1 Second Change Timing)

The HW[10:0] bits specify the second change (negation) timing of the horizontal synchronization signal STHx1, which is generated in the TCON. Set the second change timing as a distance from the first change point in terms of pixels.

HS[10:0] bits (Horizontal Synchronization Signal STHx1 First Change Timing)

The HS[10:0] bits specify the first change (assertion) timing of the horizontal synchronization signal STHx1, which is generated in the TCON. Set the change timing as a distance from the input horizontal synchronization signal (HSIN) or the reference timing based on the offset specified in the TCON_TIM.OFFSET[10:0] bit (horizontal synchronization signal generation reference timing) in terms of pixels.

Figure 64.20 shows the horizontal synchronization signal generation timing if the input horizontal synchronization signal (HSIN) is based on the negated edge reference of the HINT signal. Figure 64.21 shows the horizontal synchronization signal generation timing after offset. By setting the TCON_TIM.OFFSET[10:0] bit (horizontal synchronization generation reference timing) and these horizontal synchronization bits appropriately, it is possible to generate a signal that is asserted before HSIN and negated after HSIN, where HSIN is the horizontal synchronization signal input to the TCON.

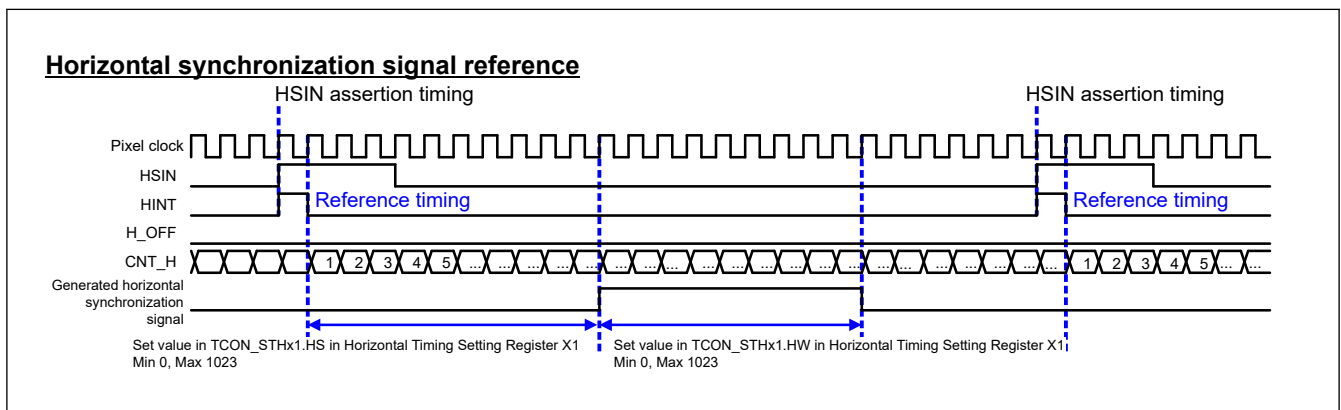


Figure 64.20 Signal generation based on input horizontal synchronization signal (HSIN)

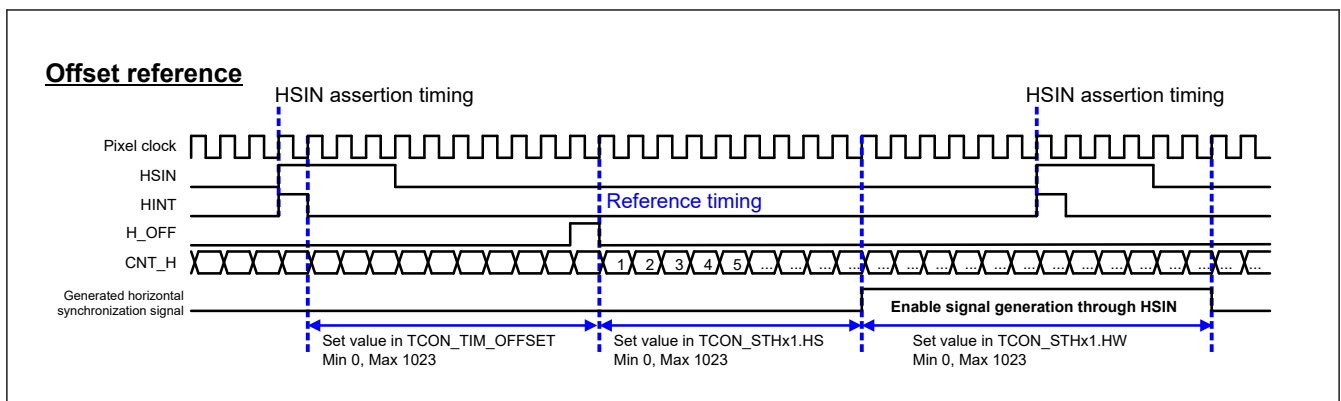


Figure 64.21 Horizontal synchronization signal generation based on offset

When generating the signal across HSIN on the offset reference, the horizontal synchronization signal of the last line of the frame spans the first line of the next frame. Even if the BG_EN.EN bit is cleared to 0 and the GLCDC operation is stopped, the horizontal synchronization signal across HSIN is not cleared at the frame end and retains the predetermined value until the second change timing set in the registers. If BG_EN.SWRST is cleared to 0, the signal returns to the initial state immediately.

64.2.54 TCON_STHx2 : TCON Horizontal Timing Setting Register x2 (x = A, B)

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x141C (TCON_STHA2)
0x1424 (TCON_STHB2)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	HSSEL	—	—	—	INV	—	SEL[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	SEL[2:0]	Output Signal Select Control for LCD_TCON2/LCD_TCON3 Pin Output signal select for LCD_TCON2 pin (controlled in TCON_STHA2 register) and LCD_TCON3 pin (controlled in TCON_STHB2 register). 0 0 0: STVA 0 0 1: STVB 0 1 0: STHA 0 1 1: STHB 1 0 0: Setting prohibited 1 0 1: Setting prohibited 1 1 0: Setting prohibited 1 1 1: DE	R/W
3	—	This bit is read as 0. The write value should be 0.	R/W
4	INV	Horizontal Synchronization Signal STHx Polarity Inversion Control 0: Do not invert 1: Invert	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W
8	HSSEL	Horizontal Synchronization Signal STHx Reference Timing Control 0: Select input horizontal synchronization signal (HSIN) as reference for signal generation 1: Select offset specified in TCON_TIM.OFFSET[10:0] (horizontal synchronization generation reference timing) as reference for signal generation	R/W
31:9	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: Rewriting these bits is prohibited during operation. Set the required settings before enabling operation. Operation is not guaranteed if the bit is rewritten during operation.

The horizontal timing setting registers (TCON_STHA1/TCON_STHB1 and TCON_STHA2/TCON_STHB2) have the same configuration, and x is either A or B in the descriptions.

SEL[2:0] bits (Output Signal Select Control for LCD_TCON2/LCD_TCON3 Pin)

The SEL[2:0] bits control the output signal select for the LCD_TCON2/LCD_TCON3 pins.

INV bit (Horizontal Synchronization Signal STHx Polarity Inversion Control)

The INV bit controls polarity inversion of the horizontal synchronization signal (STHx).

HSSEL bit (Horizontal Synchronization Signal STHx Reference Timing Control)

The HSSEL bit selects the reference timing for generating the horizontal synchronization signal STHx. For details on the signal to be generated, see [Figure 64.20](#) and [Figure 64.21](#). For the configuration of the inversion control and output signal select, see [Figure 64.19](#).

VPOSDTC bit (Specified Line Detection Control)

The VPOSDTC bit enables or disables detection of the specified line. When this bit is set to 1, the associated bit in the SYSCNT_STMON register sets to 1 on event notification from graphics 2. When it is set to 0, the associated bit in the SYSCNT_STMON register does not set to 1 even on event notification from graphics 2.

L1UNDFDTC bit (Graphics 1 Underflow Detection Control)

The L1UNDFDTC bit enables or disables detection of the graphics 1 underflow. When this bit is set to 1, the associated bit in the SYSCNT_STMON register sets to 1 when an underflow occurs in graphics 1. When it is set to 0, the associated bit in the SYSCNT_STMON register does not set to 1 even when an underflow occurs in graphics 1. The underflow state in graphics 1 is automatically cleared on assertion of the vertical synchronization signal (VS) regardless of the value of this bit, and normal operation is recovered.

L2UNDFDTC bit (Graphics 2 Underflow Detection Control)

The L2UNDFDTC bit enables or disables detection of the graphics 2 underflow. When this bit is set to 1, the associated bit in the SYSCNT_STMON register sets to 1 when an underflow occurs in graphics 2. When it is set to 0, the associated bit in the SYSCNT_STMON register does not set to 1 even when an underflow occurs in graphics 2. The underflow state in graphics 2 is automatically cleared on assertion of the vertical synchronization signal (VS) regardless of the value of this bit, and normal operation is recovered.

64.2.57 SYSCNT_INTEN : System Control Block Interrupt Request Enable Control Register

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x1444

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	L2UNDFINTEN	L1UNDFINTEN	VPOSINTEN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	VPOSINTEN	Interrupt Request Signal GLCDC_VPOS Enable Control 0: Disable GLCDC_VPOS output 1: Enable GLCDC_VPOS output	R/W
1	L1UNDFINTEN	Interrupt Request Signal GLCDC_L1UNDF Enable Control 0: Disable GLCDC_L1UNDF output 1: Enable GLCDC_L1UNDF output	R/W
2	L2UNDFINTEN	Interrupt Request Signal GLCDC_L2UNDF Enable Control 0: Disable GLCDC_L2UNDF output 1: Enable GLCDC_L2UNDF output	R/W
31:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

VPOSINTEN bit (Interrupt Request Signal GLCDC_VPOS Enable Control)

The VPOSINTEN bit enables or disables the interrupt request signal GLCDC_VPOS. When this bit is set to 1, the interrupt request signal GLCDC_VPOS is asserted when the associated status monitor flag SYSCNT_STMON[0] sets. When it is cleared to 0, the interrupt request signal GLCDC_VPOS is not asserted even when the associated status monitor flag SYSCNT_STMON[0] sets. If this bit is cleared during GLCDC_VPOS assertion, the associated status monitor flag SYSCNT_STMON[0] does not change, but the interrupt request signal GLCDC_VPOS is negated.

L1UNDFINTEN bit (Interrupt Request Signal GLCDC_L1UNDF Enable Control)

The L1UNDFINTEN bit enables or disables the interrupt request signal GLCDC_L1UNDF. When this bit is set to 1, the interrupt request signal GLCDC_L1UNDF is asserted when the associated status monitor flag SYSCNT_STMON[1] sets. When it is cleared to 0, the interrupt request signal GLCDC_L1UNDF is not asserted even when the associated status monitor flag SYSCNT_STMON[1] sets. If this bit is cleared during GLCDC_L1UNDF assertion, the associated status monitor flag SYSCNT_STMON[1] does not change, but the interrupt request signal GLCDC_L1UNDF is negated.

L2UNDFINTEN bit (Interrupt Request Signal GLCDC_L2UNDF Enable Control)

The L2UNDFINTEN bit enables or disables the interrupt request signal GLCDC_L2UNDF. When this bit is set to 1, the interrupt request signal GLCDC_L2UNDF is asserted when the associated status monitor flag SYSCNT_STMON[2] sets. When it is cleared to 0, the interrupt request signal GLCDC_L2UNDF is not asserted even when the associated status monitor flag SYSCNT_STMON[2] sets. If this bit is cleared during GLCDC_L2UNDF assertion, the associated status monitor flag SYSCNT_STMON[2] does not change, but the interrupt request signal GLCDC_L2UNDF is negated.

64.2.58 SYSCNT_STCLR : System Control Block Status Clear Register

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x1448

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	L2UN DFCL R	L1UN DFCL R	VPOS CLR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	VPOSCLR	Graphics 2 Specified Line Detection Flag Clear 0: No operation 1: Clear the graphics 2 specified line detection flag	R/W ^{*1}
1	L1UNDFCLR	Graphics 1 Underflow Detection Flag Clear 0: No operation 1: Clear the graphics 1 underflow detection flag	R/W ^{*1}
2	L2UNDFCLR	Graphics 2 Underflow Detection Flag Clear 0: No operation 1: Clears the graphics 2 underflow detection flag	R/W ^{*1}
31:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. These bits are read as 0.

VPOSCLR bit (Graphics 2 Specified Line Detection Flag Clear)

Writing 1 to the VPOSCLR bit clears the graphics 2 specified line detection flag. Clearance of the flag by this bit is only valid for the associated flag, and does not directly affect the other states and interrupt request signals. However, an interrupt request signal might be negated by clearing the detection flag.

L1UNDFCLR bit (Graphics 1 Underflow Detection Flag Clear)

Writing 1 to the L1UNDFCLR bit clears the graphics 1 underflow detection flag. Clearance of the flag by this bit is only valid for the associated flag, and does not directly affect the other states and interrupt request signals. However, an interrupt request signal might be negated by clearing the detection flag.

L2UNDFCLR bit (Graphics 2 Underflow Detection Flag Clear)

Writing 1 to the L2UNDFCLR bit clears the graphics 2 underflow detection flag. Clearance of the flag by this bit is only valid for the associated flag, and does not directly affect the other states and interrupt request signals. However, an interrupt request signal might be negated by clearing the detection flag.

64.2.59 SYSCNT_STMON : System Control Block Status Monitor Register

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x144C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	L2UNDF	L1UNDF	VPOS
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	VPOS	Graphics 2 Specified Line Detection Flag 0: Specified line notification not detected in graphics 2 1: Specified line notification detected in graphics 2	R
1	L1UNDF	Graphics 1 Underflow Detection Flag 0: No underflow detected in graphics 1 1: Underflow detected in graphics 1	R
2	L2UNDF	Graphics 2 Underflow Detection Flag 0: No underflow detected in graphics 2 1: Underflow detected in graphics 2	R
31:3	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

VPOS flag (Graphics 2 Specified Line Detection Flag)

The VPOS flag indicates that the specified line notification was detected in graphics 2. When this flag is 1, it indicates that the specified line notification was detected in graphics 2 at some time in the past. It does not necessarily mean that graphics 2 is currently processing the specified line. When this flag is 0, it indicates that no specified line notification was detected after the module operation was enabled. When the interrupt request signal is enabled and is cleared while this flag is 1, the associated interrupt request signal GLCDC_VPOS is negated, but this does not affect the state of graphics 2.

L1UNDF flag (Graphics 1 Underflow Detection Flag)

The L1UNDF flag indicates that an underflow was detected in graphics 1. When this flag is 1, it indicates that an underflow was detected in graphics 1 at some time in the past. It does not necessarily mean that graphics 1 is currently in the underflow state. When this flag is 0, it indicates that no underflow was detected after the module operation was enabled. When the interrupt request signal is enabled and is cleared while this flag is 1, the associated interrupt request signal GLCDC_L1UNDF is negated, but this does not affect the state of graphics 1.

Even if the current graphics data is not required (GR1_AB1.DISPSEL[1:0] = 0xb), if the SYSCNT_DTCTEN.L1UNDFDTC flag (graphics 1 underflow detection control) is 1 and detection is enabled, this flag is set to 1 at the start of the graphics image valid area. To avoid unnecessary detection flag settings or interrupt request signal assertions, when the display does not require the current graphics data, set the SYSCNT_DTCTEN.L1UNDFDTC flag and the SYSCNT_INTEN.L1UNDFINTEN bit (interrupt request signal GLCDC_L1UNDF enable control) to 0.

L2UNDF flag (Graphics 2 Underflow Detection Flag)

The L2UNDF flag indicates that an underflow was detected in graphics 2. When this flag is 1, it indicates that an underflow was detected in graphics 2 at some time in the past. It does not necessarily mean that graphics 2 is currently in the

underflow state. When this flag is 0, it indicates that no underflow was detected after the module operation was enabled. When the interrupt request signal is enabled and is cleared while this flag is 1, the associated interrupt request signal GLCDC_L2UNDF is negated, but this does not affect the state of graphics 2.

Even if the current graphics data is not required (GR2_AB1.DISPSEL[1:0] = 0xb), if the SYSCNT_DTCTEN.L2UNDFDTC flag (graphics 2 underflow detection control) is 1 and detection is enabled, this flag is set to 1 at the start of the graphics image valid area. To avoid the unnecessary detection flag settings or interrupt request signal assertions, when the display does not require the current graphics data, set the SYSCNT_DTCTEN.L2UNDFDTC flag and the SYSCNT_INTEN.L2UNDFINTEN bit (interrupt request signal GLCDC_L2UNDF enable control) to 0.

64.2.60 SYSCNT_PANEL_CLK : System Control Block Version and Panel Clock Control Register

Base address: GLCDC = 0x4034_2000
GLCDC_NS = 0x5034_2000

Offset address: 0x1450

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	VER[15:0]															
Value after reset:	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	PIXSEL	—	—	—	CLKSEL	—	CLKEN	DCDR[5:0]					
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
5:0	DCDR[5:0]	Clock Division Ratio Setting Control See Table 64.9 for details on these settings.	R/W
6	CLKEN	Panel Clock Output Enable Control 0: Disable panel clock output 1: Enable panel clock output Before changing the PIXSEL, CLKSEL, or DCDR bit, this bit must be set to 0.	R/W
7	—	This bit is read as 0. The write value should be 0.	R/W
8	CLKSEL	Panel Clock Supply Source Control 0: Select external clock (LCD_EXTCLK) 1: Select LCDCLK	R/W
11:9	—	These bits are read as 0. The write value should be 0.	R/W
12	PIXSEL	Pixel Clock Select Control 0: Select no frequency division, parallel RGB 1: Select quarter frequency, serial RGB This setting must have the same value as OUT_SET.FRQSEL[1].	R/W
15:13	—	These bits are read as 0. The write value should be 0.	R/W
31:16	VER[15:0]	Version Information Version information of the GLCDC.	R

Note: S-TYPE-3, P-TYPE-3

The configuration of the pixel and panel clock generator circuits are shown in [section 9, Clock Generation Circuit](#).

DCDR[5:0] bits (Clock Division Ratio Setting Control)

The DCDR[5:0] bits control the setting of the panel clock division ratio. The division ratio bit can only be set to the values listed in [Table 64.9](#). Operation is not guaranteed for values not listed.

Table 64.9 Input clock division (1 of 2)

DCDR[5:0]	Clock division ratio	CLKSEL = 0, LCD_EXTCLK (≤ 60 MHz)	CLKSEL = 1, LCDCLK (Up to 240 MHz)
000000b	1/2	LCD_EXTCLK/2	LCDCLK/2 ^{*1}

Table 64.9 Input clock division (2 of 2)

DCDR[5:0]	Clock division ratio	CLKSEL = 0, LCD_EXTCLK (≤ 60 MHz)	CLKSEL = 1, LCDCLK (Up to 240 MHz)
000001b	1/1	LCD_EXTCLK* ¹	LCDCLK* ¹
000010b	1/2	LCD_EXTCLK/2	LCDCLK/2* ¹
000011b	1/3	LCD_EXTCLK/3	LCDCLK/3* ¹
000100b	1/4	LCD_EXTCLK/4	LCDCLK/4* ¹
000101b	1/5	LCD_EXTCLK/5	LCDCLK/5
000110b	1/6	LCD_EXTCLK/6	LCDCLK/6
000111b	1/7	LCD_EXTCLK/7	LCDCLK/7
001000b	1/8	LCD_EXTCLK/8	LCDCLK/8
001001b	1/9	LCD_EXTCLK/9	LCDCLK/9
001100b	1/12	LCD_EXTCLK/12	LCDCLK/12
010000b	1/16	LCD_EXTCLK/16	LCDCLK/16
011000b	1/24	LCD_EXTCLK/24	LCDCLK/24
100000b	1/32	LCD_EXTCLK/32	LCDCLK/32

Note 1. The panel clock is output as the LCD_CLK output clock. This setting may be prohibited because LCD_EXTCLK and LCD_CLK have limited frequencies. See [section 70, Electrical Characteristics](#).

To set the panel clock:

1. After setting the input source of the panel clock in the CLKSEL bit, set the division ratio in the DCDR[5:0] bits and the pixel clock selection.
2. Set the CLKEN bit to 1.

CLKEN bit (Panel Clock Output Enable Control)

The CLKEN bit enables or disables the panel clock output. When enabling the panel clock output and operating the panel clock block, set this bit to 1. When changing the PIXSEL, CLKSEL, or DCDR bit, you must set this bit to 0 once and stop the panel clock output. Operation is not guaranteed if any setting is changed while the panel clock is being output.

CLKSEL bit (Panel Clock Supply Source Control)

The CLKSEL bit controls the selection of the panel clock supply source from either the external clock pin (LCD_EXTCLK) or LCDCLK. When the external clock is selected, set this bit to 0. When LCDCLK is selected, set this bit to 1.

PIXSEL bit (Pixel Clock Select Control)

The PIXSEL bit controls the selection of the pixel clock output. When selecting parallel RGB, set this bit to 0 to output the same frequency as the panel clock (no frequency division). When selecting serial RGB, set this bit to 1 to output the quarter frequency of the panel clock as the pixel clock.

This bit must be synchronized with the OUT_SET.FRQSEL[1:0] setting. You must set the same value as in FRQSEL[1]. Otherwise, operation is not guaranteed.

VER[15:0] bits (Version Information)

The VER[15:0] bits provide GLCDC version information.

64.3 Operation

64.3.1 Overall Control

The GLCDC consists of six modules as shown in [Figure 64.22](#), each of which functions independently. The four modules handling image data are interconnected by the vertical and horizontal synchronization signals VS, HS, VE, and HE, and image data (RGB888), as shown in [Figure 64.23](#). The processing of image data is carried out with the pixel clock (PXCLK). LCD_CLK synchronizes with PXCLK (also in phase with each other), and has the same or quadruple frequency (for serial RGB888 output). The registers controlling operation and setting parameters are connected with the PABI and operate on PCLKA. The circuits, including the data buffer and the CLUT memory, operate on PCLKA to read graphics data from

the GLCDC0 and GLCDC1 bus, access the color palette (CLUT) memory, and expand graphics data into the ARGB8888 format.

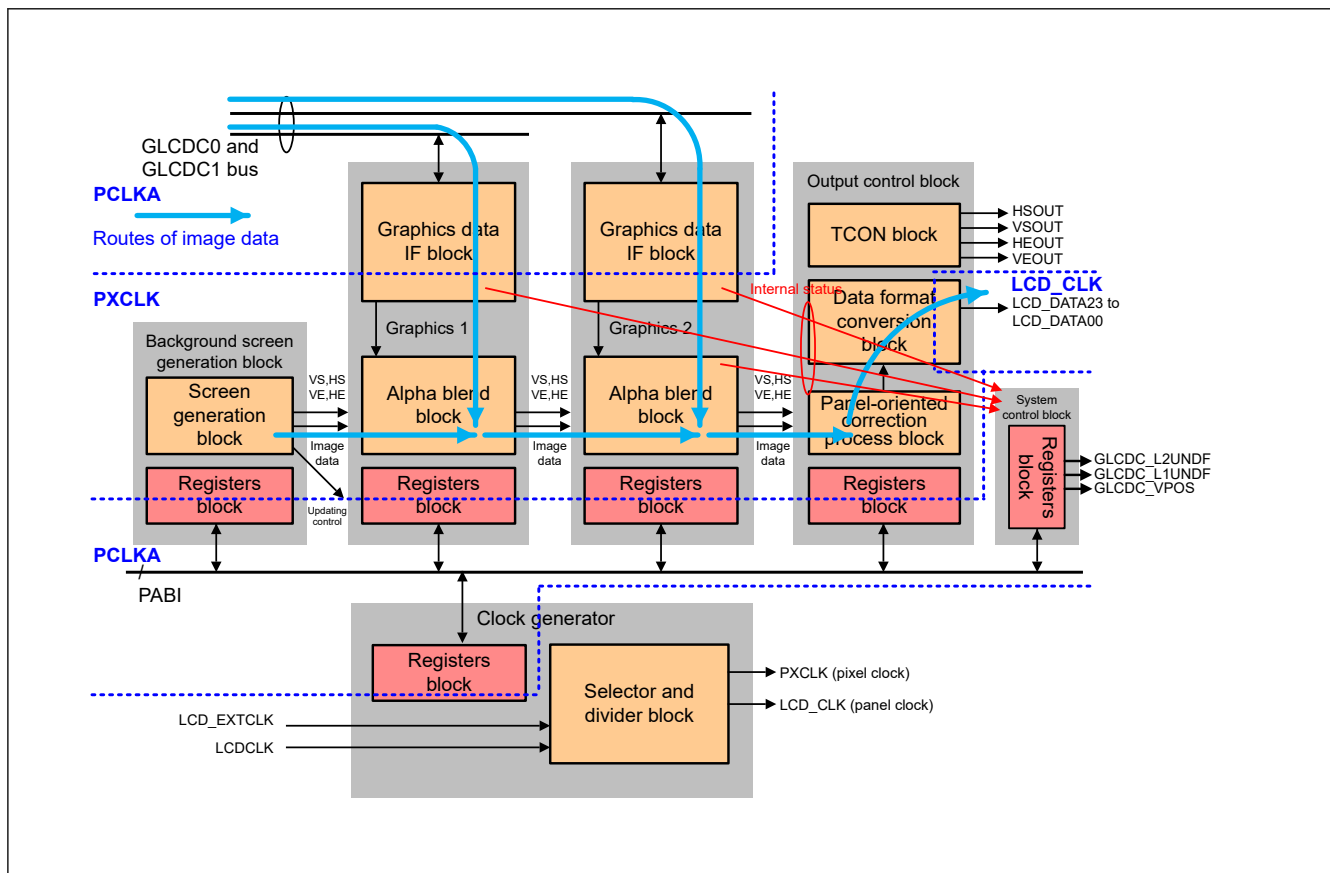


Figure 64.22 GLCDC configuration

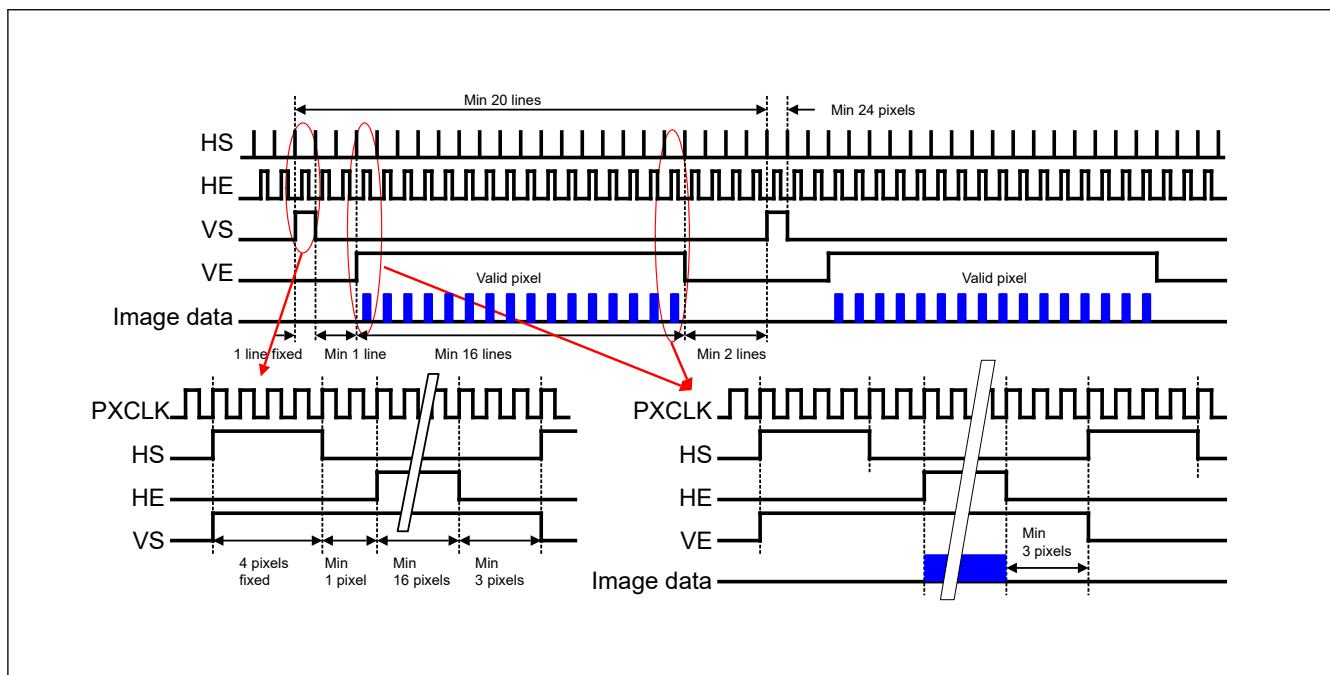


Figure 64.23 Image data carrier signals in the GLCDC

The graphics 1, graphics 2, and output control modules have no individual operation enable bits. By clearing the software reset state in the background screen generation module, these modules wait for vertical and horizontal synchronization

signals VS, HS, VE, and HE, and image data (RGB888) to be input, detect assertion of the vertical synchronization signal VS, and start operation for each frame based on the values preset in the registers. For the system control module, clearing the software reset state in the background screen generation module allows it to monitor the status of graphics 1 and 2 in accordance with the register settings and assert interrupt request signals. These status flags are intended for observing the internal status and exert no influence on the internal operation. The operation between startup and stop of the GLCDC is outlined in this section.

(1) Startup

To start GLCDC operation:

1. Confirm that PCLKA is supplied to the GLCDC and that the reset is negated. When setting LCD_CLK to the same frequency as PXCLK or LCD_CLK to the quadruple speed clock, the setting must be consistent with the value in the register specified in the subsequent stage. The quadruple speed clock is required only when the output format is serial RGB888. Operation is not guaranteed if the quadruple speed clock is supplied to LCD_CLK in other output formats than serial RGB888.
2. Set 1 to the BG_EN.SWRST bit to release the entire GLCDC from a software reset.
3. Set parameters necessary for operation in each register. Although registers of any module can be set first, while the BG_EN.VEN bit (control of GLCDC internal register value reflection to internal operations), BG_EN.EN bit (background plane operation enable) of the background screen generation module, and the VEN bit in the register update control register of each module remain cleared to 0, confirm that PCLKA and PXCLK/LCD_CLK are supplied and the BG_MON.SWRST bit (entire module software reset state monitor) is set to 1.
4. When displaying the graphics data read by graphics 1 and 2 through the GLCDC0 and GLCDC1 bus, set 1 to the GRn_FLMRD.RENB bit (graphics data read enable).
5. Write the color palette data of graphics 1 and 2 (0 and 1 planes) to the CLUT memory through the register access bus (PABI) as required. This is necessary for the LUT1, LUT4, LUT8, and ARGB1555 data formats. ARGB1555 uses addresses 0x80 and 0x00 on the two planes, LUT1 uses 0x01 and 0x00, LUT4 uses the addresses from 0x0F to 0x00, and LUT8 uses the addresses from 0xFF to 0x00.
6. Set 1 to the BG_EN.VEN bit (control of background plane register value reflection to internal operations) and the BG_EN.EN bit (background plane operation enable) of the background screen generation module at the same time. This setting allows output of the vertical and horizontal synchronization signals VS, HS, VE, and HE, and the image data (RGB888) from the background screen generation module. When the pixel data is valid (both VE and HE are 1), output from the background screen generation module uses the value in the BG_BGC register. The data value is 0x00000000 for pixels that are not valid.
7. Each module detects the assertion of the vertical synchronization signal (VS) output from its previous stage (background screen generation module for graphics 1, graphics 1 for graphics 2, and graphics 2 for the output control module) and starts operation in accordance with its register settings. All the modules control the operation in frame units. When the assertion of the vertical synchronization signal (VS) is detected, the current frame is taken as the frame head and the status is initialized. If necessary (when the VEN bits in the register update control registers are 1), the register values read through the register access bus are reflected to the internal operation.

(2) Changing parameters during operation

With the GLCDC, it is possible to update parameters of each module and reflect the updates to the internal operations during operation, without preventing graphics data to be read by the background screen generation module, graphics 1, or graphics 2. By setting 1 to the VEN bits of the modules, including the background screen generation module, almost all parameters are reflected to the internal operations at the start of the following frame (at the start of the control screen for the background screen generation module, and immediately after the VS assertion of the previous stage for the other modules). However, if the CLUT plane being used for internal operations (determined in the GRn_CLUTINT.SEL[1:0] bits) is modified, the updates are reflected to the internal operations immediately without waiting for the following VS assertion. To circumvent this immediate reflection of the CLUT plane modification to the internal operations, first write all the image data (ARGB8888) necessary for the CLUT plane that is not being used for internal operations, next modify the GRn_CLUTINT.SEL[1:0] bits, and finally set 1 to the VEN bit of the background screen generation module or the target module.

To modify parameters during operation:

1. Confirm that the VEN bit of each module is 0. Operation is not guaranteed if the target registers are modified when the VEN bit of the module to which the register values are to be reflected to the internal operations, or the VEN bit of the background screen generation module is 1.
2. Modify the value of the target registers.
3. If only a particular module is to be the target, set 1 to the VEN bit of the target module. If multiple modules or the background screen generation module are to be the targets, set 1 to the VEN bit of the background screen generation module. In this case, all the modules are included as targets, not only the background screen generation module.
4. Confirm that the VEN bit to which 1 was set is 0. If the bit is cleared to 0, the target register contents are reflected to the internal operations. If the bit remains 1, however, the target register contents might not yet be reflected to the internal operations. The VEN bit of each module is cleared to 0 immediately after the target register values are internally reflected. However, the VEN bit of the background screen generation module is not cleared to 0 until the module output VE is negated (with all the modules, sufficient delay is secured in reference to the VS assertion of the background screen generation module so that the register values are reflected to the internal operations for the same frame). Figure 64.24 shows the operation of the signals output by the background screen generation module and the monitor bits.

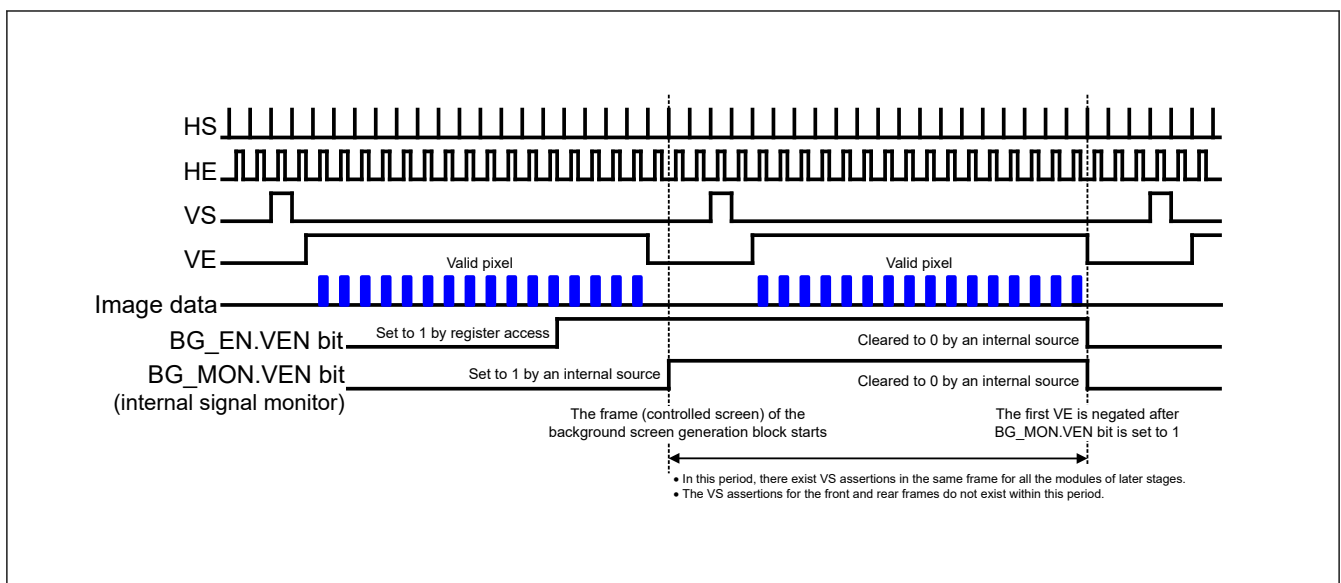


Figure 64.24 Control signals for register value reflection to internal operations

(3) Stopping and restarting in normal operation

To stop the GLCDC:

1. Confirm that the VEN bit of each module is cleared to 0.
2. Clear the BG_EN.EN bit (background plane operation enable) of the background screen generation module to 0.
3. Confirm that the BG_MON.EN bit (background plane operation monitor) of the background screen generation module has changed to 0. This bit is cleared to 0 only after operation has stopped. It is cleared at the frame end of the background screen generation module, not when the operations of all the modules are complete (not at the frame end of the output control module). If it is necessary to wait for all the modules to complete operations, a certain period (for example, a period equivalent to one line) is required.
4. Usually, a software reset can be safely applied (clearing of the BG_EN.SWRST bit) after confirming that the BG_MON.EN bit of the background screen generation module has changed to 0. (Even if the output signal returns to the initial value, no problems occur because the GLCDC has already entered the vertical blanking interval.)
5. When restarting the GLCDC by setting the relevant registers without applying a software reset, wait until sufficient time elapses after the BG_MON.EN bit becomes 0 (when the output control module output is the frame end) before starting the GLCDC. The GLCDC itself is not affected by this because the GLCDC starts operating after recognizing the assertion of the VS of the previous stage as the frame head. However, operation of some connected devices might be affected if a blanking interval or a period equivalent to one line is too short. For details, check the specifications of the connected device. When a software reset is applied, the register values are also initialized and almost all of the registers

must be set again. Only the color palette data (CLUT memory content) is retained (only for a software reset after normal end).

Figure 64.25 shows the changes in signal lines and bits for the stop and restart in normal operation. Even if the background screen generation module is stopped by a clearing of the BG_EN.EN bit to 0, the GLCDC stops because of an abnormal operation sequence, not normal operation, if the GLCDC0 and GLCDC1 bus access is not complete at the frame end because of an underflow in graphics 1 or 2, inappropriate setting of graphics data access, or other undesirable conditions.

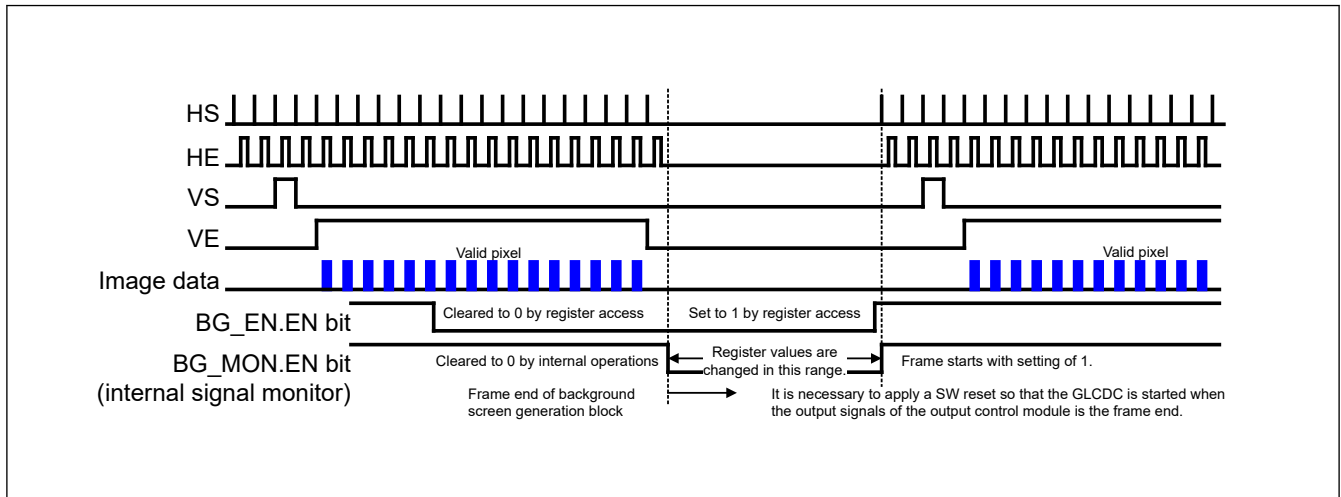


Figure 64.25 Stop and restart in normal operation

(4) Stopping and restarting in abnormal operation

Stop in abnormal operation occurs in the following cases:

- When during operation (BG_EN.EN or BG_MON.EN is set to 1), a software reset (BG_EN.SWRST bit is cleared to 0) or a reset is applied
- When unnecessary (unintended) access to the GLCDC0 and GLCDC1 bus occurs and a data cycle is not completed, even though the BG_EN.EN bit is cleared to 0 and so the BG_MON.EN bit is also cleared to 0

In both cases, the GLCDC is internally initialized and the register access bus can be accessed normally (except for a hardware reset). However, the GLCDC0 and GLCDC1 bus might write unintended graphics data to the GLCDC. Therefore, it is advisable to apply a software reset (clear the BG_EN.SWRST bit to 0) and confirm that there is no unintended access to the GLCDC0 and GLCDC1 bus, even after confirming that both the BG_EN.EN and BG_MON.EN bits are cleared to 0. Next release the GLCDC from the software reset (set the BG_EN.SWRST bit to 1), set the relevant registers, and set the BG_EN.EN and BG_EN.VEN bits in the background screen generation module to 1 to restart the GLCDC. For the procedure for checking the GLCDC0 and GLCDC1 bus state, see the *ARM® AHB Specification*. Use this procedure even after a reset when the GLCDC0 and GLCDC1 bus, a target device, or a target controller is not initialized. If these are initialized on a reset assertion for the GLCDC, a normal startup procedure can be used without checking the GLCDC0 and GLCDC1 bus state.

64.3.2 Screen Definition

The essential signals for GLCDC operations are generated by the background screen generation module, and the graphics 1 and 2 modules and output control module operate based on the sequentially transferred vertical and horizontal synchronization signals (VS and HS) and vertical and horizontal pixel enable signals (VE and HE). The reference point (frame start point) of the background screen cannot be determined by the output signals. The point shown in the figures is virtual and provided only for register settings. The reference points (frame start points) of the graphics 1 and 2 modules and output control module are the assertion of the vertical synchronization signal (VS), which is input (output) from the previous stage. Each module defines the valid pixel display area and special processing area for pixel data (rectangular areas for graphics 1 and 2) according to the position and width based on this reference point. Figure 64.26 shows the definition of the background screen, and Figure 64.27 shows the definition of the graphics 1 and 2 screen. The output control module performs correction for the entire valid pixel area (when both VE and HE are 1) output from the previous stage (graphics 2). (No registers are provided in the output control module for setting the correction area.) TCON in the output control module

generates the control signals to be output based on the internal vertical and horizontal synchronization signals (VS and HS), and the change timing of the output signals can be freely modified within the valid setting range specifiable in the registers.

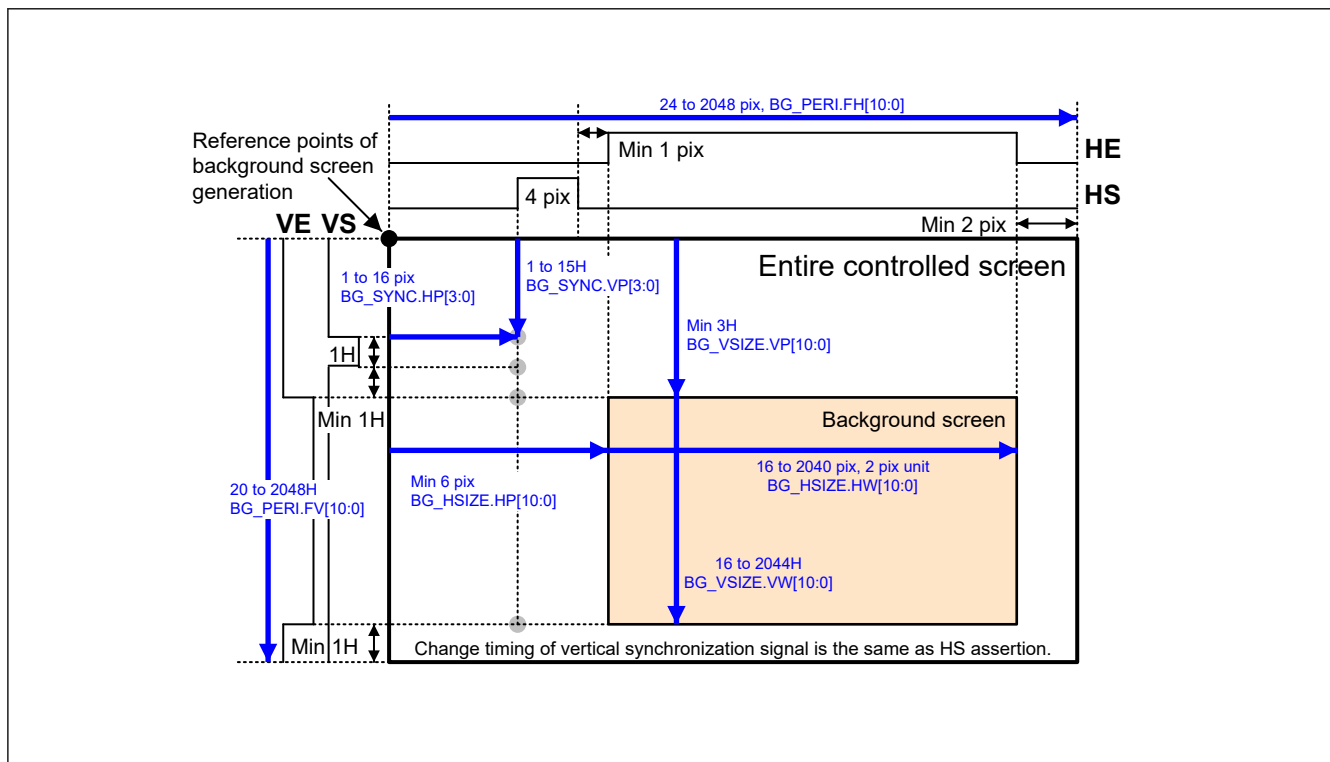


Figure 64.26 Definition of background screen

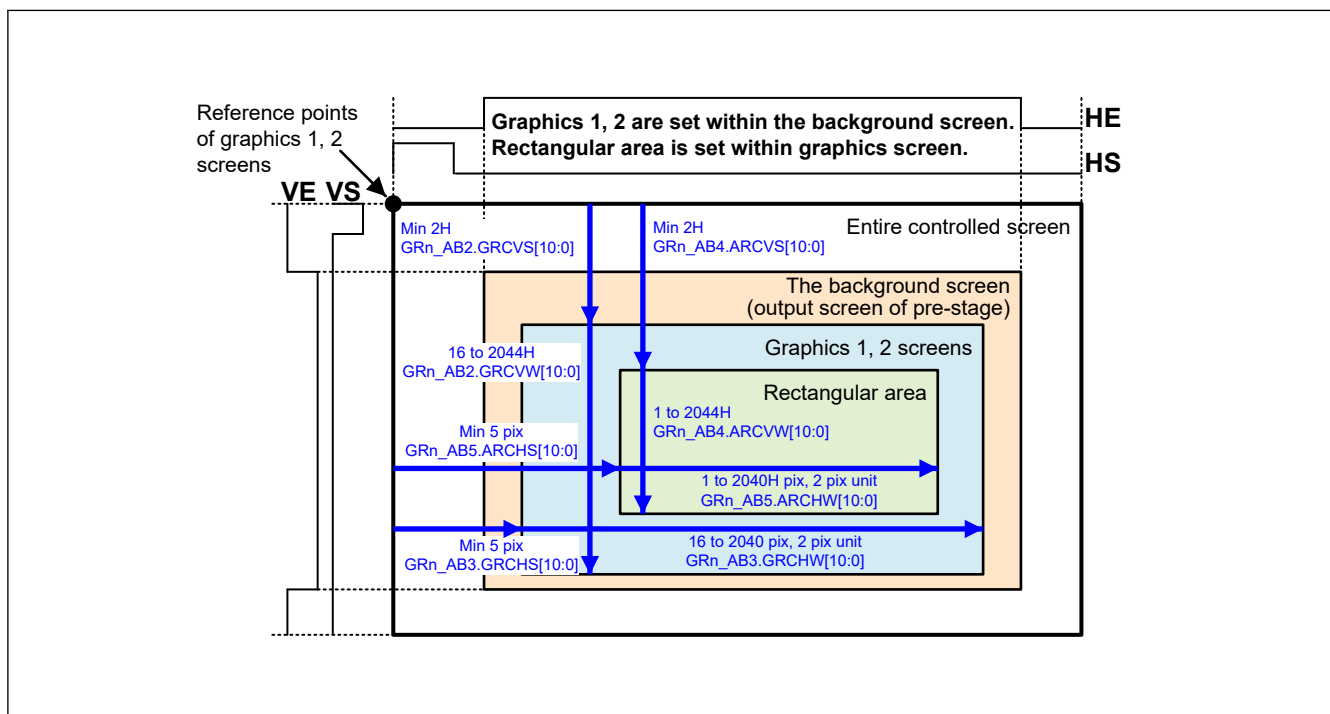


Figure 64.27 Definition of graphics 1 and 2 screens

64.3.3 Underflow and Interrupts

The GLCDC can detect the three types of status conditions described in this section. To detect each status condition, set the associated bit in the Status Detection Control Register (SYSCNT_DTCTEN) to 1.

64.3.3.1 Graphics 2 Underflow Detection

This function detects underflows in the graphics data interface block for graphics 2. The SYSCNT_STMON.L2UNDF flag sets to 1 if the graphics data cannot be read from the graphics data interface block (if valid data is not stored in the 4-stage ring buffer). The underflow is cleared as an internal state of graphics 2 on the VS (vertical synchronization) signal assertion of the previous stage. However, to clear the relevant bit in the Status Monitor Register of the system control block, the software must set the associated bit in the Status Clear Register to 1.

64.3.3.2 Graphics 1 Underflow Detection

This function detects underflows in the graphics data interface block for graphics 1. The SYSCNT_STMON.L1UNDF flag is set to 1 if the graphics data cannot be read from the graphics data interface block (if valid data is not stored in the 4-stage ring buffer). The underflow is cleared as an internal state of graphics 1 on the VS (vertical synchronization) signal assertion of the previous stage. However, to clear the relevant bit in the Status Monitor Register of the system control block, the software must set the associated bit in the Status Clear Register to 1.

64.3.3.3 Graphics 2 Line Detection

This function detects that the number of lines specified in the GR2_CLUTINT.LINE[10:0] bit was processed. The detection is performed on the HS (horizontal synchronization) signal assertion of the previous stage, not when valid pixels start to be processed. Each time the number of detected lines reach the value specified for graphics 2, the SYSCNT_STMON.VPOS flag sets to 1. As with underflow detection, to clear the relevant bit in the Status Monitor Register of the system control block, the software must set the associated bit in the Status Clear Register to 1.

64.3.3.4 Interrupts

The GLCDC provides three interrupt request output signals (GLCDC_L2UNDF, GLCDC_L1UNDF, and GLCDC_VPOS) that correspond to detection of the three status conditions. The GLCDC_L2UNDF, GLCDC_L1UNDF, and GLCDC_VPOS signals are associated with graphics 2 underflow detection, graphics 1 underflow detection, and graphics 2 line detection, respectively. Each of the interrupt request signals is asserted by setting the associated bit in the Interrupt Request Enable Register (SYSCNT_INTEN) to 1. Detecting the status and enabling the associated interrupt request can be controlled separately. Even if detection is enabled, the interrupt request signal is not asserted unless the interrupt request output signal is enabled. In addition, if the Status Monitor Register (SYSCNT_STMON) clears while the interrupt request output signal is asserted (the associated bit in the Status Clear Register (SYSCNT_STCLR) is set to 1), or if the associated bit in the Interrupt Request Enable Register (SYSCNT_INTEN) is cleared to 0, the interrupt request signal is negated. The interrupt request signal generation circuit is configured as shown in Figure 64.28. This circuit is glitch-free except for reset-induced glitches.

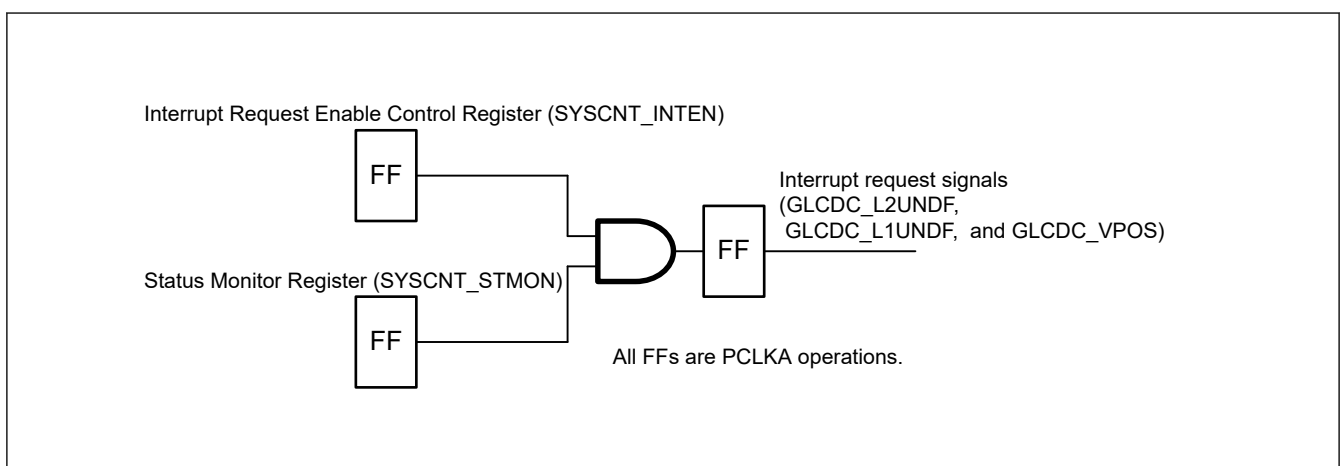


Figure 64.28 Interrupt request signal generation circuit

64.4 Usage Notes

64.4.1 Power Gating Control or Software Standby Mode

Before transitioning to graphics power domain off or software standby mode, stop the operation as described in the following sections.

See [\(3\) Stopping and restarting in normal operation](#).

After returning from graphics power domain off or software standby mode, resume operation as described in the following sections.

See [\(3\) Stopping and restarting in normal operation](#).

64.4.2 Module-Stop Function Setting

GLCDC operation can be disabled or enabled using Module Stop Control Register C (MSTPCRC). The GLCDC module is initially stopped after reset. Releasing the module-stop state enables access to the registers.